Introduction to Real Analysis
Lecture Notes

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Introduction

These notes were written to accompany an introductory course in Real Analysis given at MacEwan University in Edmonton, Alberta, Canada in the Fall of 2025. The main thrust of the exposition follows Chapters 1-8 of [13]. A number of secondary sources including [6], [8], [10], [11] were helpful, especially as sources of exercises and examples. The historical overview of the subject in this introduction is mostly based on the accounts in [1] and [9]. The sections on the mean value theorem and Taylor's theorem were partly written by N. Strungaru who was kindly covering my class at the time. Any errors which survived or were introduced when incorporating his materials into the present document are the sole responsibility of the author.

Real Analysis is, very roughly, the branch of mathematics that deals with estimation and convergence. It is also the branch of mathematics that turned Calculus into a subject that holds up to modern standards of mathematical rigour.

Calculus was discovered in the late 1600s by Newton and Leibniz. These discoveries were made independently (Newton slightly before Leibniz), but Newton accused Leibniz of stealing his ideas leading to a messy dispute the salacious details of which can be read about in [2]. Both Newton and Leibniz's approaches to Calculus involved a lot of heuristic methods. In the familiar notation $\frac{dy}{dx}$ (invented by Leibniz), the expressions dy and dx stand for infinitely-small but nonvanishing increments of the independent and dependent variables x and y. A rigorous theory of such "infinitesimal" quantities was not available at the time and so many arguments instead relied on heuristics about limits, intuition, and appeals to geometry. Early approaches to Calculus were missing two key ingredients:

- A precise framework for dealing limits. Nowadays, this is handled using the ϵ-δ definition of a limit.
- A precise understanding of "what is a real number?". Nowadays, this is achieved by axiomatizing (or constructing) the reals to have a certain "completeness property".

The following quotation from Newton's *Principia* gives some idea of the terms in which the ideas of Calculus had to be couched at the time of its discovery.

> It may also be objected, that if the ultimate ratios of evanescent quantities are given, their ultimate magnitudes will be also given: and so all quantities will consist of indivisibles ... But ... those ultimate ratios with which quantities vanish are not truly the ratios of ultimate quantities, but limits towards which

> the ratios of quantities decreasing without limit do always converge; and to which they approach nearer than by any given difference, but never go beyond, nor in effect attain to, till the quantities are diminished in infinitum.
>
> Newton (1687) [7, p. 103]

One early critic of the lack of rigour in Calculus was the Irish philosopher and clergyman Berkeley (after whom a famous university is now named). Berkeley was in fact a proponent of calculus, but argued that its foundations were not as rigorous as mathematicians of the day would like to believe. In fact, Berkeley felt that Calculus was no more a precise science than Theology.

> And what are these Fluxions? The Velocities of evanescent Increments? And what are these same evanescent Increments? They are neither finite Quantities nor Quantities infinitely small, nor yet nothing. May we not call them the Ghosts of departed Quantities? Berkeley (1734) [3, p. 59]

In spite of its lack of rigorous foundations, the newly-founded subject of Calculus flourished during an early period of 150 years or so. Calculus proved to be immensely useful to mathematicians, engineers, physicists, and other scientists. The subject expanded quickly, its development being largely spurred by the efforts by a number of "virtuoso mathematicians". In the Leibeniz camp, prominent figures included brothers Jakob and Johann Bernoulli and their illustrious student Euler. On the Newton side, one had Taylor and Maclaurin, whose names are now associated with infinite series. Other important figures include d'Lambert, Lagrange, Fourier and Riemann. This early period saw, for example:

- The invention by Newton of Taylor series.
- The establishment of notations we now take for granted such as x for a real variable, and the symbol $\int$ for an integral.
- The discovery of the constant e.
- The solution of the Basel problem $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$.
- The solution of the brachistochrone and hanging chain problem leading to the invention of the Calculus of Variations.
- Huge advances in the theory of Differential Equations.
- The invention of Fourier series.
- Higher-dimensional generalizations of the Fundamental Theorem of Calculus (Green's Theorem, Divergence Theorem, Stokes' Theorem).
- Modern approaches to Classical Mechanics (Lagrangian Mechanics).
- The Euler-Lagrange Equation and Laplace Equation.

All these things were achieved before Real Analysis became the precise science that it is today. What the subsequent introduction of mathematical rigour brought to the subject was a sort of democratizing effect. Although mathematical luminaries like the Bernoullis and Euler had been able to innovate in the subject in spite of its shaky foundations, the introduction of rigour made gaining a deep understanding of Calculus much easier to achieve. Nowadays limits, and sometimes ϵ-δ proofs, are taught even in introductory courses on Calculus. In short, with the arrival of rigorous foundations, one no longer needed to be a "virtuoso mathematician" to understand and meaningfully contribute to the development of Calculus.

The first steps towards rigorous foundations that substantially align with modern approaches were taken by Bolzano and Cauchy. Bolzano's work was not widely-known during his lifetime and only came to prominence after his death. Cauchy, on the other hand, was considered the foremost mathematician in France at the time (and a difficult person to deal with!) and his writings were more influential. The following quotations make clear that Bolzano and Cauchy were both thinking about limits in essentially the modern way.

> According to a correct definition, the expression that a function $f(x)$ varies according to the law of continuity for all values of x inside or outside certain limits means just that: if x is some such value, the difference $f(x+w) - f(x)$ can be made smaller than any given quantity provided w can be taken as small as we please. Bolzano (1817) [4]

> If the successive values attributed to the same variable approach indefinitely a fixed value, such that they finally differ from it by as little as one wishes, this latter is called the limit of all the others. Cauchy (1821) [9, p. 641]

While it is clear that Bolzano and Cauchy were on the right track, the central definitions of Real Analysis are not yet fully-fledged in their writing. Indeed, on the topic of infinite series, Cauchy's writing contains some imprecise points and (arguably) some incorrect claims concerning the continuity of sums of continuous functions. Dirichlet was the first to notice these issues and supply counterexamples.

It is only a little later, in the writings of Weierstrass, that one is able to clearly recognize the main ideas of the subject of Analysis as they are still understood today. Thus, although Bolzano and Cauchy both deserve some credit for the modern ϵ-δ approach to limits, it is usually Weierstrass who is referred to as the "father of modern analysis". One could imagine encountering the quotation below in a modern day textbook:

> That a variable quantity x becomes infinitely small simultaneously with another quantity y means: "After the assumption of an arbitrarily small quantity ϵ a bound δ for x may be found, such that for every value of x for which $|x| < \delta$, the corresponding value of $|y|$ will be less than ϵ." Weierstrass (1877) [14]

The next step in the evolution of Real Analysis was referred to by Kronecker as the "Arithmetization of Analysis". It was gradually recognized that the subject of Analysis

could not be truly precise until a precise understanding of the system of real numbers was also gained. There are two ways of dealing with this issue: a synthetic approach wherein one simply lists all the axioms one believes the real numbers should satisfy and a constructive approach wherein one constructs the system of real numbers out of more primitive objects; in the words of Kronecker "God created the natural numbers, all else is the work of man". The name most often associated with the arithmetization program is Dedekind:

> The statement is so frequently made that the differential calculus deals with continuous magnitude, and yet an explanation of this continuity is nowhere given; even the most rigorous expositions of the differential calculus . . . depend upon theorems which are never established in a purely arithmetic manner. . . . It then only remained to discover the true origin [of these theorems] in the elements of arithmetic and thus at the same time to secure a real definition of the essence of continuity. Dedekind (1858) [5]

Dedekind gave a satisfying answer to the question "what is a real number?" by constructing a model of the real numbers from the rational numbers using what are now called "Dedekind cuts". The key property of the real numbers constructed by Dedekind is a completeness property. Typically, this completeness takes the form of the "least upper bound property", a property of the ordering of the real numbers. It is interesting to note that Bolzano had already formulated a version of this property somewhat earlier.

> If a property M does not apply to all values of a variable x but does apply for all values smaller than a certain u, then there is always a quantity U, which is the greatest of those of which it can be asserted that all smaller x possess the property M. Bolzano (1817) [4, p. 41]

Dedekind's construction of the reals is not the only possible one. Another standard approach defines the reals using "Cauchy sequences" of rational numbers. This naturally leads to the reals having a different kind of a completeness property, namely metric completeness. It turns out that metric completeness and order completeness lead to equivalent theories.

The program of precisely understanding what is a real number also lead to the creation of Set Theory by Cantor. The following quotation appears in a letter written from Cantor to Dedekind.

> Take the collection of all positive whole numbers n and denote it by (n); then think of the collection of all real numbers x and denote it by (x); the question is simply whether (n) and (x) may be corresponded so that each individual of one collection corresponds to one and only one of the other? . . . As much as I am inclined to the opinion that (n) and (x) permit no such unique correspondence, I cannot find the reason. Cantor (1874)

In this letter, Cantor asks whether the set of whole numbers and the set of real numbers have the same cardinality. Later, Cantor was able to prove that the answer is "no", there

are many more real numbers than there are whole numbers. This discovery lead Cantor to found the subject of Set Theory. Nowadays, most of modern mathematics, including Real Analysis, is built on the theory of sets. Alternatives foundations to Set Theory do exist, but one is highly unlikely to need or encounter them in an undergraduate degree.

Another extremely significant milestone in the history of Real Analysis is the development of Measure Theory and the Lebesgue Integral. In the early 1900s, Lebesgue introduced a more flexible and powerful theory of integration that has essentially supplanted the earlier-defined Riemann Integral in more advanced applications. In this introductory course, we will discuss what it means for a set of reals to have "measure zero", but not delve any further into the subject of Measure Theory than that.

A final footnote in the early history of Real Analysis is a sort of redemption arc for the concept of an infinitesimal. In the 1960s, Robinson created a subject known as Nonstandard Analysis. In this theory, infinitesimal numbers (infinitely small numbers different from zero) become precise mathematical objects which can be manipulated somewhat similarly to ordinary reals. In Robinson's theory, it is possible to formulate the derivative using honest ratios of infinitesimals. One may view Nonstandard Analysis as somewhat vindicating the methods used by mathematicians who did their work in the free-wheeling, semi-rigorous world of pre-Weierstrass Calculus. It should be noted that Nonstandard Analysis has not caught on in the same way that more traditional limit-based methods have. Very few mathematicians use Nonstandard Analysis in any serious way.

Chapter 1

The real numbers

1.1 Fields

The following famous quotation:

> God created the natural numbers, all else is the work of man.
>
> Kronecker (1886)

encapsulates a certain philosophy towards foundations in mathematics, namely that we should only assume the existence of the integers $\mathbb{Z}$ and build all other objects from scratch. So, we should construct the rationals $\mathbb{Q}$ from $\mathbb{Z}$ using the algebra of fractions. We can then construct the real numbers $\mathbb{R}$ from the rationals using one of several approaches, most often Dedekind cuts or Cauchy sequences. It is actually possible to start at an even more basic level by not assuming even the existence of the integers and starting from the language of sets. One can then build the integers using what is called von Neumann cardinal assignment. In this scheme, each natural number is recursively defined to be a certain set: the number zero is defined as the empty set and the successor $x + 1$ of each nonnegative integer x is defined by $x + 1 \coloneqq x \cup \{x\}$.

$$0 \coloneqq \varnothing \qquad 1 \coloneqq \{0\} \qquad 2 \coloneqq \{0, 1\} \qquad 3 \coloneqq \{0, 1, 2\} \qquad \dots$$

This is not a course in logic and foundations so we will not follow Kronecker's advice. Instead, we adopt what is sometimes called a "synthetic approach". We will simply list all the properties that we want the real numbers to have as axioms and assume the existence of a number system called the reals which satisfies them. First we would like to be able to perform arithmetic in $\mathbb{R}$ in the manner we are accustomed to. This is accomplished by requiring that $\mathbb{R}$ be an object called a field:

Definition 1.1.1. A **field** is a set $\mathbb{F}$ equipped with two binary operations $\mathbb{F} \times \mathbb{F} \to \mathbb{F}$

$$\text{Addition: } (x, y) \mapsto x + y \qquad\qquad \text{Multiplication: } (x, y) \mapsto xy$$

such that the following nine axioms are satisfied:

A1. Commutativity of addition: For all $x, y \in \mathbb{F}$, $x + y = y + x$.

A2. Associativity of addition: For all $x, y, z \in \mathbb{F}$, $(x + y) + z = x + (y + z)$.

A3. Additive identity: There exists an element denoted $0 \in \mathbb{F}$ such that $0 + x = x + 0 = x$ for all $x \in \mathbb{F}$.

A4. Additive inverses: For every $x \in \mathbb{F}$ there exists an element $-x \in \mathbb{F}$ such that $x+(-x) = (-x) + x = 0$.

M1. Commutativity of multiplication: For all $x, y \in \mathbb{F}$, $xy = yx$.

M2. Associativity of multiplication: For all $x, y, z \in \mathbb{F}$, $(xy)z = x(yz)$.

M3. Multiplicative identity: There exists a nonzero element $1 \in \mathbb{F}$ such that $1x = x1 = x$ for all $x \in \mathbb{F}$.

M4. Multiplicative inverses: For every $x \in \mathbb{F}$ with $x \neq 0$ there exists an element denoted $x^{-1} \in \mathbb{F}$ such that $xx^{-1} = x^{-1}x = 1$.

AM1. Distributivity: For all $x, y, z \in \mathbb{F}$, $(x + y)z = xz + yz$.

Remark 1.1.2. The attentive reader may have noticed that Definition 1.1.1 is a bit incomplete because we did not spell out the order-of-operations relationship between addition and multiplication to be used in expressions involving both operations. We have left it implicit that this works in the usual way, with multiplication taking precedence over addition in the absence of parentheses. We will also use other standard notational devices without comment such sometimes using a dot instead of juxtaposition to indicate multiplication when doing so improves readability.

Example 1.1.3. Both the rationals $\mathbb{Q}$ and the real numbers $\mathbb{R}$ are fields. Another example of a field is the set $\{0, 1\}$ with $1 + 1 = 0$ and all other sums and products defined in the "obvious" way: $0 + 0 = 0$, $0 + 1 = 1$, $0 \cdot 0 = 0 \cdot 1 = 0$, $1 \cdot 1 = 1$. This is the smallest example of a "finite field", which students familiar with abstract algebra will already know about.

Starting with the axioms in Definition 1.1.1 one can prove other familiar algebraic properties. For example, if $a + b = a + c$, one is used to concluding that $b = c$ by "cancelling" a from both sides. Later in the course we will use such familiar properties without comment, but at this stage it is important to realize that this property, and many others like it, can be proven straight from the axioms. Such additional properties of fields are results, not assumptions.

Exercise 1.1.4. Let $\mathbb{F}$ be a field and let $a, b, c \in \mathbb{F}$. Prove that if $a + b = a + c$, then $b = c$.

Proof. By A4, there exists an additive inverse $-a \in \mathbb{F}$ such that $-a+a = 0$. Since $a+b = a+c$ by assumption, we have $-a + (a + b) = -a + (a + c)$. Using the associative property (A2), we may conclude that $(-a + a) + b = (-a + s) + c$. That is, $0 + b = 0 + c$. So, using the defining property of the additive identity (A3), we have $b = c$. □

Exercise 1.1.5. Let $\mathbb{F}$ be a field. Prove that $0 \cdot x = 0$ for all $x \in \mathbb{F}$.

Proof. Since 0 is an additive identity (A4), we have $0 + 0 = 0$. Combining this observation with the distributive property (AM1), we have $0 \cdot x = (0 + 0) \cdot x = 0 \cdot x + 0 \cdot x$. Using A4 again to add 0 on the left, we have

$$0 \cdot x + 0 = 0 \cdot x + 0 \cdot x.$$

Using Exercise 1.1.4 to cancel $0 \cdot x$ from both sides, we get the desired conclusion $0 = 0 \cdot x$ □

Exercise 1.1.6. Let $\mathbb{F}$ be a field. Prove that $(-1)^2 = 1$ holds in $\mathbb{F}$.

Proof. By definition, -1 is the additive inverse (A4) of 1 and so $-1+1 = 0$. By Exercise 1.1.5 above, it follows that $(-1 + 1)(-1) = 0 \cdot (-1) = 0$. Using the distributive property AM1 and the fact that 1 is the multiplicative identity, we have

$$0 = (-1 + 1)(-1) = (-1)^2 + 1 \cdot (-1) = (-1)^2 + (-1).$$

We want to deduce the conclusion $1 = (-1)^2$ from $0 = (-1)^2 + (-1)$ by "moving -1 to the other side". Technically, doing this involves the associative law (A2), the fact -1 is the additive inverse of 1 (A4), and the fact that 0 is the additive identity (A3), as shown below:

$$1 = 0 + 1 = ((-1)^2 + (-1)) + 1 = (-1)^2 + (-1 + 1) = (-1)^2 + 0 = (-1)^2.$$

□

1.2 Ordered fields

Going beyond arithmetic, another key feature we want the real numbers to have is a notion of ordering. We need to be able to compare two numbers and decide which is bigger. We also want this ordering to interact with the field operations in the way we expect. Abstractly speaking, this means we want $\mathbb{R}$ to be an object called an ordered field.

Definition 1.2.1. An **ordered field** is a field $\mathbb{F}$ equipped with a relation denoted $<$ that satisfies the following properties:

O1. Trichotomy: If $x, y \in \mathbb{F}$, exactly one of the statements $x < y$, $y < x$, $x = y$ is true.

O2. Transitivity: If $x, y, z \in \mathbb{F}$ satisfy $x < y$ and $y < z$, then $x < z$.

O3. Translation invariance: If $x, y, t \in \mathbb{F}$, then $x < y$ implies $x + t < y + t$.

O4. Positive scale invariance: If $x, y, s \in \mathbb{F}$ and $s > 0$, then $x < y$ implies $sx < sy$.

Note: a relation $<$ defined on any set satisfying O1, O2 is called a **total order**.

Example 1.2.2. The rational numbers $\mathbb{Q}$ and real numbers $\mathbb{R}$ are both examples of ordered fields. A comment for those familiar with abstract algebra: one can prove that every ordered field necessarily has characteristic zero. In particular, no finite field can be made into an ordered field. There are also some fields of characteristic zero that cannot be made into ordered fields; there is no way to order the elements of the field $\mathbb{C}$ of complex numbers which satisfies the axioms O1, O2, O3, O4.

Exercise 1.2.3. Let $\mathbb{F}$ be an ordered field. Prove that $-1 < 0 < 1$.

Proof. Since $0 \neq 1$ by the field axiom M3, the trichotomy axiom O3 implies either $0 < 1$ or $1 < 0$. Suppose for contradiction that $1 < 0$. Then, by translation invariance (O3), we have $0 = -1 + 1 < -1 + 0 = -1$. However, if $0 < -1$, then positive scale invariance gives $0 = (-1)(0) < (-1)(-1) = 1$, where Exercise 1.1.5 and Exercise 1.1.6 were both used, contradicting the supposition that $1 < 0$. Thus we have proven $0 < 1$, which was one of our desired statements. Adding -1 to both sides of this inequality and using translation invariance (O3) as above, we get the other desired statement $-1 < 0$. □

Remark 1.2.4. As usual, we will use some extra symbols besides $<$ when talking about total order. The notations $x < y$ and $y > x$ mean the same thing. The notations $x \leq y$ and $y \geq x$ mean that $x < y$ or $x = y$. The relations $\leq$ and $\geq$ don't satisfy trichotomy, of course, but one can check they are transitive and, in the ordered field context, that they satisfy translation invariance and positive scale invariance.

1.3 Upper and lower bounds

Next we review some vocabulary which will be essential for talking about total orders, and the ordering of $\mathbb{R}$ in particular.

Definition 1.3.1. Let $\mathbb{F}$ be an ordered field[1]. Let $S \subseteq \mathbb{F}$ be a subset of $\mathbb{F}$.

- We say that $M \in \mathbb{F}$ is an **upper bound** for S if $x \leq M$ for all $x \in S$. We say that $m \in \mathbb{F}$ is a **lower bound** for S if $m \leq x$ for all $x \in S$.
- We say that S is **bounded above** if it has an upper bound. We say that S is **bounded below** if it has a lower bound. We say that S is **bounded** if it is both bounded above and bounded below.
- We say that $M \in \mathbb{F}$ is the **maximum** of S and write $M = \max S$ if M is an upper bound for S and $M \in S$. We say that $m \in \mathbb{F}$ is the **minimum** of S and write $m = \min S$ if m is a lower bound for S and $m \in S$.

Example 1.3.2. Although we have not yet achieved our goal of precisely defining the ordered field of real numbers, let's nonetheless illustrate the above definitions using subsets $S \subseteq \mathbb{R}$.

[1]More generally, this definition still makes sense when $\mathbb{F}$ is any set carrying a total order. It is not essential that $\mathbb{F}$ is an ordered field.

- The set $S = \mathbb{N}$ of positive integers has no upper bound in $\mathbb{R}$ (this is called the "Archimedean property" of $\mathbb{R}$ and will be discussed a bit later). On the other hand $\mathbb{N}$ does have lower bounds. In fact, it has a smallest element, $\min S = 1$.

- The set $S = [0, 1)$ has infinitely many upper bounds in $\mathbb{R}$. Any real number $M \geq 1$ such as $M = 2, \pi, 500$ is an upper bound for S. Among all possible upper bounds, $M = 1$ is the *least* upper bound. Note however that 1 is not considered the maximum of S because $1 \notin S$ (refer to the definition!). Indeed, the set S has no maximum.

- The set $S = [0, 1]$ has both a smallest and a largest element: $\min S = 0$, $\max S = 1$.

Definition 1.3.3. Let $\mathbb{F}$ be an ordered field[2].

- Let $S \subseteq \mathbb{F}$ be a nonempty subset that is bounded above. We say that $M \in \mathbb{F}$ is the **least upper bound** or **supremum** of S and write $M = \sup S$ if M is an upper bound for S and $M \leq M'$ for every upper bound $M' \in \mathbb{F}$ of S.

- Let $T \subseteq \mathbb{F}$ be a nonempty subset that is bounded below. We say that $m \in \mathbb{F}$ is the **greatest lower bound** or **infimum** of T and write $m = \inf T$ if m is a lower bound for T and $m \geq m'$ for every lower bound $m' \in \mathbb{F}$ of T.

Whether or not a set S has an infimum/supremum can depend on the ordered field $\mathbb{F}$ that we are thinking of S as a subset of. The following example illustrates this sensitivity to the ambient field.

Example 1.3.4. When we think of

$$S = \{x \in \mathbb{Q} : x^2 < 2\} = \{x \in \mathbb{Q} : -\sqrt{2} < x < \sqrt{2}\}$$

as a subset $S \subseteq \mathbb{Q}$ of the ordered field of rational numbers, we find that S has no supremum or infimum **that belongs to** $\mathbb{Q}$. The issue is that the endpoints of S "want to be" $\pm\sqrt{2}$, but those are not rational numbers. There is no smallest element of the set of rational upper bounds for S and there is no largest element of the set of rational lower bounds for S. If we instead think of S as a subset of the larger $\mathbb{R}$ of real numbers, the problem goes away and it makes sense to write $\sup S = \sqrt{2}$, $\inf S = -\sqrt{2}$.

1.4 Completeness axiom of the real numbers

The existence of suprema (or equivalently, as it turns out, the existence of infima) is the key property that differentiates the field of rational numbers from the field of real numbers. Intuitively, one thinks of $\mathbb{Q}$ as being full of holes with many numbers missing and thinks of $\mathbb{R}$ as being a smooth connected continuum with no holes or missing numbers. We are introduced to the mental picture of $\mathbb{R}$ as an idealized "number line" early in our mathematical education. The following definition is the key to making this intuition rigorous.

[2]Again, for purposes of this definition, it is OK if $\mathbb{F}$ is just a set with a total order.

Definition 1.4.1. Let $\mathbb{F}$ be an ordered field (or more generally, any set with a total order). We say that $\mathbb{F}$ has the **least upper bound property** if every nonempty subset $S \subseteq \mathbb{F}$ that is bounded above has a least upper bound $\sup S \in \mathbb{F}$. Similarly, we say that $\mathbb{F}$ has the **greatest lower bound property** if every nonempty subset $S \subseteq \mathbb{F}$ that is bounded below has a greatest lower bound $\inf S \in \mathbb{F}$.

Remark 1.4.2. Actually, the least upper bound property and greatest upper bound property turn out to be equivalent to each other: if one property holds, then the other property holds automatically. In an ordered field, we can prove this by exploiting the negation operation $x \mapsto -x$ to "flip over" the number line. This proof may lend one the impression that the algebraic structure of an ordered field plays an important role in establishing the equivalence of the least upper bound and greatest lower bound property. However, the equivalence of these properties actually works at a more fundamental level. One can prove their equivalence for any set (not necessarily a field) equipped with a total order.

Exercise 1.4.3. Prove that a set with a total order has the least upper bound property if and only if it has the greatest lower bound property.

In this course we do not have time to develop the theory of the field of real numbers entirely from scratch, so we instead make the following assumption and move on:

Postulate 1.4.4. There exists an ordered field called the **real numbers** denoted $\mathbb{R}$ which has the least upper bound property or (equivalently) the greatest lower bound property.

To be clear, there are several aspects of the theory of the real numbers that we are skipping over. Most importantly:

- Existence: it is possible to construct a model of $\mathbb{R}$ from $\mathbb{Q}$ (using Dedekind cuts or other methods) and *prove* that it has the least upper bound property.
- Uniqueness: it is possible to prove that any other ordered field having the least upper bound property is the "same" as $\mathbb{R}$. In more technical language: if $\mathbb{F}$ is any ordered field with the least upper bound property, there is a unique order-preserving field isomorphism $\mathbb{R} \to \mathbb{F}$.

Remark 1.4.5. It is common to refer to total orders having the least upper bound/greatest lower bound property as being **Dedekind complete**, **order complete**, or even just **complete**. One should take care that there are some other (different) concepts in analysis and other areas of math that are also referred to as completeness.

Remark 1.4.6. Our starting point for a theory of the real number system is the postulate that $\mathbb{R}$ is an ordered field with the least upper bound property. Notice this does not give for free that various familiar real numbers such as $\sqrt{2}$, e and π exist. In a rigorous treatment, such constants need to first be defined and then proven to exist; one should isolate some key property that the desired constant ought to have and then prove there is a unique real

number with that property. For example, $\sqrt{2}$ is usually defined as the unique[3] positive real solution to the quadratic equation $x^2 = 2$. However, taking Postulate 1.4.4 as our starting point, it still needs to be proven that an ordered field with the least upper bound property must contain solutions to $x^2 = 2$. This will be left to the homework.

1.5 Archimedean property

Definition 1.5.1. An ordered field $\mathbb{F}$ is said to be **Archimedean** if there does not exist an upper bound $M \in \mathbb{F}$ for the set $\mathbb{N} \subseteq \mathbb{F}$ of positive integers. Equivalently (exercise) $\mathbb{F}$ is Archimedean if for every $x \in \mathbb{F}$ there exists $n \in \mathbb{N}$ such that $x \leq n$.

Remark 1.5.2. In the above definition, we implicitly assumed that the ordered field $\mathbb{F}$ contains the positive integers $\mathbb{N}$. Actually, more is true: every ordered field $\mathbb{F}$ contains the rationals $\mathbb{Q}$. We always have $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{F}$. Technically, this is an instance of what mathematicians call "abuse of notation". It would be more accurate to say that $\mathbb{F}$ contains a "canonical copy of $\mathbb{Q}$". The point is that $\mathbb{F}$ contains a multiplicative identity conventionally denoted $1 \in \mathbb{F}$. This notation already suggests it is harmless to conflate the integer 1 with the field element 1. Similarly, we conflate $-2 \in \mathbb{Z}$ with $-(1+1) \in \mathbb{F}$ and $2/3 \in \mathbb{Q}$ with $(1+1)(1+1+1)^{-1} \in \mathbb{F}$ and so on[4]. It is a harmless practice to write $\mathbb{Q} \subseteq \mathbb{F}$ even though, depending on the precise nature of $\mathbb{F}$, it may not be literally true that $\mathbb{Q}$ is a subset of $\mathbb{F}$.

Both $\mathbb{Q}$ and $\mathbb{R}$ are Archimedean. This matches our intuition for the way the integers sit in the number line. Interestingly, while the fact that $\mathbb{Q}$ is Archimedean can be proved just with a bit of algebra (clearing denominators of fractions, and so on), the fact that $\mathbb{R}$ is Archimedean requires the completeness property to be proved.

Proposition 1.5.3. *The ordered field $\mathbb{Q}$ of rational numbers is Archimedean.*

Proof. We must show that, for every $x \in \mathbb{Q}$, there exists $n \in \mathbb{N}$ such that $x \leq n$. If $x \leq 0$, then $n = 1$ does the job. Let us suppose then that $x > 0$. Write $x = a/b$ where a, b are positive integers. Then, $x \leq bx = a$, so $n = a$ does the job. □

Proposition 1.5.4. *The ordered field $\mathbb{R}$ of real numbers is Archimedean.*

Proof. Suppose for contradiction that there exists a real number which is an upper bound for $\mathbb{N}$. Then, $\mathbb{N}$ is nonempty and bounded above so, since $\mathbb{R}$ has the least upper bound propery, the supremum $M = \sup \mathbb{N} \in \mathbb{R}$ exists. Since $M - 1 < M$ and M is the least upper bound for $\mathbb{N}$, the real number $M - 1$ is not an upper bound for $\mathbb{N}$, i.e. there exists a positive

[3] If a field $\mathbb{F}$ happens to contain an element α such that $\alpha^2 = 2$, then it is fairly easy to show $x = \pm\alpha$ are the only elements of $\mathbb{F}$ that solve the quadratic equation $x^2 = 2$. Moreover, if $\mathbb{F}$ is also an ordered field, then exactly one of $\pm\alpha$ is a positive element.

[4] The fact that $\mathbb{F}$ is an ordered field does play a role here as it rules out the possibility of repeatedly adding 1 to itself and eventually getting the result zero. This cannot happen in the ordered field case because $0 < 1 < 1 + 1 < 1 + 1 + 1 < \ldots$ but it can happen in, for example, finite fields. Students with a background in abstract algebra may already be aware that every field of characteristic zero has a copy of $\mathbb{Q}$.

integer $n \in \mathbb{N}$ such that $M - 1 < n$. But then, we see that $M < n + 1$ where $n + 1 \in \mathbb{N}$, contradicting the fact that M is an upper bound for $\mathbb{N}$. We conclude that $\mathbb{N}$ has no upper bound in $\mathbb{R}$. □

The Archimedean property can be characterized in several equivalent ways.

Proposition 1.5.5. *Let $\mathbb{F}$ be an ordered field. The following statements are equivalent:*

(i) $\mathbb{F}$ is Archimedean, meaning for all $x \in \mathbb{F}$ there exists $n \in \mathbb{N}$ such that $n > x$.

(ii) For all $\epsilon > 0$ in $\mathbb{F}$ there exists $n \in \mathbb{N}$ such that $1/n < \epsilon$.

(iii) For all $a, b > 0$ in $\mathbb{F}$ there exists $n \in \mathbb{N}$ such that $na > b$.

Proof. First we assume (i) and prove (ii). Let $\epsilon \in \mathbb{F}$ be positive. By (i), there exists $n \in \mathbb{N}$ such that $n > 1/\epsilon$. Multiplying the latter inequality through be the positive element ϵ/n, we get $\epsilon > 1/n$, as needed. Next, we assume (ii) and prove (iii). Let $a, b \in \mathbb{F}$ be positive. Since a/b is positive, (ii) implies there exists $n \in \mathbb{N}$ such that $1/n < a/b$. Multiplying the latter inequality through by the positive element nb, we get $b < na$, as needed. Lastly, we assume (iii) and prove (i). Let $x \in \mathbb{R}$ be given. If $x \leq 0$, then taking $n = 1 \in \mathbb{N}$ gives $n > x$, so we may assume $x > 0$. Taking $a = 1, b = x > 0$, we have by (iii) that there is an $n \in \mathbb{N}$ such that $na = b > b = x$, as needed. □

In more informal (but also more descriptive) language, the above characterizations of the Archimedean property are roughly as follows:

(i) $\mathbb{F}$ has no "infinite" elements.

(ii) $\mathbb{F}$ has no "infinitesimal" elements.

(iii) No positive element of $\mathbb{F}$ is "infinitely smaller" than another.

Children learning about the decimal system often entertain themselves by contemplating the existence of "numbers" like these:

$$1\underbrace{00000\ldots00000}_{\text{infinitely-many 0s}} \qquad\qquad 0.\underbrace{00000\ldots00000}_{\text{infinitely-many 0s}}1$$

Later on, we come to understand that the real number system does not contain any huge "infinitely long integers" or teeny tiny decimal quantities "just next to zero". Saying that $\mathbb{R}$ is an Archimedean field is a sort of formalization of this. On a more serious note, Calculus is often explained using infinitesimal quantities dx and dy, numbers which we think of as being infinitely small yet nonzero. Armed with enough experience and intuition, one can even do many correct calculations using the heuristics of infinitesimals. For example, suppose $y = x^2$ and x is changed by the infinitesimal dx. Then, the resulting infinitesimal change in y is

$$dy = (x + dx)^2 - x^2 = x^2 + 2x \cdot dx + (dx)^2 - x^2 = 2x \cdot dx + (dx)^2.$$

Then one might assert that, since dx is already vanishingly small the number $(dx)^2$ is even more vanishingly small, so much so that it can be safely disregarded. Ignoring this term we have $dy = 2x \cdot dx$ which leads directly to the familiar expression $\frac{dy}{dx} = 2x$ for the derivative of $y = x^2$. This sort of heuristic manipulation with infinitesimals is very handy and used fairly commonly, especially in physics and engineering. However, since $\mathbb{R}$ is not an Archimedean field, it does not in fact contain any infinitesimals dx, dy. Standard approaches to Real Analysis do not use infinitesimals, but instead rely on the concept of a limit.

1.6 Density

Definition 1.6.1. A subset D of $\mathbb{R}$ is said to be **dense** if, whenever $a, b \in \mathbb{R}$ and $a < b$, there exists an element $d \in (a, b) \cap D$.

To say that D is dense is to say that its points are somehow "ubiquitous" in the real line. No open interval of $\mathbb{R}$, no matter how small or how obscurely placed, can avoid containing a point of D.

Proposition 1.6.2. *$\mathbb{Q}$ is a dense subset of $\mathbb{R}$.*

Proof. Suppose that $a, b \in \mathbb{R}$ and $a < b$. Let us first consider the special case where $a > 0$ and return to the case $a \leq 0$ at the end. Since $b-a > 0$, the Archimedean property of $\mathbb{R}$ gives that there exists a positive integer n such that $1/n < b-a$. Using the Archimedean property again (see (iii) of Proposition 1.5.5) there exists a positive integer m such that $m/n > a$. Using the well-ordering principle (see Proposition 1.7.3), we may choose m to be *smallest* positive integer such that $m/n > a$. From this choice it follows that $(m-1)/n \leq a$. We claim that $a < m/n < b$. Suppose for contradiction that $m/n \geq b$. Then, $1/n = m/n-(m-1)/n \geq b-a$, which is a contradiction. Thus, $a < m/n < b$ as desired. To conclude, we consider the case $a \leq 0$. Then there exists a positive integer N such that $a + N > 0$ (exercise: check this using the Archimedean property). By the special case just established, there exists a rational number q with $a + N < q < b + N$. Then, $q' = q - N$ is also rational and satisfies $a < q' < b$, as desired. □

It is also true that the complement of the rationals, the set of irrational numbers, is dense in $\mathbb{R}$. Let us first recall one of the simplest and best known examples of an irrational number.

Proposition 1.6.3. *$\sqrt{2}$ is not rational.*

Proof. Suppose for contradiction that $\sqrt{2}$ is rational. Then we may write it as a lowest terms fraction $\sqrt{2} = m/n$ for $m, n \in \mathbb{Z}$, $n \neq 0$. Since $2 = (\sqrt{2})^2 = m^2/n^2$, we have $2n^2 = m^2$. Thus, m^2 is even which implies m is even. We may therefore write $m = 2k$ where $k \in \mathbb{Z}$. From $2n^2 = m^2 = (2k)^2 = 4k^2$ we obtain $n^2 = 2k^2$. Thus, n^2 is even, which implies n is even. This contradicts the hypothesis that m/n is in lowest terms. □

Proposition 1.6.4. *The sum or difference of a rational number and an irrational number is irrational.*

Proof. Let $x \in \mathbb{R} \setminus \mathbb{Q}$, $q \in \mathbb{Q}$. Suppose for contradiction that $x + q$ was a rational number q'. Then, $x = x + q - q = q - q'$ is the difference of two rationals and hence rational. Similarly, $x - q$ is irrational. □

Proposition 1.6.5. *The set of irrational numbers is dense in* $\mathbb{R}$.

Proof. Fix an open interval $(a, b) \subseteq \mathbb{R}$ where $a < b$. Since $(a - \sqrt{2}, b - \sqrt{2})$ is also an open interval and $\mathbb{Q}$ is dense, there exists a rational number $q \in \mathbb{Q}$ such that $a - \sqrt{2} < q < b - \sqrt{2}$. Thus, $a < q + \sqrt{2} < b$. Since $q + \sqrt{2}$ is irrational by Proposition 1.6.4, we have found an irrational number in (a, b) as required. □

1.7 Induction

The principle of induction, which we now recall, is a fundamental proof technique. Actually we recall three principles: induction, strong induction and the well-ordering principle. These three principles are equivalent; any one of them can be used to prove the other two.

Whether induction is an axiom or a theorem depends on the context:

- The theory of the integers can be developed from a list of axioms much as we have developed the theory of the real numbers from the ordered field axioms and the completeness axiom. One common set of axioms for the positive integers is called the Peano axioms. In such approaches, the principle of induction is one of the axioms.

- It is also possible to construct the integers from more other ingredients. For instance, one can construct the integers by starting with the more primitive theory of sets. Alternatively, if the existence of the reals has already been assumed, one can define $\mathbb{N}$ to be the smallest subset of $\mathbb{R}$ containing 1 and closed under the operation of adding 1. In such approaches, the principle of induction can be proved as a theorem. Philosophically, one might complain that using the reals to construct the integers may feel a bit backward, however. Generally, one would prefer to construct more basic objects like the integers first before moving on to the reals.

We will simply state the three principles and use them freely in proofs.

Proposition 1.7.1 (Induction)**.** *Suppose that $P(n)$ is a mathematical statement for every positive integer n and both of the following are satisfied:*

- *$P(1)$ is true.*

- *If $P(n)$ is true for some $n \geq 1$, then $P(n + 1)$ is true too.*

Then $P(n)$ is true for all $n \geq 1$.

Proposition 1.7.2 (Strong induction)**.** *Suppose that $P(n)$ is a mathematical statement for every positive integer n and both of the following are satisfied:*

- *$P(1)$ is true*
- *If $n \geq 1$ and $P(k)$ is true when $1 \leq k \leq n$, then $P(n+1)$ is true too.*

Then $P(n)$ is true for all $n \geq 1$.

Proposition 1.7.3 (Well-Ordering Principle)**.** *If S is a nonempty set of positive integers, S has a smallest element.*

Example 1.7.4. We take the Fibonacci sequence $(F_n)_{n\in\mathbb{N}}$ to be defined by the recurrence:

$$F_1 = 1 \qquad F_2 = 1 \qquad F_n = F_{n-1} + F_{n-2} \text{ for } n \geq 3.$$

Conjecture a formula for $F_2 + F_4 + \ldots + F_{2n}$, the sum of the first n Fibonacci numbers of even index. Prove your formula is correct using induction.

To help with making a conjecture, we first we tabulate some data:

n	F_n	$F_2 + F_4 + \ldots + F_{2n}$
1	1	1
2	1	4
3	2	12
4	3	33
5	5	88
6	8	
7	13	
8	21	
9	34	
10	55	
11	89	

Staring at this table, one is led to suspect that $F_1 + \ldots + F_{2n}$ is always 1 less than F_{2n+1}. Thus we set out to prove by induction the statement:

$$P(n) : F_2 + F_4 + \ldots + F_{2n} = F_{2n+1} - 1 \qquad n \geq 1.$$

- Base case: when $n = 1$, we have $F_2 = 1$ and $F_{2n+1} - 1 = F_3 - 1 = 2 - 1 = 1$, as required.
- Inductive step: suppose that $n \geq 1$ and that $P(n)$ holds. We want to prove $P(n+1)$. We have:

$$\begin{aligned} F_2 + F_4 + \ldots + F_{2n} + F_{2(n+1)} &= F_{2n+1} - 1 + F_{2(n+1)} && \text{by inductive hypothesis} \\ &= F_{2n+1} + F_{2n+2} - 1 \\ &= F_{2n+3} - 1 && \text{by the Fibonacci recurrence} \\ &= F_{2(n+1)+1} - 1 \end{aligned}$$

  so $P(n+1)$ is satisfied.

By the principle of induction, $P(n)$ holds for every positive integer n.

Remark 1.7.5. Proofs by induction sometimes involve certain subtleties that are not explicitly addressed in Proposition 1.7.1 and Proposition 1.7.2 above. Firstly, induction still works if the base case starts at an integer other than $n = 1$. Another thing to keep in mind is that, especially when using strong induction, it is sometimes necessary to verify multiple base cases by hand to arrive at a valid proof.

The following example aims to illustrate the subtleties alluded to in the above remark.

Example 1.7.6. Use strong induction to show that, for any integer $n \geq 12$, it is possible to make change for n cents using only 3 and 7 cent coins.

In other words, we must prove the following statement for all integers $n \geq 12$:

$$P(n): \text{ there exist nonnegative integer } a, b \text{ such that } n = 3a + 7b.$$

- Base cases: the expressions $12 = 4 \cdot 3$, $13 = 7 + 2 \cdot 3$ and $14 = 2 \cdot 7$ show that $P(12), P(13), P(14)$ all hold.
- Suppose that n is an integer with $n \geq 14$ and $P(k)$ holds for all integers k with $12 \leq k \leq n$. From this inductive hypothesis, we want to prove $P(n+1)$ holds. Indeed, since $n \geq 14$, we also have $12 \leq n - 2 < n$ so, by the inductive hypothesis, $P(n-2)$ holds. That is $n - 2 = 3a + 7b$ for some nonnegative integers a, b. Then, $n + 1 = 3(a + 1) + 7b$, which shows that $P(n+1)$ holds.

By strong induction, $P(n)$ is true for all $n \geq 12$.

Remark 1.7.7. If you are rusty on induction, it is important to reflect on why the multiple base cases were needed in the example above. Suppose we only proved $P(12)$ as a base case. That means $P(13)$ should have instead been established by the inductive argument. However, the inductive argument refers back to $P(13 - 3) = P(10)$. Luckily, $P(10)$ is actually true (since $10 = 7 + 3$), but note this case was not checked and indeed falls outside the range of n for which we are trying to establish $P(n)$. Similarly, trying to use the inductive hypothesis to prove $P(14)$ leads to relying on the truth of $P(14 - 3) = P(11)$. Now the problem is even worse because $P(11)$ is not even true!

1.8 Absolute value and distance

Recall the definition and some basic properties of the absolute value of a real number.

Definition 1.8.1. Given a real number x, the **absolute value** of $|x|$ is defined by

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}.$$

Proposition 1.8.2. *For any $x, y \in \mathbb{R}$, the following statements hold:*

(i) $x \leq |x|$

(ii) $|x + y| \leq |x| + |y|$ (this is called the triangle inequality)

(iii) $|xy| = |x||y|$

Proof. For (i) there are two cases to consider:

- If $x \geq 0$ then $x = |x|$, so (i) is satisfied.
- If $x < 0$, then $-x > 0$ so $x < 0 < |x|$ and (i) is again satisfied.

For (ii) we again consider two cases.

- If $x + y \geq 0$ then $|x + y| = x + y \leq |x| + |y|$, since $x \leq |x|$ and $y \leq |y|$ by (i).
- If $x + y < 0$ then $|x + y| = -(x + y) = -x + (-y) \leq |-x| + |-y| = |x| + |y|$ where (i) was used again as well as $|-a| = |a|$ for all $a \in \mathbb{R}$ (exercise: prove this identity).

We leave the proof of (iii) as an exercise. □

The absolute value is closely-related to the concept of distance.

Definition 1.8.3. The **distance function** or **metric** for $\mathbb{R}$ is the function $d : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ defined by $d(x, y) = |x - y|$.

The most important properties of the distance function are:

Proposition 1.8.4. *For any $x, y, z \in \mathbb{R}$, the following statements hold:*

(i) Positive definiteness: $d(x, y) \geq 0$ and equality occurs if and only if $x = y$.

(ii) Symmetry: $d(x, y) = d(y, x)$.

(iii) Triangle inequality: $d(x, z) \leq d(x, y) + d(y, z)$.

Many spaces other than $\mathbb{R}$ come with a natural distance function d satisfying these same properties. For example, on the xy-plane $\mathbb{R}^2$, we have the usual Euclidean distance function $d : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$ defined by:

$$d((x_1, y_1), (x_2, y_2)) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}.$$

If you are familiar with the concept of a graph (a network of vertices some of which are adjacent by edges), then every connected graph $G = (V, E)$ has a path metric on V given by declaring $d(x, y)$ to be the length of the shortest walk from x to y which only takes steps between adjacent vertices.

1.9 Suprema, infima, and "epsilons of room"

Recall that, unlike the maximum, the supremum of a set of real numbers S is not required to be an element of S. What is always true, however, is that there are elements of S which are *arbitrarily close* to $\sup S$. In analysis, we are able to formulate more precisely what is meant by "arbitrarily close".

Proposition 1.9.1. *Let $S \subseteq \mathbb{R}$ be nonempty and bounded above. Let M be an upper bound for S. Then, the following are equivalent:*

(i) $M = \sup S$

(ii) For every (real number) $\epsilon > 0$, there exists $x \in S$ such that $x > M - \epsilon$.

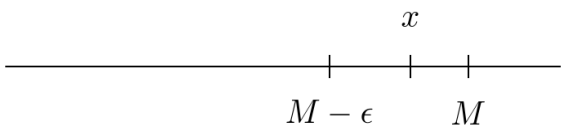


Figure 1.1: Illustration of Proposition 1.9.1.

Proof. Suppose (i) holds, i.e. $M = \sup S$. Let $\epsilon > 0$ be given. Since $M - \epsilon < M$ and M is the smallest of all upper bounds for S, $M - \epsilon$ is not an upper bound for S. This means exactly that there is some $x \in S$ such that $x > M - \epsilon$, so (ii) holds.

Conversely, suppose that (ii) holds. By assumption, M is an upper bound for S. Suppose for contradiction that $M \neq \sup S$. Then, $M > \sup S$ because the supremum is by definition the smallest of all upper bounds for S. Let $\epsilon = (\sup S) - M > 0$. By (ii), there is an $x \in S$ such that $x > M - \epsilon = M$, contradicting the fact that M is an upper bound of S. Thus, $M = \sup S$ and (i) holds. □

Exercise 1.9.2. Give a similar characterization of the infimum.

Proposition 1.9.1 is closely-related to very common proof strategy in Real Analysis: "giving oneself an epsilon of room". Indeed, this principle is so useful that Terry Tao named a whole book after it [12]. The idea is this: if one is tasked with proving some quantity x is $\leq$ some bound y, one can instead prove the weaker inequality $x < y + \epsilon$ where $\epsilon > 0$, provided one is able to do this for all positive numbers ϵ. Indeed, by the Archimedean property, it is enough to show $x < y + 1/n$ for all $n \in \mathbb{N}$.

Proposition 1.9.3. *Suppose that $x, y \in \mathbb{R}$. Then the following statements are equivalent:*

(i) $x \leq y$

(ii) $x < y + \epsilon$ *for all real numbers* $\epsilon > 0$.

Proof. If (i) holds and $\epsilon > 0$ is given, then $x \leq y < y + \epsilon$, so (ii) is satisfied. Next we assume (ii) holds. Suppose for contradiction that (i) does not hold, i.e. $x > y$. Then we may let $\epsilon = x - y > 0$. Then, by (ii), we have $x < y + \epsilon = y + (x - y) = x$, which is a contradiction, so (i) is satisfied. □

Chapter 2

Sequences

2.1 What is a sequence?

A **sequence** is simply a function whose domain is $\mathbb{N}$. Traditionally, we denote a sequence by $\{a_n\}_{n=1}^{\infty}$. This notation communicates that the value of the function at $n \in \mathbb{N}$ is a_n. We would typically say that a_n is the nth **term** of the sequence. Sometimes we write $\{a_n\}$ instead of $\{a_n\}_{n=1}^{\infty}$ to save time. This creates a minor risk of confusion since $\{a_n\}$ could also be interpreted as the 1-element set whose only element is a_n, but in practice the context usually makes it clear when a sequence is intended. It is also sometimes useful to change the domain and index a sequence by a different set of integers, such as the nonnegative integers, or integers n satisfying $5 \leq n \leq 20$ (this would be a finite sequence!). The self-explanatory notations $\{a_n\}_{n=0}^{\infty}$ or $\{a_n\}_{n=5}^{20}$ may be used in such cases.

Example 2.1.1. Here are a few interesting sequences of numbers:

- $\{1/n^2\}_{n=1}^{\infty}$, the sequence of inverse squares: $1, 1/4, 1/9, 1/25, \ldots$
- $\{F_n\}_{n=1}^{\infty}$, the Fibonacci sequence: $1, 1, 2, 3, 5, 8, \ldots$
- The "look and say sequence": $1, 11, 21, 1211, 111221, 312211, \ldots$

2.2 Countability and uncountability

Definition 2.2.1. Let A and B be sets. We say that A has the same **cardinality** as B if there exists a bijection $f : A \to B$, that is a function which is one-to-one and onto.

For finite sets, saying A and B have the same cardinality just means they have the same number of elements in the sense we are accustomed to. For example, the 3-element sets $A = \{1, 4, 10\}$ and $B = \{-1, 2, \pi\}$ have the same cardinality. There are $3! = 6$ choices for a bijection $f : A \to B$. For example one could define f by $f(1) = \pi$, $f(4) = 2$, $f(10) = -1$.

The properties in the following exercise should be recognizable to students who are familiar with equivalence relations.

Exercise 2.2.2. Let us use the shorthand notation $A \sim B$ to indicate that a set A has the same cardinality as a set B. Prove the following properties straight from Definition 2.2.1

(i) Reflexivity: $A \sim A$ for every set A

(ii) Symmetry: If $A \sim B$, then $B \sim A$.

(iii) Transitivity: If $A \sim B$ and $B \sim C$, then $A \sim C$.

After groundbreaking work of Cantor, we understand that not all infinite sets have the same cardinality. In this sense, there is more than one infinity. In fact, there are infinitely many infinities, as we we will see below. The infinite set with the smallest[1] cardinality is the set $\mathbb{N}$ of positive integers.

Definition 2.2.3. A set A is called **countable** if it is finite or if A has the same cardinality as $\mathbb{N}$. A set that is not countable is said to be **uncountable**.

Another way of expressing that A is countable is to say that there is a sequence $\{a_n\}$ which lists every element of A exactly once.

Example 2.2.4. Here are some countable sets:

- The model countable set is $\mathbb{N}$.
- The set E of even positive integers is countable. Indeed, the function $f : \mathbb{N} \to E$ defined by $f(n) = 2n$ is a bijection.
- The set $\mathbb{Z}$ of integers is countable because we can list all the integers in a sequence $\{a_n\}_{n=1}^{\infty}$ as follows:

n	1	2	3	4	5	6	7	...
a_n	0	1	-1	2	-2	3	-3	...

- The set $\mathbb{N} \times \mathbb{N}$ consisting of all pairs (m, n) of positive integers is countable. To list the elements of $\mathbb{N} \times \mathbb{N}$ in a sequence, one can first list the pairs that sum to 2, then 3, then 4, and so on. The finitely-many pairs with a given sum can be listed in increasing order according to their 1st coordinate:

$$(1,1) \ , \ (1,2),(2,1) \ , \ (1,3),(2,2),(3,1) \ , \ (1,4),(2,3),(3,2),(4,1) \ , \ \ldots$$

  It is a good idea to draw a diagram of this scheme for listing elements of $\mathbb{N} \times \mathbb{N}$; it "zigzags" through the grid points.

- The set $\mathbb{Q}$ of rational numbers is countable for similar reasons.

[1] One says the the cardinality of A is less than or equal to the cardinality of B if there exists a one-to-one function from A to B.

In general, any subset $A \subseteq \mathbb{N}$ is countable, because we can list its elements in increasing order. So, for instance, the set of prime numbers is countable. In a bit more detail:

Proposition 2.2.5. *Any subset A of $\mathbb{N}$ is countable.*

Proof. If A is finite, then it is countable by definition, so we turn our attention to the case where A is infinite. Since A is nonempty, by the well-ordering principle (Proposition 1.7.3), there is a smallest element $a_1 \in A$. Since $A \setminus \{a_1\}$ is nonempty (because A is infinite), it has a smallest element a_2. In general, if $a_1, \ldots, a_n$ are already chosen, we define a_{n+1} to be the smallest element of $A \setminus \{a_1, \ldots, a_n\}$. One may check that the resulting sequence $\{a_n\}_{n=1}^{\infty}$ lists each element of A exactly once, so A is countable. □

It was a major breakthrough when Cantor discovered a technique, now called Cantor's diagonal argument, which can be used to demonstrate the existence of uncountable sets. A nice example for learning this technique is the set of binary sequences. A **binary sequence** is a sequence $\{b_n\}_{n=1}^{\infty}$ of 0s and 1s. That is, $b_n = 0$ or 1 for all $n \in \mathbb{N}$. A simple way of generating examples of infinite binary sequences is to repeat some finite pattern:

- $0, 0, 0, 0, \ldots$
- $0, 1, 0, 1, 0, 1, 0, 1, \ldots$
- $0, 0, 1, 1, 1, 1, 0, 0, 1, 1, 1, 1, 0, 0, 1, 1, 1, 1, \ldots$

It is important, however, to realize that there are also many binary sequences which do not follow any such simple pattern. One might imagine performing an infinite sequence of random coin flips to generate a "generic" binary sequence.

Theorem 2.2.6 (Cantor)**.** *The set B of all binary sequences is uncountable.*

Proof. Suppose for contradiction that B is countable. Then exists an infinite list of binary sequences $\{b_n^{(1)}\}, \{b_n^{(2)}\}, \{b_n^{(3)}\}, \ldots$ (that is to say a sequence of sequences) such that every element of B appears on the list. We reach a contradiction by constructing a binary sequence $\{b_n\}_{n=1}^{\infty}$ that is not on the list. We define this sequence in the following way:

$$b_n = \begin{cases} 1 & \text{if } b_n^{(n)} = 0 \\ 0 & \text{if } b_n^{(n)} = 1 \end{cases}$$

By design, the sequence $\{b_n\}$ is:

- different from $\{b_n^{(1)}\}$ because $b_1^{(1)} \neq b_1$,
- different from $\{b_n^{(2)}\}$ because $b_2^{(2)} \neq b_2$ and,
- (generally) different from $\{b_n^{(k)}\}$ for any $k \in \mathbb{N}$ because $b_k^{(k)} \neq b_k$.

Thus, the sequence $\{b_n\}$ is not on our (supposedly comprehensive) list and from this contradiction it follows that B is uncountable.

This argument is called the "diagonal argument" because the sequence was constructed by taking the terms along the diagonal and flipping them:

$$\begin{array}{lccccc}
\text{sequence 1:} & \boxed{b_1^{(1)}} & b_2^{(1)} & b_3^{(1)} & b_4^{(1)} & \dots \\
\text{sequence 2:} & b_1^{(2)} & \boxed{b_2^{(2)}} & b_3^{(2)} & b_4^{(2)} & \dots \\
\text{sequence 3:} & b_1^{(3)} & b_2^{(3)} & \boxed{b_3^{(3)}} & b_4^{(3)} & \dots \\
\text{sequence 4:} & b_1^{(4)} & b_2^{(4)} & b_3^{(4)} & \boxed{b_4^{(4)}} & \dots \\
\vdots & \vdots & \vdots & \vdots & \vdots & \ddots
\end{array}$$

□

Cantor's diagonal argument is capable of proving much more: we can use it to show that, for any set (no matter how hugely infinite) there is a set with even more elements.

Definition 2.2.7. Given any set X, the **power set** $P(X)$ of X is defined as the set of all subsets of X. That is, $P(X) \coloneqq \{S : S \subseteq X\}$.

Theorem 2.2.8 (Cantor)**.** *Let X be any set. Then there does not exist a surjective function $f : X \to P(X)$.*

Proof. Suppose for contradiction that there does exist a surjective function $f : X \to P(X)$. We will reach a contradiction by constructing an example of a set $S \subseteq X$ which is not in the range of f. To construct S we need to decide, for each element $x \in X$, whether to include x in S or not. We make these decisions in the following way:

- If $x \in f(x)$, then $x \notin S$.
- If $x \notin f(x)$, then $x \in S$.

By this construction, we have $f(x) \neq S$ for every $x \in X$ because $f(x)$ and S disagree about the membership of x. Thus, a surjection from X onto $P(X)$ does not exist. □

Example 2.2.9. It is rather informative to consider how the proof of the above theorem works in the case of finite sets. Let $X = \{1, 2, 3\}$. The power set of X is:

$$P(X) = \{\varnothing, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}.$$

It is rather obvious that there is no surjection from X to $P(X)$ because X has 3 elements and $P(X)$ has 8 elements. Even so, it is instructive to see how Cantor's construction produces a set S not belonging to the range of a given function. For example, consider the function:

$$f : X \to P(X) \qquad f(1) = \{1, 2\} \qquad f(2) = \varnothing \qquad f(3) = \{2\}.$$

Since $1 \in f(1)$, $2 \notin f(2)$, $3 \notin f(3)$ we take S such that $1 \notin S$, $2 \in S$, $3 \in S$. The resulting set is $S = \{2, 3\}$ which is indeed not one of the sets in the range of f.

Remark 2.2.10. Although Theorem 2.2.6 and Theorem 2.2.8 may appear different, they are actually very closely related. This because a subset of $\mathbb{N}$ and a binary sequence are essentially the same thing. What is a subset $S \subseteq \mathbb{N}$ really but a decision for every $n \in \mathbb{N}$ of whether to include n in S or not? In other words, we can think of a subset of $\mathbb{N}$ as sequence of "yeses" and "noes". Associating "yes" with 1 and "no" with 0, we get a natural bijection between $P(\mathbb{N})$ and the the set of binary sequences.

We have already discussed that the sets $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ are all countable. Next we discuss the fact that the real numbers are uncountable. One way to prove this is to use the fact that every real number has a decimal expansion (or a binary expansion, if one prefers) and use a diagonal argument similar to the one in Theorem 2.2.6. A little bit of care is needed because some real numbers have multiple decimal representations. The example $1 = 1.000\ldots = 0.999\ldots$ is especially notorious for generating heated debates among mathematical amateurs. We will leave the task of working out a precise proof of the uncountability of the reals using decimal expansions and the diagonal argument to the reader. Instead, we give a different proof that relies on a property of the reals that is more fundamental than decimal expansions: the least upper bound property! Another reason to take this approach is that we have not at this point discussed decimal expansions in a rigorous way. We may return to decimal expansions later when we discuss infinite series.

Theorem 2.2.11. *The real numbers $\mathbb{R}$ are uncountable.*

Proof. Suppose for contradiction that the real numbers are countable. Then, there is a sequence of real numbers $\{a_n\}_{n=1}^{\infty}$ such that every real number appears somewhere on the sequence. We generate a nested sequence of intervals $(x_1, y_1) \supseteq (x_2, y_2) \supseteq (x_3, y_3) \supseteq \ldots$ in the following way:

- Choose x_1, y_1 such that $x_1 < y_1$ and $a_1 \notin (x_1, y_1)$.
- Choose x_2, y_2 such that $x_1 < x_2 < y_2 < y_1$ and $a_2 \notin (x_2, y_2)$.
- In general, if $n \geq 3$ and x_i, y_i are already chosen for $i < n$, we choose x_n, y_n such that $x_{n-1} < x_n < y_n < y_{n-1}$ and $a_n \notin (x_n, y_n)$.

The reader should draw a picture to convince themself this recursive procedure can always continue. Observe the set $X \coloneqq \{x_i : i \in \mathbb{N}\}$ is bounded above. Indeed,

$$x_1 < x_2 < x_3 < \ldots < y_3 < y_2 < y_1$$

so every element of $Y \coloneqq \{y_i : i \in \mathbb{N}\}$ is an upper bound for X. Thus, since $\mathbb{R}$ has the least upper bound property, it makes sense to consider $z = \sup\{x_i : i \in \mathbb{N}\}$. We claim that $z \neq a_n$ for any $n \in \mathbb{N}$. Indeed, we have $z \in (x_n, y_n)$ for all $n \in \mathbb{N}$ because:

- $z \geq x_{n+1} > x_n$, using that z is an upper bound for X.
- $z \leq y_{n+1} < y_n$, using that $y_{n+1} \in Y$ is an upper bound for X and z is the least upper bound.

Since $z \in (x_n, y_n)$ and $a_n \notin (x_n, y_n)$ we conclude $z \neq a_n$. Since $n \in \mathbb{N}$ is arbitrary here, z does not appear on the sequence and this contradiction shows that $\mathbb{R}$ is uncountable. □

Corollary 2.2.12. *For any real $a < b$, the interval (a, b) is uncountable.*

Proof. One may check that the function $f : \mathbb{R} \to (-1, 1)$ defined by $f(x) = \frac{x}{\sqrt{1+x^2}}$ is a bijection (the inverse function sends x to $\frac{x}{\sqrt{1-x^2}}$), so the interval $(-1, 1)$ has the same cardinality as $\mathbb{R}$. Also, given any real $a < b$, the function $g : (0, 1) \to (a, b)$ defined by $g(x) = a + (b - a)x$ is a bijection, so all open intervals of $\mathbb{R}$ have the same cardinality. □

Since every real interval is uncountable and the set of rationals is countable, one may argue that every real interval contains an uncountable number of irrational numbers. This gives us a second way of proving Proposition 1.6.5 which stated that the irrational numbers are dense in $\mathbb{R}$.

Corollary 2.2.13. *For any real interval $(a, b) \subseteq \mathbb{R}$, there are uncountably many irrational numbers in (a, b).*

2.3 Limits of sequences

The following definition is one of the most important definitions in the whole course. Newton and Leibniz did not use this formulation of convergence. It took a long time and the work of many people to arrive at the modern definition of a limit.

Definition 2.3.1. We say that a real sequence $\{a_n\}_{n=1}^{\infty}$ **converges** to a real number L if, for every real $\epsilon > 0$, there exists a positive integer N such that $n \geq N$ implies $|a_n - L| < \epsilon$.

Remark 2.3.2. As usual, we deploy various other terminologies and notations relating to limits that the reader is likely familiar with from calculus. The following are synonymous:

- $\{a_n\}$ converges to L
- The limit of $\{a_n\}$ is L.
- $\lim_{n\to\infty} a_n = L$.
- $a_n \to L$ as $n \to \infty$.

Though a bit casual, it is a useful and common practice to still write the expression $\lim_{n\to\infty} a_n$ even when we are not yet sure whether the sequence $\{a_n\}$ is convergent or not. It is only for sequences that do converge that we interpret the otherwise formal symbol $\lim_{n\to\infty} a_n$ as being equal a real number, namely the numerical value of limit. If the sequence does not converge, we would say "$\lim_{n\to\infty} a_n$ does not exist".

Here are some basic examples of convergent sequences and proofs that they converge to their limit.

Example 2.3.3. Prove that

$$\lim_{n\to\infty} \frac{n^2}{2n^2+1} = \frac{1}{2}.$$

Before launching directly into a proof, it is often helpful to do a bit of scratch work. We want to figure out how to make $|\frac{n^2}{2n^2+1} - \frac{1}{2}|$ small, so it makes sense to work out an upper bound for this quantity that we can exert control over. For instance:

$$\left|\frac{n^2}{2n^2+1} - \frac{1}{2}\right| = \left|\frac{n^2 - \frac{1}{2}(2n^2+1)}{2n^2+1}\right| = \left|\frac{-1/2}{2n^2+1}\right| = \frac{1/2}{2n^2+1} \leq \frac{1}{2n^2+1} \leq \frac{1}{2n^2}.$$

Now, if we want to make the left hand side smaller than a given ϵ, it is enough to make sure the right hand side is smaller than ϵ. It is fairly straightforward to check that $\frac{1}{2n^2} < \epsilon$ if and only if $n > \frac{1}{\sqrt{2\epsilon}}$. This looks promising because, no matter how small the given $\epsilon > 0$, we will still have $n > \frac{1}{\sqrt{2\epsilon}}$ for all sufficiently big enough n. Now let's write up a formal proof!

Proof. We claim that $\lim_{n\to\infty} \frac{n^2}{2n^2+1} = \frac{1}{2}$. To this end, let $\epsilon > 0$ be any fixed positive real. We must demonstrate the existence of a positive integer N such that $n \geq N$ implies $|\frac{n^2}{2n^2+1} - \frac{1}{2}| < \epsilon$. By the Archimedean property, we can choose N to be an integer such that $N > \frac{1}{\sqrt{2\epsilon}}$ (we made this choice strategically based on our scratch work). The latter inequality is equivalent to $\frac{1}{2N^2} < \epsilon$. Now, provided $n \geq N$, we calculate that

$$\left|\frac{n^2}{2n^2+1} - \frac{1}{2}\right| \leq \underset{\text{here we can copy down our scratchwork}}{\ldots} \leq \frac{1}{2n^2} \leq \frac{1}{2N^2} < \epsilon,$$

as required. □

Example 2.3.4. Prove that

$$\lim_{n\to\infty} \frac{1}{n} = 0.$$

Proof. This limit is a bit simpler to verify than the last one. It follows straight from the Archimedean property of $\mathbb{R}$. Given any $\epsilon > 0$, the Archimedean property guarantees we can choose a positive integer N such that $\frac{1}{N} < \epsilon$. Then, for all $n \geq N$, we have

$$|\tfrac{1}{n} - 0| = \tfrac{1}{n} \leq \tfrac{1}{N} < \epsilon,$$

as required. □

It is also important to examine some examples of sequences which do not converge to any real limit.

Example 2.3.5. The sequence $\{(-1)^n\}_{n=1}^{\infty}$ does not converge to any real number $L \in \mathbb{R}$.

Proof. First we prove that the sequence does not converge to $L = 1$. Indeed, consider $\epsilon = 1/2$. We claim there is no positive integer N such that $n \geq N$ implies $|(-1)^n - 1| < 1/2$. Indeed, no matter the selection of N, there exists an odd number $n > N$ which then satisfies $|(-1)^n - 1| = |-1-1| = 2 > 1/2 = \epsilon$.

Next we prove the sequence does not converge to any fixed real number $L \neq 1$. Let $\epsilon = |1 - L|/2$. Since $L \neq 1$, we have $\epsilon > 0$. We claim that there is no positive integer N such that $n \geq N$ implies $|(-1)^n - L| < \epsilon$. Indeed, no matter the selection of N, there exists an even number $n > N$ which then satisfies $|(-1)^n - L| = |1 - L| > |1 - L|/2 = \epsilon$. □

2.4 Bogus definitions of convergence

Learning to parse mathematical statements involving quantifiers[2] and implications is an important skill in all parts of math, but especially in analysis. Ideally, one should aim to be able to read a mathematical statement and get a feel for what property that statement is capturing. It takes time and practice before one is able to this efficiently.

The definition of convergence of a sequence (Definition 2.3.1) is an important example of a mathematical statement that is very sensitive to the ordering of the quantifiers. It is a valuable exercise to take this definition and consider minor variations, for example by making changes to the ordering of the quantifiers. Even seemingly small changes to the definition can result in statements which have drastically different meanings.

Example 2.4.1 (Malformed definitions of convergence)**.** Let $\{a_n\}$ be a real sequence and L a real number. Below are five attempts to define what it means for $\{a_n\}$ to converge to L. Only one of these statements is the correct definition of $a_n \to L$.

(A) For all $N \in \mathbb{N}$ there exists $\epsilon > 0$ such that $n \geq N$ implies $|a_n - L| < \epsilon$.

(B) For all $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that $n \geq N$ implies $|a_n - L| < \epsilon$.

(C) There exists $N \in \mathbb{N}$ such that $n \geq \mathbb{N}$ implies $|a_n - L| < \epsilon$ for all $\epsilon > 0$.

(D) For all $N \in \mathbb{N}$ there exists $\epsilon > 0$ such that $|a_n - L| < \epsilon$ implies $n \geq N$.

(E) For all $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that $|a_n - L| < \epsilon$ implies $n \geq N$.

Exercise 2.4.2. Identify the correct definition of convergence of a sequence among the statements given above. For each of the four "bogus definitions" give an example of a sequence $\{a_n\}$ and a choice of L such that either:

[2]Mainly "for all" and "there exists".

- $a_n \to L$ but the statement doesn't hold, or else
- $\{a_n\}$ does not converge to L, but the statement does hold.

(A) This definition is bogus. Let $\{a_n\} = \{\sin(n)\}$ (any bounded sequence will work) and let $L = 10$ (any real number will work). Note $\{\sin(n)\}$ does not converge to 10. However, the statement (A) is true. Indeed, challenged with any $N \in \mathbb{N}$, we can respond with $\epsilon = 100$. Then $|\sin(n) - 10| \leq 11 < 100$ for any $n \in \mathbb{N}$, so in particular for all $n \geq N$.

(B) This is the correct definition of $a_n \to L$.

(C) This definition is bogus. Let $\{a_n\} = \{1/n\}$ and $L = 0$. Note $\{1/n\}$ does converge to 0. However, the statement (C) is false. Indeed, no matter which $N \in \mathbb{N}$ we choose, we have $|a_N - L| = \frac{1}{2N}$ so, taking $\epsilon = \frac{1}{2N} > 0$ we see that N does not satisfy $|a_N - L| < \epsilon$.

Discussion. Actually "$|a_n - L| < \epsilon$ for all $\epsilon > 0$" holds if and only if $a_n = L$. This means (C) is actually equivalent to "there exists $N \in \mathbb{N}$ such that $n \geq N$ implies $a_n = L$". In other words, the bogus definition (C) only thinks a sequence $\{a_n\}$ converges to L if the sequence is eventually equal to L i.e. has the form $a_1, a_2, \ldots, a_{N-1}, L, L, L, L, \ldots$ for some $N \in \mathbb{N}$.

(D) This definition is bogus. Let $\{a_n\} = \{1 + 1/n\}$ and $L = 0$. Note $\{1 + 1/n\}$ does not converge to 0. However, the statement (D) is true. Indeed, challenged with any $N \in \mathbb{N}$, we can respond with an $\epsilon > 0$ chosen so that $1 + 1/N < \epsilon < 1 + 1/(N-1)$. This choice of ϵ ensures that

$$\begin{cases} |a_n - L| = 1 + 1/n < \epsilon & \text{for all } n \geq N \\ |a_n - L| = 1 + 1/n > \epsilon & \text{for all } n < N. \end{cases}$$

In particular, if $|a_n - L| < \epsilon$ then $n \geq N$.

(E) This definition is bogus. Actually, (D) is always true no matter what choices are made for $\{a_n\}_{n=1}^{\infty}$ and L. Indeed, challenged with any $\epsilon > 0$, we may respond with $N = 1$. Since the sequence is indexed by $\mathbb{N}$, we have $n \geq 1$ for all indices n regardless of whether or not $|a_n - L| < \epsilon$ is satisfied. In particular, $|a_n - L| < \epsilon$ implies $n \geq 1$.

2.5 Bounded sequences

A sequence is bounded if all of its terms are contained in some fixed finite interval. A bit more precisely:

Definition 2.5.1. A real sequence $\{a_n\}$ is said to be **bounded** if there exists a real number $M > 0$ such that $-M \leq a_n \leq M$ for all $n \in \mathbb{N}$ or, equivalently, $|a_n| \leq M$ for all $n \in \mathbb{N}$.

Example 2.5.2. The sequence $\{\sin(n)\}$ bounded. All of its terms belong to the finite interval $[-1, 1]$.

It is quite intuitive that any convergent sequence must be bounded; if the terms are permitted to get arbitrarily large, then they cannot approach a finite value. Even so, this statement is useful and important enough that we state and prove it as a theorem.

Theorem 2.5.3. *Let $\{a_n\}$ be a real sequence. If $\{a_n\}$ is convergent, then it is bounded.*

Proof. Suppose $\{a_n\}$ is convergent. Let $L \in \mathbb{R}$ be its limit. Taking $\epsilon = 1$ in the definition of $a_n \to L$, we get that there exists $N \in \mathbb{N}$ such that $n \geq N$ implies $|a_n - L| < 1$. Now define

$$M \coloneqq \max\{|a_n| : n < N\} \cup \{|L| + 1\}.$$

We claim that $|a_n| \leq M$ for all $n \in \mathbb{N}$. Indeed, if $n < N$, then $|a_n| \leq M$ because M is defined as the maximum of $|a_n|$ and other quantities. If $n \geq N$, then $|a_n - L| < 1$ and so

$$|a_n| = |(a_n - L) + L| \leq |a_n - L| + |L| < 1 + |L| \leq M$$

by the triangle inequality and the definition of M as a maximum of $|L| + 1$ and other quantities. □

It is sometimes useful to talk about real sequences which are bounded above or below, but not necessarily from both sides. This parallels the earlier discussion for sets of real numbers. Indeed, the various notions of boundedness of a real sequence $\{a_n\}$ may be alternatively understood by considering the boundedness of its set of terms $\{a_n : n \in \mathbb{N}\}$.

Definition 2.5.4. Let $\{a_n\}$ be a real sequence. We say $\{a_n\}$ is **bounded above** if there is a real number M such that $a_n \leq M$ for all $n \in \mathbb{N}$. We say $\{a_n\}$ is **bounded below** if there is a real number m such that $m \leq a_n$ for all $n \in \mathbb{N}$.

One may check that a sequence is bounded if and only if it is both bounded above and bounded below.

Lastly, we look at special classes of unbounded sequences. Recall a sequence $\{a_n\}$ is **convergent** if it converges to some real number L. A sequence which is not convergent is **divergent**. Some divergent sequences diverge in a specific way, towards ∞ or $-\infty$. For example, $\lim_{n\to\infty} 5n^2 = \infty$ and $\lim_{n\to\infty} -\sqrt{n+3} = -\infty$. The precise definitions are:

Definition 2.5.5. Let $\{a_n\}$ be a real sequence.

- We say $\{a_n\}$ **diverges to** ∞ if for all $M > 0$ (no matter how large) there exists $N \in \mathbb{N}$ such that $n \geq N$ implies $a_n > M$. In symbols, we write $\lim\limits_{n\to\infty} a_n = \infty$ or $a_n \to \infty$.
- Similarly, $\{a_n\}$ **diverges to** $-\infty$ if for all $M > 0$ there exists $N \in \mathbb{N}$ such that $n \geq N$ implies $a_n < -M$. In symbols, we write $\lim\limits_{n\to\infty} a_n = -\infty$ or $a_n \to -\infty$.

Example 2.5.6. Consider the sequence $\{a_n\} = \{(-1)^n n^2\}$. Which of the following are true?

- $a_n \to \infty$. False: Challenged with $M = 100$, there is no response $N \in \mathbb{N}$ such that $n \geq N$ implies $a_n > 100$. All the odd-index terms are negative and fail this bound.
- $a_n \to -\infty$. False: challenged with $M = 100$, there is no response $N \in \mathbb{N}$ such that $n \geq N$ implies $a_n < -100$. All the even-index terms are positive and fail this bound.
- $\{a_n\}$ is divergent. True: this sequence does not converge to any real number.

2.6 Limit laws

Recall from your calculus education how limit laws are used to calculate more complicated limits in terms of simpler ones.

Example 2.6.1. In calculus, one might want to calculate $\lim_{n\to\infty} \frac{3n^2+2}{4n^2-n}$. A typical way of doing this might be to first rewrite the sequence by dividing both numerator and denominator by n^2 (the highest power of n present) in order to obtain

$$\frac{3n^2+2}{4n^2-n} = \frac{3+2(\frac{1}{n^2})}{4-\frac{1}{n}}.$$

Then, knowing the following simple limits:

$$\lim_{n\to\infty} 3 = 3 \qquad \lim_{n\to\infty} \frac{1}{n^2} = 0 \qquad \lim_{n\to\infty} 4 = 4 \qquad \lim_{n\to\infty} \frac{1}{n} = 0$$

limit laws allow us to combine them to compute the desired limit:

$$\lim_{n\to\infty} \frac{3n^2+2}{4n^2-n} = \lim_{n\to\infty} \frac{3+2\frac{1}{n^2}}{4-\frac{1}{n}} = \frac{(3)+2(0)}{(4)-(0)} = \frac{3}{4}.$$

The limit laws provide rigorous underpinning for the type of calculation shown above. We only prove a couple of them. The remaining proofs are exercises/reading assignments.

Theorem 2.6.2 (Limit laws)**.** *Suppose $\{a_n\}$ and $\{b_n\}$ are convergent real sequences and $\lambda \in \mathbb{R}$. Denote the limits by $\lim_{n\to\infty} a_n = A$ and $\lim_{n\to\infty} b_n = B$. Then the following limits exist:*

(i) $\lim_{n\to\infty} a_n + b_n = A + B$

(ii) $\lim_{n\to\infty} a_n - b_n = A - B$

(iii) $\lim_{n\to\infty} \lambda a_n = \lambda A.$

(iv) $\lim_{n\to\infty} a_n b_n = AB.$

Assuming $b_n \neq 0$ for all $n \in \mathbb{N}$ and $B \neq 0$, we also have:

(v) $\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{A}{B}.$

Proof of ((i)). Let $\epsilon > 0$ be given. We must show that there exists $N \in \mathbb{N}$ such that $n \geq N$ implies $|a_n + b_n - (A+B)| < \epsilon$. For any $n \in \mathbb{N}$, the triangle inequality gives:

$$|a_n + b_n - (A+B)| = |(a_n - A) + (b_n - B)| \leq |a_n - A| + |b_n - B|.$$

Since $a_n \to A$ and $\epsilon/2 > 0$, we can find N_1 such that $n \geq N_1$ implies $|a_n - A| < \epsilon/2$. Similarly, since $b_n \to B$, we can find $N_2 \in \mathbb{N}$ such that $n \geq N_2$ implies $|b_n - B| < \epsilon/2$. Now define $N \coloneqq \max\{N_1, N_2\}$. By design, if $n \geq N$, then $n \geq N_1, N_2$ and we obtain

$$|a_n + b_n - (A + B)| \leq |a_n - A| + |b_n - B| < \epsilon/2 + \epsilon/2 = \epsilon,$$

as required. □

Remark 2.6.3. We will also prove (iv), but let us first give a bit of discussion. We want to control $|a_n b_n - AB|$ by bounding it above in terms of other quantities that we already have control over, such as $|a_n - A|$ and $|b_n - B|$. This motivates us to add and subtract things in a strategic way to make these quantities appear. For any $n \in \mathbb{N}$, we have:

$$\begin{aligned} |a_n b_n - AB| &= |(a_n b_n - A b_n) + (A b_n - AB)| \\ &= |(a_n - A)b_n + A(b_n - B)| \\ &\leq |(a_n - A)b_n| + |A(b_n - B)| && \text{(triangle inequality)} \\ &= |a_n - A||b_n| + |A||b_n - B|. \end{aligned}$$

This looks promising because $|a_n b_n - AB|$ is now bounded in terms of four quantities that we have control over:

- $|a_n - A|$ gets small as $n \to \infty$
- $|b_n|$ is bounded as $n \to \infty$ (see Theorem 2.5.3).
- $|b_n - B|$ gets small as $n \to \infty$
- $|A|$ is also bounded; it is a constant.

Proof of ((iv)). Let $\epsilon > 0$ be given. We must show that there exists $N \in \mathbb{N}$ such that $n \geq N$ implies $|a_n b_n - AB| < \epsilon$. Repeating the calculation above, for any $n \in \mathbb{N}$ we have:

$$\begin{aligned} |a_n b_n - AB| &= |(a_n b_n - A b_n) + (A b_n - AB)| \\ &= |(a_n - A)b_n + A(b_n - B)| \\ &\leq |(a_n - A)b_n| + |A(b_n - B)| && \text{(triangle inequality)} \\ &= |a_n - A||b_n| + |A||b_n - B|. \end{aligned}$$

By Theorem 2.5.3, $\{b_n\}$ is bounded so there exists an $M > 0$ such that $|b_n| < M$ for all $n \in \mathbb{N}$. The possibility that $|A| = 0$ adds an annoying and slightly distracting technicality to the proof, but note $|A| \leq |A| + 1$ which is positive. Summing up, for all $n \in \mathbb{N}$, we have

$$|a_n b_n - AB| \leq |a_n - A| \cdot M + |b_n - B| \cdot (|A| + 1).$$

Now, since $\frac{\epsilon}{2M} > 0$ and $a_n \to A$ we can find $N_1 \in \mathbb{N}$ such that $n \geq N_1$ implies $|a_n - A| < \frac{\epsilon}{2M}$. Similarly, since $b_n \to B$ and $\frac{\epsilon}{2(|A|+1)} > 0$, we can find $N_2 \in \mathbb{N}$ such that $n \geq N_2$ implies $|b_n - B| < \frac{\epsilon}{2(|A|+1)}$. Now put $N \coloneqq \max\{N_1, N_2\}$. By design, if $n \geq N$, then $n \geq N_1, N_2$ and

$$|a_n b_n - AB| \leq |a_n - A| \cdot M + |b_n - B| \cdot (|A| + 1) < \frac{\epsilon}{2M} \cdot M + \frac{\epsilon}{2(|A| + 1)} \cdot (|A| + 1) = \epsilon,$$

as required. □

Next we look at a few more limit laws which describe what happens to inequalities when we take limits.

Theorem 2.6.4. *Suppose $\{a_n\}$ and $\{b_n\}$ are convergent sequences.*

(i) If $a_n \leq b_n$ for all $n \in \mathbb{N}$ then $\lim_{n\to\infty} a_n \leq \lim_{n\to\infty} b_n$.

(ii) If $x, y \in \mathbb{R}$ and $x \leq a_n \leq y$ for all $n \in \mathbb{N}$ then $x \leq \lim_{n\to\infty} a_n \leq y$.

Proof. Write $A \coloneqq \lim_{n\to\infty} a_n$ and $B \coloneqq \lim_{n\to\infty} b_n$ so that the goal of (i) is to prove that $A \leq B$. It is enough to prove that $A < B + \epsilon$ for all $\epsilon > 0$ (this uses the technique of giving ourselves an "epsilon of room", see Proposition 1.9.3). Let $\epsilon > 0$ be arbitrary. There exists $N_1 \in \mathbb{N}$ such that $n \geq N_1$ implies $|a_n - A| < \epsilon/2$. There also exists $N_2 \in \mathbb{N}$ such that $n \geq N_2$ implies $|b_n - B| < \epsilon/2$. Now choose $N \in \mathbb{N}$ so that $N \geq N_1, N_2$. With this choice, we have

$$\begin{aligned} A &= (A - a_N) + (a_N - b_N) + (b_N - B) + B \\ &\leq |A - a_N| + 0 + |b_n - B| + B && a_N \leq b_N, \text{ so } a_N - b_N \leq 0 \\ &< \epsilon/2 + \epsilon/2 + B \\ &= \epsilon + B. \end{aligned}$$

Thus $A < B + \epsilon$ for all $\epsilon > 0$, and it follows that $A \leq B$.

One may obtain (ii) as a corollary of (i) by viewing x and y as the limits of constant sequences. Nonetheless, just for practice, let's prove (ii) in a more direct manner. Suppose $x \leq a_n \leq y$ for all $n \in \mathbb{N}$. Our goal is to prove that $x \leq A \leq y$. It is enough to prove that $x - \epsilon < A < y + \epsilon$ for all $\epsilon > 0$. Again, this uses the technique of giving ourselves an "epsilon of room". Since $a_n \to A$, there exists $N \in \mathbb{N}$ such that $|a_N - A| < \epsilon$ (and, in fact, N can be taken such that $n \geq N$ implies $|a_n - A| < \epsilon$). Equivalently, $a_N - \epsilon < A < a_N + \epsilon$ and so, since $x \leq a_N \leq y$, we get $x - \epsilon < A < y + \epsilon$, as needed.

□

Theorem 2.6.5 (Squeeze law)**.** *Let $\{x_n\}$, $\{y_n\}$ and $\{a_n\}$ be sequences satisfying:*

(i) $\{x_n\}$ and $\{y_n\}$ converge to the same limit $L \in \mathbb{R}$,

(ii) $x_n \leq a_n \leq y_n$ for all $n \in \mathbb{N}$.

Then $\{a_n\}$ also converges to this same limit L.

Proof. Let $\epsilon > 0$ be given. We must find an $N \in \mathbb{N}$ such that $n \geq N$ implies

$$L - \epsilon < a_n < L + \epsilon.$$

Since $x_n \to L$, there exists $N_1 \in \mathbb{N}$ such that $n \geq N_1$ implies $L - \epsilon < x_n < L + \epsilon$. Since $y_n \to L$, there exists $N_2 \in \mathbb{N}$ such that $n \geq N_2$ implies $L - \epsilon < y_n < L + \epsilon$. Put $N \coloneqq \max\{N_1, N_2\}$. Then, if $n \geq N$, we have $L - \epsilon < x_n \leq a_n \leq y_n < L + \epsilon$ as required. □

Example 2.6.6. Consider the sequence $\left\{\frac{\sin n}{n}\right\}_{n=1}^{\infty}$. Even though the values of $\sin n$ jump around somewhat wildly,

n	$\sin n$
1	$0.841\ldots$
2	$0.909\ldots$
3	$0.141\ldots$
4	$-0.756\ldots$
5	$-0.958\ldots$

they always fall in the interval $[-1, 1]$. Thus $-\frac{1}{n} \leq \frac{\sin n}{n} \leq \frac{1}{n}$ for all n. Since $\lim_{n\to\infty} \frac{-1}{n} = \lim_{n\to\infty} \frac{1}{n} = 0$, the squeeze law gives that $\lim_{n\to\infty} \frac{\sin n}{n} = 0$ as well.

We will now discuss a highly useful sufficient condition for convergence of sequences: the convergence of bounded monotone sequences. First we define the relevant terminology.

Definition 2.6.7. We say that a sequence of real numbers $\{a_n\}$ is:

- **strictly increasing** if $a_1 < a_2 < a_3 < \ldots$
- **strictly decreasing** if $a_1 > a_2 > a_3 > \ldots$
- **increasing** if $a_1 \leq a_2 \leq a_3 \leq \ldots$
- **decreasing** if $a_1 \geq a_2 \geq a_3 \geq \ldots$
- **monotone** if it is increasing or decreasing.

Theorem 2.6.8 (Convergence of montone sequences)**.** *Every bounded monotone sequence is convergent. More specifically:*

(i) If $\{a_n\}$ is bounded and increasing, then $\lim_{n\to\infty} a_n = \sup\{a_n : n \in \mathbb{N}\}$.

(ii) If $\{a_n\}$ is bounded and decreasing, then $\lim_{n\to\infty} a_n = \inf\{a_n : n \in \mathbb{N}\}$.

Proof. We only consider the case of an increasing sequence, as the other case is similar. Let $\{a_n\}$ be a bounded increasing sequence. Since the sequence is bounded, the least upper bound property of the real line allows us to define:

$$M \coloneqq \sup\{a_n : n \in \mathbb{N}\}.$$

We claim that $\lim_{n\to\infty} a_n = M$. To this end, fix an arbitrary $\epsilon > 0$. By Proposition 1.9.1, there must be an $N \in \mathbb{N}$ such that $a_N \in S$ satisfies $M - \epsilon < a_N$. Then, for any $n \geq N$, we have $a_N \leq a_n$ because the sequence is increasing and also $a_n \leq M$ because M is an upper bound for S. Thus, we have

$$M - \epsilon < a_N \leq a_n \leq M$$

for all $n \geq N$ which in particular implies $|a_n - M| < \epsilon$ for all $n \geq N$.

□

2.7 Subsequences and the Bolzano-Weierstrass theorem

Informally, a subsequence of a sequence is a new sequence obtained by "skipping over" some of the terms of the original sequence. So, for example, the sequence of odd primes is a subsequence of the sequence of odd numbers because it is obtained by skipping over all the non-primes. We formulate the concept of a subsequence more precisely in the definition below. The "double subscript" notation takes a bit of getting used to but is, for better or for worse, a fairly standard notation.

Definition 2.7.1. Let $\{a_n\}_{n=1}^{\infty}$ be a sequence. Let $\{n_k\}_{k=1}^{\infty}$ be a strictly increasing sequence of positive integers. Then, the sequence $\{a_{n_k}\}_{k=1}^{\infty}$ is called a subsequence of $\{a_n\}_{n=1}^{\infty}$.

Example 2.7.2. The sequence $\{1/2^k\}_{k=1}^{\infty}$ is a subsequence of $\{1/n\}_{n=1}^{\infty}$.

$$1, \boxed{\frac{1}{2}}, \frac{1}{3}, \boxed{\frac{1}{4}}, \frac{1}{5}, \frac{1}{6}, \frac{1}{7}, \boxed{\frac{1}{8}}, \frac{1}{9}, \frac{1}{10}, \dots$$

A subsequence of a convergent sequence always converges to the same limit as the original sequence. In fact, the subsequence can only converge to the limit "faster" than the original sequence. Case in point, both $\{1/n\}$ and its subsequence $\{1/2^k\}$ converge to 0, but the subsequence converges to 0 at a faster rate.

Proposition 2.7.3. *If $\{a_n\}$ is a sequence converging to a limit $L \in \mathbb{R}$ and $\{a_{n_k}\}$ is a subsequence of $\{a_n\}$, then $\{a_{n_k}\}$ converges to L too.*

Proof. Exercise. □

Sometimes a divergent sequence has a convergent sequence hidden inside it. For example, the sequence $\{(-1)^n\}$ does not converge, but by passing to the subsequence of even-index terms, we get the constant sequence $1, 1, 1, \dots$ which is convergent. The Bolzano-Weierstrass theorem says that in fact every bounded sequence admits a convergent subsequence. This theorem is intimately related the topological concept of compactness, which some students will see in other courses.

Theorem 2.7.4 (Bolzano-Weierstrass theorem)**.** *Every bounded sequence of real numbers $\{a_n\}$ has a convergent subsequence $\{a_{n_k}\}$.*

The following proof of this theorem is sometimes compared to a highly impractical method for "hunting lions in the Sahara". Suppose there is at least one lion somewhere in the Sahara. To capture a lion in this vast desert, we first build a fence splitting the desert in half. One half of this divided desert must contain a lion. We can split that half in half again with another fence to get two smaller regions, one of which must contain a lion. Continuing in this way, we create smaller and smaller fenced regions of desert containing a lion until capture becomes possible.

Proof. Since $\{a_n\}$ is bounded, there exists a finite interval $[x_0, y_0]$ containing all the terms of the sequence. Beginning with this interval, construct an infinite sequence of nested intervals

$$[x_0, y_0] \supseteq [x_1, y_1] \supseteq [x_2, y_2] \supseteq \dots$$

each half the length of the preceding one, and subject to the following requirements:

(i) For each $k \geq 1$, the interval $[x_k, y_k]$ is equal to either the left half[3] or the right half of the preceding interval $[x_{k-1}, y_{k-1}]$.

(ii) For each $k \geq 1$, infinitely-many terms of the sequence $\{a_n\}$ fall in $[x_k, y_k]$. That is, $\{n \in \mathbb{N} : a_n \in [x_k, y_k]\}$ is infinite.

The reason we can keep making these choices is a version of the pigeonhole principle:

- $\{a_n\}$ has infinitely-many terms and they all lie in $[x_0, y_0]$. Therefore infinitely-many terms must belong to either the left or the right half (or possibly both halves) of $[x_0, y_0]$. Define $[x_1, y_1]$ to be a half with infinitely-many terms.
- Similarly, infinitely-many terms of $\{a_n\}$ must belong to either the left or the right half of $[x_1, y_1]$. Define $[x_2, y_2]$ to be a half with infinitely-many terms.
- etc.

Next we choose a sequence of positive integers $n_1 < n_2 < n_3 < \dots$ in the following way:

- n_1 is chosen such that $a_{n_1} \in [x_1, y_1]$
- n_2 is chosen such that $a_{n_2} \in [x_2, y_2]$ and $n_2 > n_1$.
- Generally, if $n_1 < \dots < n_{k-1}$ are already chosen for $k \geq 2$, we choose n_k in such a way that $a_{n_k} \in [x_k, y_k]$ and $n_k > n_{k-1}$. This is always possible because there are infinitely-many choices of n such that $a_n \in [x_k, y_k]$ and only finitely-many of those choices are excluded by the requirement that $n_k > n_{k-1}$.

To complete the proof, we argue that $\{x_k\}$ and $\{y_k\}$ both converge to the same limit (drawing a picture makes this very plausible). Since $x_k \leq a_{n_k} \leq y_k$ for all k, the Squeeze Theorem 2.6.5 will then imply that $\{a_{n_k}\}_{k=1}^{\infty}$ is also convergent to this limit. Because the intervals $[x_k, y_k]$ are nested, we have

$$x_0 \leq x_1 \leq x_2 \leq x_3 \leq \dots \leq y_3 \leq y_2 \leq y_1 \leq y_0.$$

In particular, the sequences $\{x_k\}_{k=0}^{\infty}$ and $\{y_k\}_{k=0}^{\infty}$ are monotone and bounded. Thus, by Theorem 2.6.8, they are convergent. Let us say that their limits are $L_1 = \lim_{k\to\infty} x_k$ and $L_2 = \lim_{k\to\infty} y_k$. Note that the interval $[x_k, y_k]$ has length $\frac{1}{2^k}$ times the length of the interval $[x_0, y_0]$. Thus, using the limit law for differences (Theorem 2.6.2), we have

$$L_2 - L_1 = \lim_{k\to\infty} y_k - \lim_{k\to\infty} x_k = \lim_{k\to\infty} (y_k - x_k) = \lim_{k\to\infty} \frac{y_0 - x_0}{2^k} = 0$$

proving $L_1 = L_2$. □

[3]The left and right halves of an interval $[c, d]$ are $[c, \frac{c+d}{2}]$ and $[\frac{c+d}{2}, d]$.

Example 2.7.5. The sequence $\{\sin(n)\}_{n=1}^{\infty}$ is bounded; each of its terms belongs to the interval $[-1, 1]$. Thus, by the Bolzano-Weierstrass theorem, there exists a subsequence $\{\sin(n_k)\}_{k=1}^{\infty}$ which converges. Actually, with more work, one can show that, for any choice of $x \in [-1, 1]$, there exists a subsequence which converges to x.

2.8 The Cauchy criterion

Before we move on to the Cauchy criterion for convergence, it is a good idea to do the following exercise. This exercise may be interpreted as saying that, if the terms of a sequence are getting closer and closer to some number L, then the terms of the sequence must also get closer and closer to one another. Cauchy's criterion shows that the converse is also true.

Exercise 2.8.1. Let $\{a_n\}$ be a convergent sequence. Prove that, for every $\epsilon > 0$, there exists a positive integer N such that $i, j \geq N$ implies that $|a_i - a_j| < \epsilon$.

Proof. Let $L = \lim_{n\to\infty} a_n$. We know there exists $N \in \mathbb{N}$ such that $n \geq N$ implies $|a_n - L| < \epsilon/2$. Then, if $i, j \geq N$, we have by the triangle inequality

$$|a_i - a_j| = |a_i - L + L - a_j| \leq |a_i - L| + |a_j - L| < \epsilon/2 + \epsilon/2 = \epsilon.$$

□

Sequences with the property in the above exercise are called Cauchy sequences.

Definition 2.8.2. A sequence of real numbers $\{a_n\}$ is a **Cauchy sequence** if, for every $\epsilon > 0$, there exists a positive integer N such that $i, j \geq N$ implies that $|a_i - a_j| < \epsilon$.

Cauchy's criterion says that, in the real line, Cauchy sequences and convergent sequences are one and the same. This result is intimately-related to the metric concept of completeness, which some students will encounter in other courses.

Theorem 2.8.3 (Cauchy criterion)**.** *A sequence of real numbers is convergent if and only if it is a Cauchy sequence.*

Proof. We already know every convergent sequence is Cauchy (Exercise 2.8.1 above) so it remains to show every Cauchy sequence is convergent. Let $\{a_n\}_{n=1}^{\infty}$ be a Cauchy sequence. We will structure the proof as follows:

1. Use the Cauchy property to show $\{a_n\}$ is bounded[4] so that, by the Bolzano-Weierstrass theorem, there is a convergent subsequence $\{a_{n_k}\}$.

2. Use the Cauchy property again to show that the whole sequence $\{a_n\}$ gets dragged off to the same limit as the subsequence $\{a_{n_k}\}$.

[4]Convergent sequences are bounded so, unless this theorem is false, it must be possible to prove this.

Here are the details. Applying the definition of a Cauchy sequence with $\epsilon = 1$, we get that there exists $N_1 \in \mathbb{N}$ such that $i, j \geq N_1$ implies $|a_i - a_j| < 1$. Now, put

$$M \coloneqq \max\{|a_1|, |a_2|, \ldots, |a_{N_1-1}|, |a_{N_1}| + 1\}.$$

We claim that $|a_n| \leq M$ for all $n \in \mathbb{N}$. Indeed, if $n < N_1$, then $|a_n| \leq M$ because M is the maximum of $|a_n|$ and other quantities. If, on the other hand, $n \geq N_1$, then

$$|a_n| = |a_n - a_{N_1} + a_{N_1}| \leq |a_n - a_{N_1}| + |a_{N_1}| < 1 + |a_{N_1}| \leq M.$$

We have shown $\{a_n\}_{n=1}^{\infty}$ is bounded so, by Theorem 2.7.4 (the Bolzano-Weierstrass theorem), there is a convergent subsequence $\{a_{n_k}\}_{k=1}^{\infty}$, say with limit $L \in \mathbb{R}$. We claim the whole sequence $\{a_n\}$ also converges to L. To this end, let $\epsilon > 0$ be fixed and arbitrary. Using the Cauchy property, there exists $N \in \mathbb{N}$ such that $i, j \geq N$ implies $|a_i - a_j| < \epsilon/2$. Next, since $\lim_{k\to\infty} a_{n_k} = L$, there exists some $K \in \mathbb{N}$ such that $k \geq K$ implies $|a_{n_k} - L| < \epsilon/2$. Choose a specific integer $k \geq K$ such that $n_k \geq N$ also holds (this is possible because only finitely-many $k \in \mathbb{N}$ satisfy $n_k < N$). Now we see that, if $n \geq N$, then

$$|a_n - L| = |a_n - a_{n_k} + a_{n_k} - L| \leq |a_n - a_{n_k}| + |a_{n_k} - L| < \epsilon/2 + \epsilon/2 = \epsilon.$$

Here, $|a_n - a_{n_k}| < \epsilon/2$ because $n, n_k \geq N$, and $|a_{n_k} - L| < \epsilon/2$ because $k \geq K$. □

Example 2.8.4. An important application of Cauchy's criterion is the "Banach fixed point theorem". A function $f : \mathbb{R} \to \mathbb{R}$ is a called a **contraction** if there is a number $r \in [0, 1)$ such that $|f(x) - f(y)| \leq r|x - y|$ for all $x, y \in \mathbb{R}$. This means applying f always strictly decreases the distance between two points by a factor of r. One can check (exercise) that if f is a contraction then, for any choice of initial point $x_0 \in \mathbb{R}$, the sequence of points $\{x_n\}_{n=0}^{\infty}$ defined recursively by $x_{n+1} = f(x_n)$, $n \geq 0$ is a Cauchy sequence and therefore, by Cauchy's criterion, has a limit $p = \lim_{n\to\infty} x_n$. Using other limit laws, one can also show that $f(p) = p$, i.e. that p is a "fixed point" for the function f. This fixed point theorem is very important in many areas of mathematics. For example, versions of it are used in the theory of ordinary differential equations to demonstrate the existence and uniqueness of solutions.

Here is a quick illustration: the function $f(x) = x/2$ is a contraction. For any choice of initial point, the sequence $\{x_n\}$ is $\{x_0/2^n\}$ which converges to 0, the unique fixed point of this function.

2.9 Examples of limits of monotone sequences

Example 2.9.1. Consider the sequence $\{a_n\}$ defined recursively by $a_1 = 0$, $a_n = \sqrt{a_{n-1} + 2}$ for all $n \geq 2$. The first few terms are:

$$a_1 = 0 \qquad a_2 = \sqrt{2} \approx 1.41 \qquad a_3 = \sqrt{2 + \sqrt{2}} \approx 1.85 \qquad \ldots$$

It is easy to show by induction that the sequence $\{a_n\}$ is strictly increasing. Indeed, the values above give the base case of the induction and, if we assume $a_{n-1} < a_n$ for some $n \geq 2$,

then it follows that $a_{n+1} = \sqrt{2+a_n} > \sqrt{2+a_{n-1}} = a_n$. We can also show the sequence is bounded above, say by 7. Again, the values shown above give the base case and, if we assume that $a_n < 7$ for some $n \geq 1$, then it follows that $a_{n+1} = \sqrt{a_n+2} < \sqrt{7+2} = 3 < 7$ as well. Thus by the Theorem 2.6.8 (the monotone sequence convergence theorem), there exists a finite limit $L = \lim_{n\to\infty} a_n$. Now, since $a_{n+1}^2 = a_n + 2$, we can use limit laws to conclude that the number L satisfies

$$L^2 = \lim_{n\to\infty} a_{n+1}^2 = \lim_{n\to\infty} a_n + 2 = L + 2.$$

The real solutions of the equation $x^2 = x + 2$ are $x = -1$ and $x = 2$. Clearly the limit L cannot be -1, so we conclude that $L = 2$.

Remark 2.9.2. One sometimes sees the above limit, and similar ones, expressed using whimsical equations like the following:

$$\sqrt{2+\sqrt{2+\sqrt{2+\sqrt{2+\ldots}}}} = 2.$$

Actually, somewhat entertainingly, in this specific case it is possible to get a convenient closed form for the general term of this specific sequence of nested roots:

$$2\cos(\tfrac{\pi}{4}) = \sqrt{2} \qquad 2\cos(\tfrac{\pi}{8}) = \sqrt{2+\sqrt{2}} \qquad 2\cos(\tfrac{\pi}{16}) = \sqrt{2+\sqrt{2+\sqrt{2}}} \qquad \cdots$$

This curious fact was pointed out to me by N. Strungaru. Since $\lim_{n\to\infty} \cos(\frac{\pi}{2^n}) = \cos(0) = 1$, this provides an alternative way of calculating the limit. Note, however, the approach of looking for a closed form is not viable for variants of this sequence.

Example 2.9.3. The famous mathematical constant e was first discovered by Jakob Bernoulli while he was studying compound interest. In this context, e naturally arises as the limit of the sequence $\{(1+1/n)^n\}_{n=1}^{\infty}$. The first few terms of this sequence are approximately given by:

$$2 \quad 2.250 \quad 2.370 \quad 2.441 \quad 2.488 \quad 2.521 \quad 2.546 \quad 2.565 \quad 2.581 \quad 2.593 \quad \ldots$$

The above approximations suggest this sequence may be strictly increasing and bounded. Provided we can prove those properties hold, Theorem 2.6.8 gives that the sequence converges. It turns out these properties do hold and the sequence converges (quite slowly) to a transcendental number $e \approx 2.718$. We will prove $\{(1+1/n)^n\}$ is strictly increasing and bounded using an approach taken from [8]. Given real numbers $0 \leq a < b$ and $n \in \mathbb{N}$, the geometric series formula gives

$$\frac{b^{n+1} - a^{n+1}}{b-a} = \sum_{k=0}^{n} a^{n-k}b^k < \sum_{k=0}^{n} b^{n-k}b^k = (n+1)b^n$$

which can be rerranged into the inequality:

$$[(n+1)a - nb] \cdot b^n < a^{n+1}. \tag{2.1}$$

To see our sequence is strictly increasing, we substitute $a = 1 + 1/(n+1)$ and $b = 1 + 1/n$ in (2.1). After a bit of calculation, the expression in square braces simplifies to 1 which exactly gives that our sequence is strictly increasing:

$$\left(1 + \frac{1}{n}\right)^n < \left(1 + \frac{1}{n+1}\right)^{n+1} \qquad \text{for all } n \in \mathbb{N}.$$

Next, we substitute $a = 1$ and $b = 1 + 1/(2n)$ in (2.1). This time, the expression in square braces simplifies to $1/2$ and, after a bit more algebra, leads to the inequality

$$\left(1 + \frac{1}{2n}\right)^{2n} < 4 \qquad \text{for all } n \in \mathbb{N}.$$

This shows the even-numbered terms of our sequence are bounded by 4 which, since the sequence is strictly increasing, implies all the terms of the sequence are bounded by 4.

Chapter 3

Series

3.1 Infinite series - basic notions

An **infinite series** (or just **series** for short) is the infinite summation of a sequence of real numbers $\{a_n\}_{n=1}^{\infty}$. One typically denotes an infinite series by:

$$\sum_{n=1}^{\infty} a_n \qquad \text{or sometimes} \qquad a_1 + a_2 + a_3 + \dots.$$

At the moment, $\sum_{n=1}^{\infty} a_n$ is just a formal expression which evokes the idea of attempting to add together infinitely-many terms. However, the most interesting situation is when the series converges to a real value S and in such cases we will also interpret the series as being equal to the value it converges to, writing $\sum_{n=1}^{\infty} a_n = S$. The precise definition of convergence of a series will be given shortly. First note there are two important sequences associated to every series:

- The sequence of **terms**: $\{a_n\}_{n=1}^{\infty}$, the numbers we are attempting to add.
- The sequence of **partial sums**: $\{s_n\}_{n=1}^{\infty}$, where s_n denotes the sum of the first n terms of the series: $s_n \coloneqq \sum_{k=1}^{n} a_k = a_1 + a_2 + \ldots + a_n$.

Definition 3.1.1. We say that $\sum_{n=1}^{\infty} a_n$ **converges** if its sequence of partial sums $\{s_n\}$ converges to a real number S. In this case, we write

$$\sum_{n=1}^{\infty} a_n = \lim_{n\to\infty} s_n = S.$$

Otherwise, we say $\sum_{n=1}^{\infty} a_n$ **diverges**.

Remark 3.1.2. As in the case of sequences, we will feel free to modify the indexing when convenient considering series like $\sum_{n=3}^{\infty} a_n$ or $\sum_{j=0}^{\infty} a_j$, for example.

The easiest series to get a handle on are those admitting a convenient closed form for the partial sums. The following is an example of such a series.

Example 3.1.3 (Telescoping series)**.** Consider the series

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)}.$$

In order to calculate the sequence of partial sums for this series, it is helpful to first observe that the nth term can be rewritten as:

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}.$$

This identity is easy to confirm by bringing the right side to a common denominator. We remark that this identity is a particular instance of a "partial fraction decomposition" which is often taught to calculus students as a useful technique for finding antiderivatives. Now, for any $n \in \mathbb{N}$, we calculate:

$$\begin{aligned} s_n &= \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \ldots + \frac{1}{n(n+1)} \\ &= \left(\frac{1}{1} - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \ldots + \left(\frac{1}{n} - \frac{1}{n+1}\right) \\ &= 1 - \frac{1}{n+1}. \end{aligned}$$

So, for this series, the sequence of partial sums is $\{1 - \frac{1}{n+1}\}_{n=1}^{\infty}$, which converges to 1. By definition, this means the series also converges to 1:

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \lim_{n\to\infty} s_n = \lim_{n\to\infty} \left(1 - \frac{1}{n+1}\right) = 1.$$

Convergent series for which we are able to compute the sum are fairly uncommon. In practice, it is more likely that one will be able to prove that a series converges but be unable to calculate the value of the sum exactly.

Example 3.1.4 (Basel sum)**.** An especially famous infinite series is the Basel sum:

$$\sum_{n=1}^{\infty} \frac{1}{n^2}.$$

It is fairly easy to show that this series converges, but much less straightforward to calculate the value of the sum. The correct value of $\pi^2/6$ was first calculated by Euler using not-so-rigorous methods. There are now many known proofs of this result. Although we will not calculate the value of the sum here, we will at least argue that the Basel sum converges by comparing it to the telescoping series considered in Example 3.1.3. Note that the sequence of partial sums of the Basel sum, or indeed of any series whose terms are nonnegative, is increasing. Thus, by Theorem 2.6.8 (bounded montone sequences are convergent), to show

the Basel sum converges, we just need to show its sequence of partial sums is bounded. Indeed, denoting the nth partial sum of the Basel sum by t_n, we have:

$$\begin{aligned} t_n &= 1 + \frac{1}{2^2} + \frac{1}{3^2} + \ldots + \frac{1}{n^2} \\ &\leq 1 + \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \ldots + \frac{1}{(n-1) \cdot n} \qquad \text{(since } \tfrac{1}{k^2} \leq \tfrac{1}{(k-1)k} \text{ for } k \geq 2\text{)} \\ &\leq 1 + \sum_{n=1}^{\infty} \frac{1}{n(n+1)} \\ &= 2 \end{aligned}$$

It follows that the Basel sum converges and that $\sum_{n=1}^{\infty} \frac{1}{n^2} = \lim_{n\to\infty} t_n \leq 2$. Indeed, the true value of the sum is known to be $\pi^2/6 \approx 1.64$.

The approach used above to prove that the Basel series converges by comparing it to a known convergent series is very simple, but so useful that it is worth our while to formalize it as a "test" for convergence.

Theorem 3.1.5 (Comparison test)**.** *Let $\{a_n\}$ and $\{b_n\}$ be real sequences with $0 \leq a_n \leq b_n$ for all $n \in \mathbb{N}$. If $\sum_{n=1}^{\infty} b_n$ converges, then so does $\sum_{n=1}^{\infty} a_n$. Moreover, $0 \leq \sum_{n=1}^{\infty} a_n \leq \sum_{n=1}^{\infty} b_n$.*

Proof. Let $\{s_n\}$ and $\{t_n\}$ be the sequences of partial sums for $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ respectively. That is, $s_n = \sum_{k=1}^{n} a_k$ and $t_n = \sum_{k=1}^{n} b_k$. Since these $a_n, b_n \geq 0$ for all $n \in \mathbb{N}$, the sequences of partial sums are increasing. Since $\{t_n\}$ converges by assumption, $\{t_n\}$ is a bounded sequence (Theorem 2.5.3). Since $0 \leq a_n \leq b_n$ for all $n \in \mathbb{N}$ by assumption, it follows that $0 \leq s_n \leq t_n$ for all $n \in \mathbb{N}$. Thus, the sequence $\{s_n\}$ is also bounded and, by Theorem 2.6.8, we have

$$\sum_{n=1}^{\infty} a_n = \lim_{n\to\infty} s_n = \sup\{s_n : n \in \mathbb{N}\} \leq \sup\{t_n : n \in \mathbb{N}\} = \lim_{n\to\infty} t_n = \sum_{n=1}^{\infty} b_n$$

as claimed. □

Next let's look at an example of a divergent series.

Example 3.1.6. Does the sequence

$$\sum_{n=0}^{\infty} (-1)^n = 1 - 1 + 1 - 1 + 1 - 1 + \ldots$$

converge or diverge? Beginners may be tempted to say that each 1 is cancelled by a -1 so that the whole sum should converge to 0. Recall however that the precise definition is that

the series converges when the sequence of partial sums $\{s_n\}$ converges. Let us calculate the first few partial sums and extract the trend:

$$\begin{aligned}
\text{sum of first term} &\qquad s_1 = 1 \\
\text{sum of first two terms} &\qquad s_2 = 1 - 1 = 0 \\
\text{sum of first three terms} &\qquad s_3 = 1 - 1 + 1 = 1 \\
\text{sum of first four terms} &\qquad s_4 = 1 - 1 + 1 - 1 = 0 \\
\text{sum of first five terms} &\qquad s_5 = 1 - 1 + 1 - 1 + 1 = 1 \\
\vdots &\qquad \vdots
\end{aligned}$$

We see that the sequence of partial sums is $0, 1, 0, 1, 0, 1, 0, \ldots$ which oscillates back and forth between two values. In particular, the sequence of partial sums is divergent so, by definition, the series $\sum_{n=0}^{\infty}(-1)^n$ also diverges.

Now we look at an important family of infinite series whose partial sums can be explicitly calculated.

Example 3.1.7 (Geometric series)**.** For fixed $r \in \mathbb{R}$, the associated **geometric series** is:

$$\sum_{n=0}^{\infty} r^n = 1 + r + r^2 + r^3 + \ldots$$

Let $\{s_n\}_{n=0}^{\infty}$ be the sequence of partial sums, $s_n = 1 + r + \ldots + r^n$. If $r = 1$, then we have $s_n = n + 1$ and the series diverges. If $r \neq 1$, we have the familiar closed formula:

$$s_n = \sum_{k=0}^{n} r^k = \frac{1 - r^{n+1}}{1 - r} \qquad\qquad r \neq 1.$$

To obtain this closed form, one may subtract the equations:

$$s_n = 1 + r + r^2 + \ldots + r^n \qquad\qquad rs_n = r + r^2 + r^3 + \ldots + r^{n+1}$$

and solve for s_n. The sequence of partial sums converges if and only if $|r| < 1$ so that $\lim_{n\to\infty} r^n = 0$. We get:

$$\sum_{n=0}^{\infty} r^n = \lim_{n\to\infty} \frac{1 - r^{n+1}}{1 - r} = \frac{1}{1 - r} \qquad\qquad |r| < 1. \tag{3.1}$$

The case $r = 1/4$ has a particularly fun illustration. Figure 3.1 suggests that it is possible to divide the unit square into three regions of area $\sum_{n=1}^{\infty}(1/4)^n$. Thus, this area should be equal to $1/3$. This finding is consistent with one what gets by applying (3.1).

$$\sum_{n=1}^{\infty}(1/4)^n = \sum_{n=0}^{\infty}(1/4)^{n+1} = \sum_{n=0}^{\infty}(1/4)(1/4)^n = (1/4)\sum_{n=0}^{\infty}(1/4)^n = (1/4)\frac{1}{1 - (1/4)} = \frac{1}{3}.$$

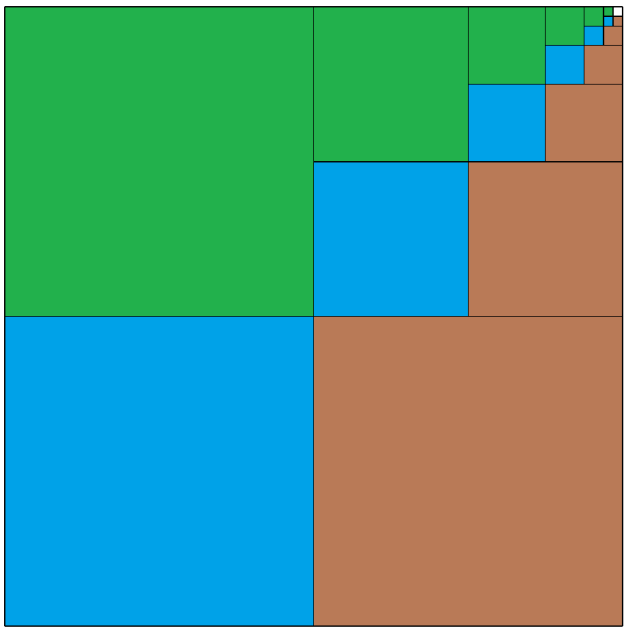


Figure 3.1: "Proof without words" that $\sum_{n=1}^{\infty}(1/4)^n$.

In the calculation above we used a few computational tricks: first we re-indexed the sum and then pulled out a constant. "Pulling out a constant" is a valid operation for convergent series. Ultimately, this boils down to the fact that the sum of a series is defined as the limit of its sequence of partial sums, and we have already established a variety of limit laws in the context of sequences.

Theorem 3.1.8 (Limit laws for series)**.** *Suppose that* $\sum_{n=1}^{\infty} a_n$ *and* $\sum_{n=1}^{\infty} b_n$ *are convergent series. Then,*

(i) $\sum_{n=1}^{\infty}(a_n + b_n) = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n$.

(ii) $\sum_{n=1}^{\infty} \lambda a_n = \lambda \sum_{n=1}^{\infty} a_n$ *for any* $\lambda \in \mathbb{R}$.

Proof. Denote the sequences of partial sums of the three infinite series $\sum_{n=1}^{\infty} a_n$, $\sum_{n=1}^{\infty} b_n$ and $\sum_{k=1}^{\infty}(a_k + b_k)$ by $\{\alpha_n\}$, $\{\beta_n\}$ and $\{s_n\}$, respectively. That is:

$$\begin{aligned} \alpha_n &= a_1 + a_2 + \ldots + a_n \\ \beta_n &= b_1 + b_2 + \ldots + b_n \\ s_n &= a_1 + b_1 + a_2 + b_2 + \ldots + a_n + b_n, \qquad\qquad n \in \mathbb{N}. \end{aligned}$$

Since $s_n = \alpha_n + \beta_n$, the limit law for sums of convergent sequences (Theorem 2.6.2) gives:

$$\sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n = \lim_{n\to\infty} \alpha_n + \lim_{n\to\infty} \beta_n = \lim_{n\to\infty} (\alpha_n + \beta_n) = \lim_{n\to\infty} s_n = \sum_{n=1}^{\infty}(a_n + b_n).$$

The proof of (ii) is similar. □

Remark 3.1.9. Note that the limit law for products of sequences does not yield a simple law for products of series. In general, $\sum_{n=1}^{\infty} a_n b_n \neq (\sum_{n=1}^{\infty} a_n)(\sum_{n=1}^{\infty} b_n)$. Indeed, we don't even have such a simple rule for finite sums! Generally, $a_1 b_1 + a_2 b_2 \neq (a_1 + a_2)(b_1 + b_2)$ because of the cross terms. The issue is similar in the context of infinite series.

3.2 Additional convergence tests for series

There are many so-called convergence tests one can deploy in an attempt to decide whether or not a given series converges. Since the convergence of a series is defined in terms of the convergence of its sequence of partial sums, the Cauchy criterion for convergence of a sequence can be translated quite directly into a convergence criterion for series.

Theorem 3.2.1 (Cauchy criterion for series)**.** *An infinite series $\sum_{k=1}^{\infty} a_k$ converges if and only if the following condition is satisfied: for all $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that $N \leq m \leq n$ implies $|\sum_{k=m}^{n} a_k| < \epsilon$.*

A colloquial reading of the above theorem is that a series converges if and only if, by going out far enough, all the "tails" of the series can be made as small as desired.

Proof. By definition, the series $\sum_{k=1}^{\infty} a_k$ converges if and only if the sequence $\{s_k\}$ converges where $s_k \coloneqq a_1 + a_2 + \ldots + a_k$ is the kth partial sum. By the Cauchy criterion for sequences (Theorem 2.8.3), $\{s_k\}$ converges if and only if it is a Cauchy sequence. This exactly means that, for all $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that $i, j \geq N$ implies $|s_j - s_i| < \epsilon$. Note however that this condition is not changed if we also insist $i < j$. Indeed, if $i = j$, then $|s_j - s_i| = 0 < \epsilon$ holds automatically and, if $i > j$, then we may interchange the roles of i and j without affecting the truth value of the inequality $|s_j - s_i| < \epsilon$. Note as well that when $i < j$, we have

$$s_j - s_i = \sum_{k=1}^{j} a_k - \sum_{k=1}^{i} a_k = \sum_{k=i+1}^{j} a_k.$$

Thus, relabelling $m = i$ and $n = j - 1$ so that $N \leq i < j \Leftrightarrow N \leq m \leq n$, we obtain that $\sum_{k=1}^{\infty} a_k$ converges if and only if the desired condition holds: for all $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that $N \leq m \leq n$ implies $|\sum_{k=m}^{n} a_k| < \epsilon$. □

The Cauchy criterion for series convergence has a couple corollaries worth taking note of.

Corollary 3.2.2. *Consider an infinite series $\sum_{n=1}^{\infty} a_n$ and fix $M \in \mathbb{N}$. Then, $\sum_{n=1}^{\infty} a_n$ converges if and only if $\sum_{n=M}^{\infty} a_n$ converges.*

Proof. Exercise. □

A colloquial read of the above corollary is that any initial segment of an infinite series is irrelevant to the convergence or nonconvergence of the series. Changing only finitely-many terms of a series does not have an impact on whether the series converges or not (though it can change the numerical value of the sum).

Corollary 3.2.3 (nth term test)**.** *If* $\sum_{k=1}^{\infty} a_k$ *is convergent, then* $\lim_{k\to\infty} a_k = 0$.

Proof. Suppose that $\sum_{k=1}^{\infty} a_k$ converges. Fix $\epsilon > 0$. By Theorem 3.2.1, there exists $N \in \mathbb{N}$ such that $N \leq m \leq n$ implies $|\sum_{k=m}^{n} a_k| < \epsilon$. In particular, taking $m = n$, we get that $n \geq N$ implies $|a_n| < \epsilon$, which shows $\lim_{n\to\infty} a_n = 0$. □

A colloquial read of the above corollary is that, in order for a series $\sum_{n=1}^{\infty} a_n$ to have any hope of converging, it **must** hold that the terms of the sequences are approaching 0 in size. So, for example, the following series cannot possibly be convergent because, in all cases, the terms of the sequence do not approach 0:

- $\sum_{n=1}^{\infty} n$ (the magnitude of the nth term is unbounded)
- $\sum_{n=1}^{\infty} (-1)^n \sqrt{n}$ (the magnitude of the nth term is unbounded)
- $\sum_{n=1}^{\infty} \frac{n}{n+1}$ (the nth term approaches $1 \neq 0$).

The converse to Corollary 3.2.3 is false. Not every series whose sequence of terms tends towards 0 must converge. An important example of this phenomenon is the following:

Example 3.2.4 (Harmonic series)**.** The sum of the reciprocals of the positive integers is known as harmonic series:

$$\sum_{n=1}^{\infty} \frac{1}{n}.$$

Note that $\lim_{n\to\infty} 1/n = 0$ i.e. the terms of this sequence tend towards 0. However, in spite of this, the partial sums of the series still diverge (extremely slowly) towards ∞. The sequence of partial sums of this series is called the sequence of harmonic numbers $\{h_n\}$:

$$\begin{aligned}
h_1 &= 1 \\
h_2 &= 1 + 1/2 = 1.5 \\
h_3 &= 1 + 1/2 + 1/3 \approx 1.83 \\
h_4 &= 1 + 1/2 + 1/3 + 1/4 \approx 2.08 \\
h_5 &= 1 + 1/2 + 1/3 + 1/4 + 1/5 \approx 2.28
\end{aligned}$$

Clearly this is a monotone increasing series. The growth is extremely slow. For example:

$$h_{2000000000000000000000000000000000000000000000} \approx 100.28$$

However, in spite of the slow growth, the series still diverges to infinity. One way to see this is to use calculus to underestimate $\sum_{k=1}^{n} \frac{1}{k}$ by $\int_1^n \frac{dx}{x} = \log(n)$ and recall $\lim_{x\to\infty} \log(x) = \infty$. A different proof, which in some ways gives more insight into why the series diverges so slowly,

is note that one can strategically organize the terms of the harmonic series into groups of size $2, 4, 8, 16, \ldots$ in such a way that each group sums to more than $1/2$. For example:

$$\begin{aligned} h_{15} &= 1 + \underbrace{\frac{1}{2} + \frac{1}{3}} + \underbrace{\frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7}} + \underbrace{\frac{1}{8} + \frac{1}{9} + \frac{1}{10} + \frac{1}{11} + \frac{1}{12} + \frac{1}{13} + \frac{1}{14} + \frac{1}{15}} \\ &> 1 + \underbrace{\frac{1}{4} + \frac{1}{4}}_{>1/2} + \underbrace{\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}}_{>1/2} + \underbrace{\frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16}}_{>1/2} \\ &= 1 + 3/2. \end{aligned}$$

In general, the same trick shows that for any $n \in \mathbb{N}$, one has

$$h_{2^{n+1}-1} > 1 + \underbrace{1/2 + 1/2 + \ldots + 1/2}_{n \text{ times}} = 1 + n/2.$$

This bound shows the subsequence $h_1, h_3, h_7, h_{15}, h_{31}, \ldots$ of the sequence of harmonic numbers diverges to infinity and so, since $\{h_n\}$ is monotone increasing, $\sum_{n=1}^{\infty} \frac{1}{n} = \lim_{n\to\infty} h_n = \infty$ as well.

The trick we just used to show the harmonic series diverges can be formalized into a useful convergence test which can be applied to many different series.

Theorem 3.2.5 (2^n test)**.** *Let $\{a_n\}_{n=1}^{\infty}$ be a decreasing sequence of nonnegative reals (meaning $a_1 \geq a_2 \geq a_3 \geq \ldots \geq 0$). Then $\sum_{n=1}^{\infty} a_n$ converges if and only if $\sum_{n=1}^{\infty} 2^n a_{2^n}$ converges.*

Proof. We assume that $\sum_{n=1}^{\infty} 2^n a_{2^n}$ converges and prove that $\sum_{n=1}^{\infty} a_n$ converges. The proof of the reverse implication is similar and left to the reader as an exercise. The convergence of $\sum_{n=1}^{\infty} a_n$ by is equivalent, by definition, to the convergence of the sequence of partial sums $s_n = \sum_{k=1}^{n} a_k$. Since the terms of the series are nonnegative, the sequence $\{s_n\}_{n=1}^{\infty}$ is increasing. Because of this monotonicity we need only prove that $\{s_n\}$ is bounded above (by Theorem 2.6.8). Indeed, notice that

$$\begin{aligned} a_1 &\leq a_1 \\ a_2 + a_3 &\leq 2a_2 \\ a_4 + a_5 + a_6 + a_7 &\leq 4a_4 \end{aligned}$$

and, in general,

$$\sum_{k=2^n}^{2^{n+1}-1} a_k \leq 2^n a_{2^n}$$

for any $n \in \mathbb{N}$. Summing these inequalities we have

$$s_{2^{n+1}-1} = \sum_{k=1}^{2^{n+1}-1} a_k \leq \sum_{k=0}^{n} 2^k a_{2^k} \leq \sum_{k=0}^{\infty} 2^k a_{2^k} < \infty.$$

We have so far only shown a subsequence of $\{s_n\}$ to be bounded above. However, using the montonicity of $\{s_n\}$, the whole sequence $\{s_n\}$ must also be bounded above (why?). □

3.3 Absolute convergence

Next we study a stronger convergence condition for series known as absolute convergence.

Definition 3.3.1. A series $\sum_{n=1}^{\infty} a_n$ is said to **converge absolutely** if $\sum_{n=1}^{\infty} |a_n|$ converges.

Example 3.3.2. Recall that a geometric series $\sum_{n=0}^{\infty} r^n$ converges to $\frac{1}{1-r}$ if $|r| < 1$ (see Example 3.1.7). In particular, taking $r = -1/2$, we have a convergent series:

$$\sum_{n=0}^{\infty}(-1/2)^n = 1 - 1/2 + 1/4 - 1/8 + 1/16 + \ldots = \frac{1}{1-(-1/2)} = 2/3. \tag{3.2}$$

If in (3.2) above we replace each term by its absolute value, we get a different geometric series, this time with ratio $r = 1/2$:

$$\sum_{n=0}^{\infty} |(-1/2)^n| = \sum_{n=0}^{\infty}(1/2)^n = \frac{1}{1-(1/2)} = 2. \tag{3.3}$$

This means the series $\sum_{n=0}^{\infty}(-1/2)^n$ is both convergent in the usual sense (3.2) as well as being absolutely convergent (3.3). Take note that the sum $2/3$ of the original series (which had a mixture of positive and negative terms) was smaller than the sum of 2 that was obtained after taking the absolute value of every term. This is typical. Taking the absolute value results in a series all of whose terms "cooperate" because they all have the same sign. On the other hand, before the taking the absolute value, the positive and negative terms of the series "compete" with one another leading to some amount of cancellation and a sum of smaller magnitude. Sciencey types may find themselves reminded of constructive/destructive interference of wave-like phenomena.

As the discussion in the above example suggests, if a series is absolutely convergent, it should also be convergent. The idea is that the partial sums of $\sum_{n=1}^{\infty} |a_n|$ can only be larger in magnitude than the partial sums of $\sum_{n=1}^{\infty} a_n$. We state this as a theorem, though it is more or less a corollary of the Cauchy criterion for convergence of a series.

Theorem 3.3.3. *If $\sum_{n=1}^{\infty} a_n$ converges absolutely, then it converges in the usual sense.*

Proof. We assume $\sum_{k=1}^{\infty} |a_k|$ converges and prove that $\sum_{k=1}^{\infty} a_k$ also converges. To do this we make use of the Cauchy criterion for series convergence (Theorem 3.2.1). Fix $\epsilon > 0$. We must show there exists an $N \in \mathbb{N}$ such that $N \leq m \leq n$ implies $|\sum_{k=m}^{n} a_k| < \epsilon$. Applying the same criterion to the series of absolute values $\sum_{k=1}^{\infty} |a_k|$ (which we assumed converges), we know we can choose $N \in \mathbb{N}$ such that $N \leq m \leq n$ implies $\sum_{k=m}^{n} |a_k| < \epsilon$. However, by the triangle inequality, the same choice of N works for the original series as well, i.e. $N \leq m \leq n$ implies

$$\left| \sum_{k=m}^{n} a_n \right| \leq \sum_{k=m}^{n} |a_k| < \epsilon.$$

□

The converse to the above theorem is false. There exist convergent series that are not absolutely convergent.

Example 3.3.4 (Alternating harmonic series)**.** The series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = 1 - 1/2 + 1/3 - 1/4 + 1/5 - \dots$$

is convergent. However, it is not absolutely convergent. The corresponding series of absolute values is the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ which we know diverges (see Example 3.2.4). Rather than just prove the specific alternating series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ converges, we will look at a more general class of alternating series and prove those converge in Theorem 3.4.1 below.

Remark 3.3.5. Using some properties of power series that we are not likely to cover, one can show that the value of the alternating harmonic series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ is $\log(2)$, the natural logarithm of 2. Formally speaking, the idea is pretty simple: by integrating the geometric series $\frac{1}{1+x} = \sum_{n=0}^{\infty}(-1)^n x^n$ term-by-term, one obtains $\log(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^n$. Then we simply plug in $x = 1$. This sketch glosses over some finer analytical details, however. The first series only converges for $x \in (-1, 1)$, while the second converges for $x \in (-1, 1]$.

We end this section with the so-called "root test". This simple but useful test for absolute convergence of a series $\sum a_n$ is based on the natural idea of investigating for which constants $r > 1$ does one have $|a_n| \leq r^n$ (perhaps only for sufficiently large n). The root test really comes into its own in the context of power series where it may be used to determine the radius of convergence. For more information, search for the Cauchy-Hadamard theorem.

Theorem 3.3.6 (Root test)**.** *Given a series $\sum_{n=1}^{\infty} a_n$, define $L = \limsup_{n\to\infty} |a_n|^{1/n}$, with the understanding that $L = \infty$ if this sequence is unbounded. If $L < 1$, then $\sum_{n=1}^{\infty} a_n$ converges absolutely. If $L > 1$, then $\sum_{n=1}^{\infty} a_n$ diverges.*

Proof. If $L < 1$ we may choose r such that $L < r < 1$. We may then find (why?) a positive integer N such that $n \geq N$ implies $|a_n|^{1/n} \leq r$. That is, $|a_n| \leq r^n$ for $n \geq N$. Since a series converges if and only one of its tails converges (Corollary 3.2.2) and the tail $\sum_{n=N}^{\infty} |a_n|$ converges by comparison with the convergent geometric series $\sum_{n=N}^{\infty} r^n$ (Theorem 3.1.5 and Example 3.1.7), we have that $\sum_{n=1}^{\infty} |a_n|$ converges. That is, the original series converges absolutely. The $L > 1$ case is left as an exercise for the reader. □

Remark 3.3.7. Two cautionary notes are appropriate at this point. In the exercises below you are asked to explore these issues in more detail.

- When $L = 1$ the root test is "indeterminate"; the series may or may not converge.
- Note from the proof above that, for any fixed $r > L$, one always has $|a_n| \leq r^n$ for all sufficiently large n. It is tempting to think that perhaps $|a_n| \leq L^n$ holds for sufficiently large n, but this is not always the case.

Exercise 3.3.8. Give two examples $\{a_n\}$, $\{b_n\}$ of positive sequences both of which satisfy $\limsup_{n\to\infty} a_n^{1/n} = \limsup_{n\to\infty} b_n^{1/n} = 1$ such that $\sum_{n=1}^{\infty} a_n$ converges and $\sum_{n=1}^{\infty} b_n$ diverges.

Exercise 3.3.9. Give an example of a positive sequence $\{a_n\}$ with $\limsup_{n\to\infty} a_n^{1/n} = 1/2$ for which there do not exist constants $M, N > 0$ such that $a_n \leq M(1/2)^n$ for all $n \geq N$.

Exercise 3.3.10. Establish the validity of the so-called "ratio test":

> **Theorem** (Ratio test)**.** Consider a series $\sum_{n=1}^{\infty} a_n$, and assume that the limit $L = \lim_{n\to\infty} \frac{|a_{n+1}|}{|a_n|}$ exists. If $L < 1$, then $\sum_{n=1}^{\infty} a_n$ converges absolutely. If $L > 1$, then $\sum_{n=1}^{\infty} a_n$ diverges.

Remark 3.3.11. Take note that, just as in the case of the root test, the ratio test is indeterminate if $L = 1$. In a technical sense, the root test is strictly more powerful than the ratio test because one can prove (exercise) that if the limit in the ratio test exists, then the limit superior in the root test also exists and has the same value. However, in a practical sense, there are some examples where the limit in the ratio test is more conveniently evaluated than the limit superior in the root test.

Exercise 3.3.12. Give an example of a positive sequence $\{a_n\}_{n=1}^{\infty}$ with $\lim_{n\to\infty} |a_n|^{1/n} = 1/2$ such that $\lim_{n\to\infty} \frac{|a_{n+1}|}{|a_n|}$ does not exist.

3.4 Alternating series

The convergence of the alternating harmonic series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ is a special case of the following theorem (take $a_n = \frac{1}{n}$).

Theorem 3.4.1 (Alternating series test)**.** *If $\{a_n\}_{n=1}^{\infty}$ is a decreasing sequence of nonnegative reals (i.e. $a_1 \geq a_2 \geq a_3 \geq \ldots \geq 0$) and $\lim_{n\to\infty} a_n = 0$, then the series*

$$\sum_{n=1}^{\infty} (-1)^{n+1} a_n = a_1 - a_2 + a_3 - a_4 + a_5 - \ldots$$

is convergent.

Before proceeding to the proof of the theorem, it is an extremely good idea to convince oneself by sketching a picture that the hypotheses of the theorem cause the partial sums of the series to be arranged as follows:

$$s_2 \leq s_4 \leq s_6 \leq \ldots \leq s_5 \leq s_3 \leq s_1.$$

One should think of jumping from one partial sum to the next by adding the next term of the series:

$$s_1 = a_1 \qquad s_2 = s_1 - a_2 \qquad s_3 = s_2 + a_3 \qquad s_4 = s_3 - a_4 \qquad \ldots$$

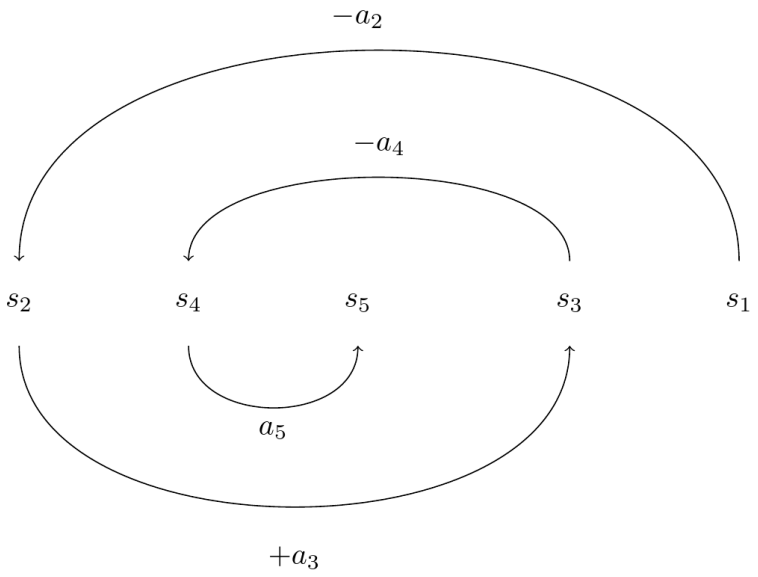


Figure 3.2: The alternating series test.

When n is odd, we jump to the right by adding a_n. When n is even, we jump to the left by subtracting a_n. Furthermore, the sizes of the jumps are getting smaller. This behaviour is illustrated in Figure 3.4.1.

Now let us translate this picture into a formal proof.

Proof. We must show the sequence $\{s_n\}_{n=1}^{\infty}$ of partial sums $s_n = \sum_{k=1}^{n}(-1)^{k+1}a_k$ is convergent. We claim that, for every $n \in \mathbb{N}$, one has

$$s_{2n} \underset{(i)}{\leq} s_{2n+2} \underset{(ii)}{\leq} s_{2n+1} \underset{(iii)}{\leq} s_{2n-1}.$$

Indeed, the explanations for these three inequalities are the following:

(i) $s_{2n+2} = s_{2n} + a_{2n+1} - a_{2n+2}$ and $a_{2n+1} - a_{2n+2} \geq 0$ because $a_{2n+1} \geq a_{2n+2}$.

(ii) $s_{2n+2} = s_{2n+1} - a_{2n+2}$ and $a_{2n+2} \geq 0$.

(iii) $s_{2n+1} = s_{2n-1} - a_{2n} + a_{2n+1} = s_{2n-1} - (a_{2n} - a_{2n+1})$ and $a_{2n} \geq a_{2n+1}$.

These inequalities imply that the subsequence $\{s_{2k}\}_{n=1}^{\infty}$ of even partial sums is increasing and bounded above (any odd partial sum is an upper bound). Symmetrically, the subsequence $\{s_{2n-1}\}_{n=1}^{\infty}$ of odd partial sums is decreasing and bounded below (any even partial sum is an lower bound). Thus (Theorem 2.6.8) each of these subsequences is convergent, say to $L_0 \coloneqq \lim_{n\to\infty} s_{2n}$ and $L_1 \coloneqq \lim_{n\to\infty} s_{2n-1}$. Furthermore, since $s_{2n} = s_{2n-1} - a_{2n}$ for all $n \in \mathbb{N}$ and $\lim_{n\to\infty} a_{2n} = 0$ (see Theorem 2.7.3), the limit laws (Theorem 2.6.2) give that

$$L_0 = \lim_{n\to\infty} s_{2n} = \lim_{n\to\infty}(s_{2n-1} - a_{2n}) = \lim_{n\to\infty} s_{2n-1} - \lim_{n\to\infty} a_{2n} = L_1 - 0 = L_1.$$

Thus, both subsequences converge to the same limit $L := L_0 = L_1$. Only one technical detail remains: since the subsequences of even and odd partial sums "cover" the sequence of partial sums $\{s_n\}$ and converge to the same limit L, the whole sequence $\{s_n\}$ also goes to L. Indeed, given any $\epsilon > 0$, there exists $N_0, N_1 \in \mathbb{N}$ such that $n \geq N_0$ implies $|s_{2n} - L| < \epsilon$ and $n \geq N_1$ implies $|s_{2n-1} - L| < \epsilon$. Put $N = 2\max\{N_1, N_2\}$ and suppose $n \geq N$. If n is even, we may write $n = 2m$ where $m = n/2 \geq N/2 \geq N_0$. If n is odd, we may write $n = 2m - 1$ where $m = (n+1)/2 \geq n/2 \geq N/2 \geq N_1$. In both cases, we may conclude $|a_n - L| < \epsilon$. $\square$

Exercise 3.4.2. Show the monotonicity of the sequence $\{a_n\}$ in the alternating series test (Theorem 3.4.1) is an essential hypothesis by showing the following does not hold:

> **Fake theorem.** *If $\{a_n\}_{n=1}^{\infty}$ is a sequence of positive numbers with $\lim_{n\to\infty} a_n = 0$, then $\sum_{n=1}^{\infty}(-1)^{n+1}a_n$ converges.*

By interlacing the terms of the harmonic sequence $1, 1/2, 1/3, 1/4, \ldots$ with the geometric sequence $1/10, 1/10^2, 1/10^3, \ldots$ we get the following sequence of positive numbers $\{a_n\}$ converging to 0:

$$a_1 = 1 \qquad a_2 = \frac{1}{10} \qquad a_3 = \frac{1}{2} \qquad a_4 = \frac{1}{10^2} \qquad a_5 = \frac{1}{3} \qquad a_6 = \frac{1}{10^3} \ldots$$

Note that $\{a_n\}$ is not a monotone sequence and the alternating series cannot be applied. Indeed, the corresponding alternating series:

$$\sum_{n=1}^{\infty}(-1)^{n+1}a_n = 1 - \frac{1}{10} + \frac{1}{2} - \frac{1}{10^2} + \frac{1}{3} - \frac{1}{10^3} + \frac{1}{4} - \frac{1}{10^4} + \ldots$$

diverges (why?).

Chapter 4

The topology of the real line

4.1 Open and closed sets

Next we are going to discuss the something called the topology of the real line. Topology is a huge subject in mathematics which some students will study in detail in other courses. In topology, one studies the "shape" of objects called "topological spaces". The real line $\mathbb{R}$ is just one animal in the zoo of all topological spaces. We have actually already encountered various concepts from topology before this point:

- Dense sets (Definition 1.6.1) are a concept from topology.
- Convergence (Definition 2.3.1) is a concept from topology.
- The Bolzano-Weierstrass theorem (Theorem 2.7.4) relates to a concept from topology called "compactness".
- Cauchy sequences (Definition 2.8.2) relate to a concept called "completeness" from (metric) topology. This concept is different from (but not totally unrelated to) the concept of completeness based on the least upper bound property (Definition 1.4.1).

Open and closed sets are core concepts in topology.

Definition 4.1.1. A set $F \subseteq \mathbb{R}$ is said to be **closed** if, whenever $\{x_n\}$ is a sequence in F converging to a limit $x \in \mathbb{R}$, it follows that $x \in F$.

Example 4.1.2. Every closed interval $[a, b]$ is (as the name would suggest) a closed set. It is not possible to converge to a point outside of $[a, b]$ using a sequence in $[a, b]$. Note the same is not true for other kinds of intervals. We can approach the right endpoint b of $[a, b)$ using a sequence of points in $[a, b)$, but $b \notin [a, b)$. This means the interval $[a, b)$ is not a closed set.

Example 4.1.3. The whole real line $\mathbb{R}$ is closed because the limit of any convergent sequence of real numbers is, by definition, a real number.

Example 4.1.4. Every finite subset of $\mathbb{R}$ is closed:

- The empty set $F = \varnothing$ is closed for reasons of "vacuous logic". There are no sequences of elements of F **at all**, let alone any convergent sequences of elements of F converging to a point outside of F.

- Any 1-element set $F = \{x\} \subseteq \mathbb{R}$ is closed for a different silly reason: the only sequence of elements of F is the constant sequence $x, x, x, \ldots$ which of course converges to the point $x \in F$.

- Suppose that $F \subseteq \mathbb{R}$ is a finite set with at least two elements. Note that the minimum distance $\delta := \min\{|x - y| : x, y \in F \text{ and } x \neq y\}$ between pairs of points in F is a positive number, $\delta > 0$. Suppose $\{x_n\}$ is a convergent sequence of points in F. Since $\{x_n\}$ is a Cauchy sequence (see Theorem 2.8.3), there exists $N \in \mathbb{N}$ such that $i, j \geq N$ implies $|x_i - x_j| < \delta$. By the definition of δ, this implies $x_i = x_j$ for all $i, j \geq N$. Thus, there is a specific element $x \in F$ such that $x_n = x$ for all $n \geq N$, i.e. the sequence $\{x_n\}$ is eventually constant and $x_n \to x \in F$.

Exercise 4.1.5. Does there exist a subset of $\mathbb{R}$ which is both closed and dense (see Definition 1.6.1)?
Answer: Yes, but there is only one example. The only closed dense subset of $\mathbb{R}$ is $\mathbb{R}$ itself. Indeed, if $D \subseteq \mathbb{R}$ is dense, then for every $x \in \mathbb{R}$ there exists a sequence $\{d_n\}$ in D such that $d_n \to x$. If D is also closed, this implies that $x \in D$ for all $x \in \mathbb{R}$ which means $D = \mathbb{R}$. For thoroughness, let us check the claim that if D is dense then every real number can be realized as the limit of a sequence in D. Fix $x \in \mathbb{R}$. By definition of density, for every $n \in \mathbb{N}$, the open interval $(x - \frac{1}{n}, x + \frac{1}{n})$ contains an element $d_n \in D$. We claim the sequence $\{d_n\}$ constructed in this way converges to x. Indeed, given $\epsilon > 0$, we can choose $N \in \mathbb{N}$ such that $\frac{1}{N} < \epsilon$. Then, provided $n \geq N$, we have $|d_n - x| < \frac{1}{n} \leq \frac{1}{N} < \epsilon$.

Exercise 4.1.6. We have seen that closed sets can be finite (indeed, all finite sets are closed) and uncountable (closed intervals are closed and uncountable). Do there also exist closed subsets of $\mathbb{R}$ which are countably infinite?
Answer: Yes. For example $F = \{0\} \cup \{\frac{1}{n} : n \in \mathbb{N}\}$ is closed and countably infinite. To prove that F is closed, one needs to check that any convergent sequence of elements of F converges to either 0 or $\frac{1}{n}$ for some $n \in \mathbb{N}$. We leave the verification of this as an exercise to the reader. One recommendation is to use the Cauchy criterion to show that any sequence of elements of F which is not eventually constant must converge to 0. Of course, using eventually constant sequences such as $\frac{1}{2}, \frac{1}{2}, 0, 0, 0, \frac{1}{25}, \frac{1}{10}, \frac{1}{10}, \frac{1}{10}, \frac{1}{10}, \ldots$ it is possible to converge to other elements of F.

The complementary concept to a closed set is an open set.

Definition 4.1.7. A set $U \subseteq \mathbb{R}$ is said to be **open** if, for all $x \in U$, there exists $\epsilon > 0$ such that $(x - \epsilon, x + \epsilon) \subseteq U$.

Informally, every point of an open set U is required to have a positive (but potentially very small) amount of "wiggle room" within the set. The set $[4, 5)$ is not open because the point $4 \in [4, 5)$ does not have any wiggle room. No matter how small $\epsilon > 0$ is, the interval $(4 - \epsilon, 4 + \epsilon)$ contains points outside of $[4, 5)$.

Remark 4.1.8. In everyday usage, one usually thinks "open" and "closed" as opposite concepts. However, take care that in math there exist sets which are both open and closed and also sets which are neither. In the words of J. Munkres, "sets are not doors". The slightly garish portmanteau "clopen" is sometimes used for sets that are both open and closed. Clopen sets are more important in the broader context of general topological spaces (which is outside the scope of this course) than in the specific context of the real line. The real line has exactly two subsets that are clopen: the empty set $\varnothing$ and the whole real line $\mathbb{R}$ are both open and closed. The absence of any other clopen sets in $\mathbb{R}$ exactly means that $\mathbb{R}$ is "connected" in the sense of topological spaces.

Example 4.1.9. Every open interval (a, b) is (as the name would suggest) an open set. Indeed, given any $x \in (a, b)$, we can choose $\epsilon \coloneqq \frac{1}{2}\min\{|x - a|, |x - b|\}$. This definition ensures that ϵ is smaller than both the distance from x to a and the distance from x to b. Accordingly, $(x - \epsilon, x + \epsilon) \subseteq (a, b)$. Unbounded intervals such as (a, ∞), $(-\infty, b)$ are also open, as is the whole real line $\mathbb{R} = (-\infty, \infty)$.

Let's ask ourselves the same questions about open sets that we asked ourselves above about closed sets above.

Exercise 4.1.10. Does there exist an open set $U \subseteq \mathbb{R}$ which is dense in $\mathbb{R}$?
Answer: Yes. Here are three examples of open dense sets:

$$\mathbb{R} \setminus \mathbb{Z} = \bigcup_{n\in\mathbb{Z}} (n, n + 1) \qquad \mathbb{R} \setminus \{0\} = (-\infty, 0) \cup (0, \infty) \qquad \mathbb{R} = (-\infty, \infty).$$

Exercise 4.1.11. Does there exist a subset of $\mathbb{R}$ that is both open and countably infinite?
Answer: No. Suppose $U \subseteq \mathbb{R}$ is open and countably infinite. Since U is infinite, it is certainly nonempty, and therefore there exists an element $x \in U$. By the definition of an open set, there exists $\epsilon > 0$ such that $(x-\epsilon, x+\epsilon) \subseteq U$. Since the open interval $(x-\epsilon, x+\epsilon)$ already contains uncountably many points, the set U must be uncountable as well.

4.2 Complements, intersections and unions

Theorem 4.2.1. *A set $A \subseteq \mathbb{R}$ is open if and only if its complement A^c is closed.*

Proof. First we assume A is open and argue by contradiction that A^c is closed. If A^c were not closed, there would be a convergent sequence $\{x_n\}$ in A^c whose limit x is not in A^c. That is, $x \in A$. Since A is open, there exists an $\epsilon > 0$ such that $(x - \epsilon, x + \epsilon) \subseteq A$. Since $x_n \to x$ and $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that $n \geq N$ implies $x_n \in (x - \epsilon, x + \epsilon) \subseteq A$. This is a contradiction because $x_n \in A^c$ for all $n \in \mathbb{N}$.

Next we assume A^c is closed and argue by contradiction that A is open. If A is not open, then there exists a point $x \in A$ such that no open interval centred on x is contained in A. In particular, for every $n \in \mathbb{N}$, there exists a point $x_n \in (x-1/n, x+1/n)$ such that $x_n \notin A$. That is $\{x_n\}$ is a sequence in A^c such that $|x_n - x| < 1/n$ for all $n \in \mathbb{N}$. By construction $x_n \to x$ (check this claim if it seems unclear to you) so, since A^c was assumed closed, we must have $x \in A^c$. This is a contradiction because $x \in A$. □

Example 4.2.2. The interval $(1, \infty)$ is open and its complement $(-\infty, 1]$ is closed.

Remark 4.2.3. The next two theorems involve a concept which, depending on the student's background, might be unfamiliar: the idea of taking the union or intersection of arbitrarily-many sets. The student should already know how to take the union or intersection of an infinite sequence $S_1, S_2, S_3, \ldots$ of sets:

$$\bigcup_{n=1}^{\infty} S_n \coloneqq \{x : x \in S_{n_0} \text{ for some } n_0 \in \mathbb{N}\} \qquad \bigcap_{n=1}^{\infty} S_n \coloneqq \{x : x \in S_n \text{ for all } n \in \mathbb{N}\}. \tag{4.1}$$

However, some students may not have had the need to take the union or intersection of an arbitrarily large (possibly uncountable) collection of sets. When uncountably-many sets are present, the above notation is inadequate because we cannot list the sets in a sequence. One notational device to overcome this is based on taking the union of a possibly uncountable set $\mathscr{C}$ whose elements are themselves sets. Given any "set of sets" $\mathscr{C}$, we can write:

$$\bigcup \mathscr{C} \coloneqq \{x : x \in S_0 \text{ for some } S_0 \in \mathscr{C}\} \qquad \bigcap \mathscr{C} \coloneqq \{x : x \in S \text{ for all } S \in \mathscr{C}\}. \tag{4.2}$$

An alternative formalism to (4.2) above which is a bit more directly analogous to the case of sequences of sets (4.1) is based on the idea of an index set. Basically, instead of labelling our sets with natural numbers, we label them with elements of some arbitrary set I, the "index set". That is, we consider a family of sets S_i for each $i \in I$. What exactly the elements of I are is not important here. What is important is that, since I could have potentially any cardinality, we permit the possibility that our collection of sets S_i, $i \in I$ is uncountable. In terms of this index set formalism, we would write:

$$\bigcup_{i \in I} S_i \coloneqq \{x : x \in S_{i_0} \text{ for some } i_0 \in I\} \qquad \bigcap_{i \in I} S_i \coloneqq \{x : x \in S_i \text{ for all } i \in I\}. \tag{4.3}$$

Theorem 4.2.4. *The intersection of finitely-many open sets is open and the union of arbitrarily-many open sets is open.*

Proof. For the first assertion, supposing $U_1, \ldots, U_n \subseteq \mathbb{R}$ are open, we must prove their intersection is open. Indeed, suppose that $x \in U_1 \cap \ldots \cap U_n$. For each $i \in \{1, \ldots, n\}$ we have $x \in U_i$ and U_i open. Thus, there exists $\epsilon_i > 0$ such that $(x - \epsilon_i, x + \epsilon_i) \subseteq U_i$. Put $\epsilon \coloneqq \min\{\epsilon_1, \ldots, \epsilon_n\}$. Then, $(x - \epsilon, x + \epsilon) \subseteq U_i$ for $i = 1, \ldots, n$ which implies that $(x - \epsilon, x + \epsilon) \subseteq U_1 \cap \ldots \cap U_n$.

For the second assertion, supposing $\mathscr{U}$ is an arbitrary collection of open sets[1], we must show the union of the collection is open. Indeed, suppose that $x \in \bigcup \mathscr{U}$. Then there exists some open set $U_0 \in \mathscr{U}$ such that $x \in U_0$. Since U_0 is open, there exists $\epsilon > 0$ such that $(x - \epsilon, x + \epsilon) \subseteq U_0$, which implies $(x - \epsilon, x + \epsilon) \subseteq \bigcup \mathscr{U}$ as well. □

[1]The word "collection" here means exactly the same thing as the word "set". However, some people find it improves readability a bit to talk about a "collection of sets" instead of a "set of sets'.

There is a similar theorem describing what happens when we take unions and intersections of closed sets. Note that this second theorem is the same as the first except that the words "open" and "closed" and the words "union" and "intersection" are swapped.

Theorem 4.2.5. *The union of finitely-many closed sets is closed and the intersection of arbitrarily-many closed sets is closed.*

Exercise 4.2.6. The proof of Theorem 4.2.5 is left as an exercise. In fact, the reader is encouraged to give two proofs: a direct proof based only on the definition of a closed set (Definition 4.1.1) and another proof using Theorem 4.2.4 and De Morgan's Law (the complement of a union is the intersection of the complements and the complement of an intersection is the union of the complements).

Example 4.2.7. Given an example of an infinite sequence of open sets whose intersection is not open.
Solution: $\bigcap_{n=1}^{\infty}(-\frac{1}{n}, \frac{1}{n}) = \{0\}$.

Example 4.2.8. Given an example of an infinite sequence of closed sets whose union is not closed.
Solution: $\bigcup_{n=1}^{\infty}[-1+\frac{1}{n}, 1-\frac{1}{n}] = (-1, 1)$.

4.3 The structure of open sets

What do open subsets of $\mathbb{R}$ look like? Since open intervals are open and arbitrary unions of open sets are open, one way produce examples of open sets is to take the union of some collection of open intervals. Let's look at an interesting example of an open set constructed in this way.

Example 4.3.1. Fix in advance some tiny number $\epsilon > 0$. Fix also some sequence $\{q_n\}_{n=1}^{\infty}$ listing all the rational numbers (this is possible because $\mathbb{Q}$ is countable) and define:

$$U \coloneqq \bigcup_{n=1}^{\infty}\left(q_n - \frac{\epsilon}{2^n}, q_n + \frac{\epsilon}{2^n}\right).$$

Since arbitrary unions of open sets are open (Theorem 4.2.4), the set U defined in this way is open. It is also clear that U contains all the rational numbers, suggesting that U is a "big set". Indeed, one might even suspect at first that, because U contains a little interval around every rational, it must be the case that $U = \mathbb{R}$. In fact this is false, U is not equal to $\mathbb{R}$. We get some suggestion (but not a proof) that U cannot equal $\mathbb{R}$ by calculating the total length of the intervals we used to define it:

$$\sum_{n=1}^{\infty} 2\frac{\epsilon}{2^n} = 2\epsilon \qquad \text{(geometric series)}.$$

Since 2ϵ is finite and very small, this suggests that U doesn't take up very much of the line at all, i.e. U is a "small set". Turning this heuristic argument into a rigorous proof is best done with help of something called called measure theory which is not part of this course.

The next theorem says that it is always possible to take any open set and break it down as the union of countably-many non-overlapping open intervals.

Theorem 4.3.2. *Let $U \subseteq \mathbb{R}$ be any open set. Then there exists a unique*[2] *sequence*[3] *of disjoint open intervals*[4] *$I_1, I_2, \ldots$ such that $U = \bigcup_n I_n$.*

Proof. We start by defining an equivalence relation $\sim$ on U. Given $x, y \in U$, we define:

$$x \sim y \iff \text{there exists an open interval } J \text{ such that } x, y \in J \subseteq U.$$

- Clearly $\sim$ is symmetric.
- To see $\sim$ is reflexive, note that, since U is open, for any $x \in U$ there is an $\epsilon > 0$ such that $x \in (x - \epsilon, x + \epsilon) \subseteq U$.
- To see $\sim$ is transitive note that if $x \sim y$ and $y \sim z$ then there exist open intervals $J_1, J_2 \subseteq U$ with $x, y \in J_1$ and $y, z \in J_2$. Note $J := J_1 \cup J_2$ is also an open interval because $y \in J_1 \cap J_2$ and the union of two overlapping open intervals is also an open interval. Since $x, z \in J$, we have $x \sim z$ as needed.

By standard facts about equivalence relations and partitions, U is the disjoint union of the equivalence classes of $\sim$. We claim every equivalence class of $\sim$ is an open interval. Indeed, fix $x \in U$ and let I_x be the equivalence class of x. Observe that the definition of $\sim$ gives that I_x is the union of all open intervals J such that $x \in J \subseteq U$. Since the union of any collection of open intervals containing a common point is again an open interval (for a proof of this technical point, see Lemma 4.3.3 below), this observation gives that I_x is an open interval.

Next we claim that the number of equivalence classes of $\sim$ is countable. Indeed, the equivalence classes are a disjoint collection of open intervals in $\mathbb{R}$ and any such collection must be countable for the following reason: $\mathbb{Q}$ is dense in $\mathbb{R}$, so we can select for each open interval I in the collection a rational number $q_I \in I$. Since the intervals in the collection are disjoint, the function which sends $I \mapsto q_I$ is injective. This shows our collection of intervals has the same cardinality as a subset of $\mathbb{Q}$ and is therefore countable.

Lastly let us discuss the uniqueness of this decomposition into disjoint open intervals, leaving some of the steps as exercises to the reader. Suppose that $(a_1, b_1), (a_2, b_2), \ldots$ is any collection of pairwise disjoint intervals with the property that $U = \bigcup_n (a_n, b_n)$. We must argue that the intervals $(a_1, b_1), (a_2, b_2), \ldots$ are just the equivalence classes of $\sim$ so that this is the same decomposition obtained above. Firstly, we claim that $a_n, b_n \notin U$ for all $n \in \mathbb{N}$. Indeed, if $a_n \in U$, then we would have $a_n \in (a_m, b_m)$ for some $m \neq n$ which leads (exercise) to the contradiction that (a_m, b_m) and (a_n, b_n) are not disjoint. Secondly, if $x \in (a_n, b_n) \subseteq U$, then the fact that $a_n, b_n \notin U$ implies (exercise) that (a_n, b_n) is the largest open interval J such that $x \in J \subseteq U$ holds. Using this, we see the equivalence class of x for the relation $\sim$ is exactly (a_n, b_n). □

[2] Of course, the ordering of the intervals is not unique; we can switch which interval is called I_1 and which is called I_2 etc. without affecting the union. It is only the set of intervals appearing which is unique.

[3] It may be that only finitely-many intervals are needed so the word "sequence" here should be interpreted as meaning either an infinite sequence $\{I_n\}_{n=1}^{\infty}$ or a finite sequence $\{I_n\}_{n=1}^{N}$.

[4] Unbounded open intervals of the form (a, ∞), $(-\infty, a)$ or $(-\infty, \infty)$ may be required.

The following lemma was used in the proof of the above theorem. Note that Theorem 4.2.4 already shows that the union of any collection of open intervals is an open set. The lemma below says that, if we assume the intervals in the collection all share some common point, then their union is also an open interval.

Lemma 4.3.3. *Fix $x \in \mathbb{R}$. The union of any nonempty collection of open intervals all of which contain x is an open interval containing x.*

Proof. Let $\mathscr{I}$ be any nonempty collection of open intervals all of which contain x. We must show $\bigcup \mathscr{I}$ is also an open interval. Let A denote the set of left endpoints of intervals belonging to $\mathscr{I}$. Similarly, let B denote the set of right endpoints of intervals belonging to I. Put $\alpha := \inf A$ and $\beta := \sup B$ with the understanding that $\alpha = -\infty$ if A is not bounded below and $\beta = \infty$ if B is not bounded above. We claim that $\bigcup \mathscr{I} = (\alpha, \beta)$. Indeed, if $y \in \bigcup \mathscr{I}$, then $y \in (a, b)$ for some $(a, b) \in \mathscr{I}$ and we have $\alpha \leq a < y < b \leq \beta$ so that $y \in (\alpha, \beta)$. On the other hand, suppose $y \in (\alpha, \beta)$. Let us first consider the case $y \geq x$. Since $y < \beta$ and $\beta = \sup B$, there exists $(a, b) \in \mathscr{I}$ such that $y < b \leq \beta$. Also, since $x \in (a, b)$, we have $a < x \leq y$ so that $y \in (a, b) \subseteq \bigcup \mathscr{I}$. Similarly, suppose $y \leq x$. Since $\alpha < y$, there exists $(a, b) \in \mathscr{I}$ such that $\alpha \leq a < y$. Also, since $x \in (a, b)$, we have $y \leq x < b$ so that $y \in (a, b) \subseteq \bigcup \mathscr{I}$. □

4.4 The Heine-Borel property

Next we study an important property called the Heine-Borel property which is enjoyed by all closed intervals $[a, b]$ but not by open intervals (a, b). This property will be extremely useful to us very soon when we transition towards the study of continuous/differentiable functions of a real variable. In the more general context of topological spaces, the "Heine-Borel property" property is usually called "compactness". Some students will learn more about it in other courses.

Theorem 4.4.1 (The Heine-Borel property)**.** *Fix a closed bounded interval $[a, b]$. Let $\mathscr{U}$ be any collection of open sets such that $\bigcup \mathscr{U} \supseteq [a, b]$. Then, there exist finitely many open sets $U_1, \ldots, U_n \in \mathscr{U}$ such that $U_1 \cup \ldots \cup U_n \supseteq [a, b]$.*

One usually paraphrases this result in something like the following way: "every open cover of $[a, b]$ (no matter how complicated) admits a finite subcover".

Example 4.4.2. Open intervals do not have the Heine-Borel property. Some open covers of $(0, 1)$ have finite subcovers but others do not. For example we can consider $\mathscr{U} = \{(-5, 5)\}$. This is a trivial open covering of $(0, 1)$ by just one open set. This covering certainly admits finite subcovers because there is only one set in the cover to begin with. However, there also exist more complicated infinite coverings such as $\mathscr{U} = \{(\frac{1}{n}, 5) : n \in \mathbb{N}\}$ of $(0, 1)$ which do not admit finite subcovers. Any finite set of intervals of the form $(\frac{1}{n}, 5)$ will fail to cover $(0, 1)$; some small small interval of numbers to the right of 0 will be missed.

On the other hand, the closed interval $[0,1]$ does have the Heine-Borel property. This means that every open cover of $[0,1]$, no matter how complicated or ingeniously designed, necessarily has a finite subcover.

Proof of Theorem 4.4.1. We begin by defining a set $S \subseteq [a,b]$ as follows:

$$S \coloneqq \{x \in [a,b] : \text{there exists } n \in \mathbb{N} \text{ and } U_1, \ldots, U_n \in \mathscr{U} \text{ such that } [a,x] \subseteq U_1 \cup \ldots \cup U_n\}.$$

To prove the theorem we must prove that $b \in S$. First, note $a \in U$ for some $U \in \mathscr{U}$. Since U is open, there is an $\epsilon > 0$ such that $(a-\epsilon, a+\epsilon) \subseteq U$. Every x satisfying $(a < x < a+\epsilon)$ belongs to S because $[a,x]$ is covered by the single open set $U \in \mathscr{U}$. Since S is nonempty (and bounded) it makes sense to talk about $s \coloneqq \sup S$. Moreover, from the discussion above, we know that $a < s \leq b$.

Next, we prove $s \in S$. We know $s \in W$ for some $W \in \mathscr{U}$. Since W is open there exists $\epsilon > 0$ such that $(s-\epsilon, s+\epsilon) \subseteq W$. Since $s = \sup S$, there exists $x \in S$ such that $s - \epsilon < x \leq s$. Since $x \in S$, there exists $n \in \mathbb{N}$ and $U_1, \ldots, U_n \in \mathscr{U}$ such that $U_1 \cup \ldots \cup U_n \supseteq [a,x]$. Then, since $W \supseteq [x,s]$, we have $U_1 \cup \ldots \cup U_n \cup W \supseteq [a,s]$ so that $s \in S$.

Lastly, we want to prove that $s = b$ so that $b \in S$ as desired. Indeed, suppose for contradiction that $s < b$. We have $U_1, \ldots, U_n, W \in \mathscr{U}$ such that $U_1 \cup \ldots \cup U_n \cup W \supseteq [a,s]$ and $s \in W$. Since W is open, there exists $\epsilon > 0$ such that $(s-\epsilon, s+\epsilon)$. Choose any x such that $s < x < \min\{s+\epsilon, b\}$. Then, $U_1 \cup \ldots \cup U_n \cup W \supseteq [a,x]$ as well, which means $x \in S$. This is a contradiction because $x > s = \sup S$. □

4.5 Compactness

Last time we saw that closed bounded intervals have the Heine-Borel property: every open cover of $[a,b]$ admits a finite subcover. We also saw that open intervals do not: some open covers of (a,b) do not admit a finite subcovers. In general, sets which have the Heine-Borel property are called "compact". Compactness is an important concept in topology which some students will study in more detail in other courses.

Definition 4.5.1. A set $X \subseteq \mathbb{R}$ is said to be **compact** if it has the Heine-Borel property: for all collections of open sets $\mathscr{U}$ satisfying $\bigcup \mathscr{U} \supseteq X$, there exist finitely-many open sets $U_1, \ldots, U_n \in \mathscr{U}$ such that $U_1 \cup \ldots \cup U_n \supseteq X$.

The Heine-Borel property may appear quite abstract the first time one encounters it. Fortunately, we have the following equivalent description of the compact subsets of $\mathbb{R}$ which is somewhat more concrete:

Theorem 4.5.2. *Given a set $X \subseteq \mathbb{R}$, the following are equivalent:*

(i) X is compact (meaning it has the Heine-Borel property),

(ii) X is closed and bounded.

Proof. First we show $(ii \Rightarrow i)$. Suppose that X is closed and bounded. Let $\mathscr{U}$ be any open cover of X, i.e. a collection of open sets $\mathscr{U}$ satisfying $\bigcup \mathscr{U} \supseteq X$. We must show there exist finitely-many open sets $U_1, \ldots, U_n \in \mathscr{U}$ such that $U_1 \cup \ldots \cup U_n \supseteq X$.

Since X is bounded, there exists a closed interval $[a, b]$ such that $X \subseteq [a, b]$. Since X is open, its complement X^c is open. Note that the collection of open sets $\mathscr{W} = \mathscr{U} \cup \{X^c\}$ covers $[a, b]$. Indeed, $\bigcup \mathscr{W} = \mathbb{R}$, since $\mathscr{U}$ covers X and X^c covers the rest of $\mathbb{R}$. We already know that $[a, b]$ has the Heine-Borel property (Theorem 4.4.1), so there must exist finitely-many open sets $U_1, \ldots, U_n \in \mathscr{W}$ such that $U_1 \cup \ldots \cup U_n \supseteq [a, b]$. There are two cases to consider: if all of $U_1, \ldots, U_n$ actually belong to $\mathscr{U}$ then we are finished because they also cover $X \subseteq [a, b]$. Otherwise, it must be that X^c is among $U_1, \ldots, U_n$. Let us say without loss of generality that $U_n = X^c$ and $U_1, \ldots, U_{n-1} \in \mathscr{U}$. Then, $U_1 \cup \ldots \cup U_{n-1}$ contains $[a, b] \setminus X^c = X$, and we are again finished.

Next we should prove $(i \Rightarrow ii)$. Suppose that $X \subseteq \mathbb{R}$ has the Heine-Borel property. We should show that X is closed and bounded. We only show that X is bounded, leaving the proof that X is closed to the homework. Consider $\mathscr{U} \coloneqq \{(-n, n) : n \in \mathbb{N}\}$. Note that $\mathscr{U}$ is an open cover of X. Indeed, $\bigcup \mathscr{U} = \mathbb{R}$, so $\mathscr{U}$ covers all of $\mathbb{R}$. Since X has the Heine-Borel property, only finitely many of these intervals are needed to cover X. This implies (why?) that there exists $N \in \mathbb{N}$ such that $X \subseteq (-N, N)$, so X is bounded as required. $\square$

Chapter 5

Continuous functions

5.1 Continuity

For rest of the course, our main concern will be the study of real-valued functions of a real variable. For brevity, we will refer to such functions, i.e. functions whose domain and range are subsets of $\mathbb{R}$, as **real functions**.

The reader should already be familiar with various standard operations for producing new real functions from old ones. One has the "pointwise" operations of addition, multiplication, scalar multiplication, and addition as well as the operation of function composition. In particular, given real functions f and g, the reader should know how (and why) the domains $\operatorname{dom} f$ and $\operatorname{dom} g$ of the ingredient functions determine the domains of the various functions we get by combining them:

$$\begin{aligned}
\operatorname{dom}(f+g) &= \operatorname{dom} f \cap \operatorname{dom} g \\
\operatorname{dom}(f \cdot g) &= \operatorname{dom} f \cap \operatorname{dom} g \\
\operatorname{dom}(\lambda f) &= \operatorname{dom} f, \text{ for a given constant } \lambda \in \mathbb{R} \\
\operatorname{dom}\left(\frac{f}{g}\right) &= \{x \in \mathbb{R} : x \in \operatorname{dom} f \cap \operatorname{dom} g \text{ and } g(x) \neq 0\} \\
\operatorname{dom}(g \circ f) &= \{x \in \mathbb{R} : x \in \operatorname{dom} f \text{ and } f(x) \in \operatorname{dom} g\}.0
\end{aligned}$$

Example 5.1.1. Define real functions $f, g : \mathbb{R} \to \mathbb{R}$ by $f(x) = x^2 - 1$ and $g(x) = x - 1$. Notice that f/g has what is usually called a "removable singularity" because $(f/g)(x) = x+1$ everywhere that f/g is defined. Nonetheless, unless some explicit redefinition is made, the domain of f/g is still $(-\infty, 1) \cup (1, \infty)$. We do not by default extend f/g to the whole line by imposing the "natural value" $(f/g)(1) = 2$.

We begin with the definition of a continuous function. Note that this definition is very much in the same spirit as Definition 2.3.1.

Definition 5.1.2. Let f be a real function. We say f is **continuous at a point** $c \in \operatorname{dom} f$ if, for every $\epsilon > 0$, there exists a $\delta > 0$ such that $|x - c| < \delta$ implies $|f(x) - f(c)| < \epsilon$. We say that f is **continuous** if it is continuous at every point in its domain.

Remark 5.1.3. Note it is considered implicit in the above definition that $x \in \operatorname{dom} f$, i.e. we only pass f inputs belonging to its domain.

Remark 5.1.4. It can be psychologically helpful to think of the $\epsilon > 0$ in the above definition as being given to you by an adversary. A function is continuous at c if, no matter how small of a tolerance $\epsilon > 0$ is prescribed to you by an adversary, it is possible for you to respond with a number $\delta > 0$ such that all inputs x that are δ-close to c yield outputs $f(x)$ which are ϵ-close to $f(c)$.

As the following theorem catalogues, the basic operations on real functions preserve continuity. The proofs of these statements are very similar to the proofs of the limit laws seen earlier (Theorem 2.6.2).

Theorem 5.1.5. *Let f and g be continuous real functions. Then the following functions are also continuous:*

(i) $f + g$,

(ii) fg,

(iii) λf *for any scalar* $\lambda \in \mathbb{R}$,

(iv) $\dfrac{f}{g}$.

Proof. We just prove (i) and a weaker version of (iv) wherein $f \equiv 1$ so that $\frac{f}{g} = \frac{1}{g}$. The remainder of the proofs are an exercise for the reader.

For part (i), fix arbitrarily $c \in \operatorname{dom} f \cap \operatorname{dom} g$. We must prove that $f+g$ is continuous at c. To this end, let $\epsilon > 0$ be specified. Since f is continuous at c, there exists $\delta_1 > 0$ such that $|x - c| < \delta_1$ implies $|f(x) - f(c)| < \epsilon/2$. Since g is continuous at c, there exists $\delta_2 > 0$ such that $|x - c| < \delta_2$ implies $|g(x) - g(c)| < \epsilon/2$. Now put $\delta \coloneqq \min\{\delta_1, \delta_2\}$. Then, if $|x - c| < \delta$, it follows that

$$\begin{aligned} |(f+g)(x) - (f+g)(c)| &= |f(x) - f(c) + g(x) - g(c)| \\ &\leq |f(x) - f(c)| + |g(x) - g(c)| \\ &\leq \epsilon/2 + \epsilon/2 \\ &= \epsilon. \end{aligned}$$

Next we prove (iv) under the additional hypothesis that $f \equiv 1$. That is, we prove $1/g$ is continuous at every point $c \in \operatorname{dom} g$ with $g(c) \neq 0$. Let's do a bit of scratch work first. Provided that $g(x) \neq 0$ as well, we have:

$$\left| \frac{1}{g(x)} - \frac{1}{g(c)} \right| = \frac{|g(x) - g(c)|}{|g(x)g(c)|}.$$

By making x sufficiently close to c, we can make the numerator $|g(x) - g(c)|$ as small as we desire. However, we must also take care that the denominator $|g(x)g(c)|$ does not get too

small for two reasons: firstly, we cannot divide by zero and, secondly, even if it is small and positive we can get into trouble because its reciprocal will be big!

With this scratch work out of the way, we return to the formal proof. Since we assume $g(c) \neq 0$, the quantity $\frac{1}{2}|g(c)|$ is positive. Since g is continuous at c, we can choose $\delta_1 > 0$ such that $|x-c| < \delta_1$ implies $|g(x)-g(c)| \leq \frac{1}{2}|g(c)|$ which then gives $|g(x)| \geq \frac{1}{2}|g(c)|$ (why?). Thus, if $|x-c| < \delta_1$, we have:

$$\left|\frac{1}{g(x)} - \frac{1}{g(c)}\right| = \frac{|g(x)-g(c)|}{|g(x)g(c)|} \leq \frac{|g(x)-g(c)|}{\frac{1}{2}|g(c)|^2}.$$

Again using the continuity of g at c, there exists $\delta_2 > 0$ such that $|x-c| < \delta_2$ implies that $|g(x)-g(c)| < \frac{1}{2}|g(c)|^2\epsilon$. Putting $\delta \coloneqq \min\{\delta_1, \delta_2\}$ we have that $|x-c| < \delta$ implies:

$$\left|\frac{1}{g(x)} - \frac{1}{g(c)}\right| \leq \frac{|g(x)-g(c)|}{\frac{1}{2}|g(c)|^2} < \frac{\frac{1}{2}|g(c)|^2\epsilon}{\frac{1}{2}|g(c)|^2} = \epsilon.$$

□

Example 5.1.6. All polynomial functions are continuous. For instance, $p : \mathbb{R} \to \mathbb{R}$ defined by $p(x) = x^5 - \frac{1}{3}x^2 + 8$ is continuous. To prove this we only need to check that the two specific polynomials x and 1 are continuous, which is simple to do. Every other polynomial can, by definition, be built up out of these two simple polynomials using using addition, multiplication and scalar multiplication. Theorem 5.1.5 tells us that combining continuous functions using these operations always yields continuous functions.

Similarly, if we also use the division property Theorem 5.1.5 (*iv*), we obtain that every rational function (roughly speaking, a quotient of polynomials) is continuous. Here one must appropriately restrict the domain to inputs for which the denominator does not vanish. Thus, for example, the function $r : \mathbb{R} \setminus \{1, -1\} \to \mathbb{R}$ defined by $r(x) = \frac{x^5+7}{x^2-1}$ is continuous.

Another important operation for combining functions is composition. Compositions of continuous functions are also continuous.

Theorem 5.1.7. *If f and g are continuous real functions, then the composition $g \circ f$ is also continuous.*

Proof. Fix $c \in \operatorname{dom} f$ with $f(c) \in \operatorname{dom} g$. We must show $g \circ f$ is continuous at c. To this end, let $\epsilon > 0$ be given. We must prove that there exists $\delta > 0$ such that $|x-c| < \delta$ implies $|g(f(x)) - g(f(c))| < \epsilon$. First, since g is continuous at $y_0 \coloneqq f(c)$, there exists $\delta_1 > 0$ such that $|y-y_0| < \delta_1$ implies $|g(y)-g(y_0)| < \epsilon$. Next, because f is continuous at c, there exists $\delta_2 > 0$ such that $|x-c| < \delta_2$ implies $|f(x)-f(c)| < \delta_1$. We take $\delta \coloneqq \delta_2$. Then, by design, if $x \in \operatorname{dom}(g \circ f)$ satisfies $|x-c| < \delta = \delta_2$, then $y = f(x)$ and $y_0 = f(c)$ satisfy $|y-y_0| < \delta_1$ so that

$$|g(f(x)) - g(f(c))| = |g(y) - g(y_0)| < \epsilon,$$

as required. □

5.2 The maximum and minimum theorem

Earlier we introduced the Heine-Borel property, but did not apply it to any useful end. We remedy this now by using the Heine-Borel property to prove that every continuous function on a closed interval $[a, b]$ has a maximum value and a minimum value.

Definition 5.2.1. We say that a function $f : X \to \mathbb{R}$ is **bounded** if there exist $m, M \in \mathbb{R}$ such that $m \leq f(x) \leq M$ for all $x \in X$.

Remark 5.2.2. Near the beginning of this course, we talked about sets of real numbers being bounded (or bounded above, or bounded below). Later, when we referred to a sequence of real numbers $\{a_n\}_{n=1}^{\infty}$ as being bounded (or bounded above or bounded below) we meant that its set of terms $\{a_n : n \in \mathbb{N}\}$ had that property. The idea is the same for functions: when we say that a function $f : X \to \mathbb{R}$ is bounded (or bounded above or bounded below), we mean that its set of range values $\{f(x) : x \in X\}$ has that property.

It is clear that some continuous functions are not bounded, for example the function $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x$. Even if the domain of a continuous function is bounded, the function can still be unbounded. To get examples of this behaviour, we can consider a function with a vertical asymptote such as the function $f : (-1, 1) \to \mathbb{R}$ defined by $f(x) = \frac{1}{1-x^2}$. The latter function becomes unbounded towards the ends of its domain. However, it turns out that the fact that the interval $(-1, 1)$ is open is crucial here. A continuous function on a closed interval $[a, b]$ is always bounded. Before proving this, let us first prove a lemma to the effect that every continuous function is at least "locally bounded", i.e. bounded on a little interval around every point in its domain.

Lemma 5.2.3. *Suppose $f : X \to \mathbb{R}$ is continuous, where $X \subseteq \mathbb{R}$. Then, for each $x \in X$ there exists $\delta_x > 0$ and $m_x, M_x \in \mathbb{R}$ such that $m_x \leq f(y) \leq M_x$ for all $y \in (x-\delta_x, x+\delta_x) \cap X$.*

Proof. Setting $\epsilon = 1 > 0$ and using the continuity of f at x, we have that there exists $\delta_x > 0$ such that $|y-x| < \delta_x$ implies $|f(y)-f(x)| < 1$. In other words, for all $y \in (x-\delta_x, x+\delta_x) \cap X$, we have $f(x) - 1 < f(y) < f(x) + 1$. Putting $m_x \coloneqq f(x) - 1$ and $M_x \coloneqq f(x) + 1$, we are finished. □

Theorem 5.2.4. *A continuous function on a closed interval is bounded. That is, given a continuous function $f : [a, b] \to \mathbb{R}$, there exist real constants m, M such that $m \leq f(x) \leq M$ for all $x \in [a, b]$.*

Proof. By the above lemma, we can choose for each $x \in [a, b]$ some $\delta_x > 0$ and $m_x, M_x \in \mathbb{R}$ such that $m_x \leq f(y) \leq M_x$ for all $y \in (x - \delta_x, x + \delta_x) \cap [a, b]$. The collection of intervals

$$\mathscr{U} \coloneqq \{(x - \delta_x, x + \delta_x) : x \in [a, b]\}$$

is an open cover of $[a, b]$. Since $[a, b]$ has the Heine-Borel property (Theorem 4.4.1), there must exist finitely many points $x_1, \ldots, x_n \in [a, b]$ such that

$$[a, b] \subseteq (x_1 - \delta_{x_1}, x_1 + \delta_{x_1}) \cup \ldots \cup (x_n - \delta_{x_n}, x_n + \delta_{x_n}).$$

Now put $M = \max\{M_{x_1}, \ldots, M_{x_n}\}$ and $m \coloneqq \min\{m_{x_1}, \ldots, m_{x_n}\}$. With these definitions, we have $m \leq f(x) \leq M$ for all $x \in [a,b]$. Indeed, for every $x \in [a,b]$, we must have $x \in (x_i - \delta_{x_i}, x_i + \delta_{x_i})$ for some $i \in \{1, \ldots, n\}$ whence $m \leq m_{x_i} \leq f(x) \leq M_{x_i} \leq M$. $\square$

Remark 5.2.5. The above theorem could be proved just as easily for an arbitrary compact set $X \subseteq \mathbb{R}$ in place of a closed interval $[a,b]$.

The definitions of the supremum and infimum of a set (Defintion 1.3.3) apply to a function $f : X \to \mathbb{R}$ in a straighforward manner by applying them to the set of range values:

$$\sup f \coloneqq \sup\{f(x) : x \in X\} \qquad\qquad \inf f \coloneqq \inf\{f(x) : x \in X\}.$$

If a function $f : X \to \mathbb{R}$ is bounded, then $\inf f$ and $\sup f$ are its sharpest possible lower and upper bounds. However, it sometimes occurs that the supremum and infimum of a bounded function f are not actually "achieved" in the sense that they do not occur as values of the function.

Example 5.2.6. Consider $f : (0,1) \to \mathbb{R}$ defined by $f(x) = 2x+6$. It is important to realize that the specification of the domain $(0,1)$ is part of the definition of this function. Even though the formula $f(x) = 2x + 6$ in principle could be applied for all $x \in \mathbb{R}$, we have the right to specify the domain of a function to be anything we want. Here, when we specify the domain to be $(0,1)$, we are enforcing that $f(x)$ is undefined for all $x \notin (0,1)$. Keeping in mind the domain we have chosen, it is clear that $\sup f = 8$ and $\inf f = 2$. However, there are no inputs $x \in (0,1)$ which achieve either of these values. By our choice of domain, $f(0)$ and $f(1)$ are undefined and $2 < f(x) < 8$ for all $x \in (0,1)$.

Now we improve Theorem 5.2.4 by proving that, for a continuous function on a closed interval $[a,b]$, the supremum and infimum of the function are actually achieved.

Theorem 5.2.7. *Let $f : [a,b] \to \mathbb{R}$ be a continuous function on a closed interval $[a,b]$. Then, there exist $c, d \in [a,b]$ such that $f(c) \leq f(x) \leq f(d)$ for all $x \in [a,b]$.*

Proof. Put $M \coloneqq \sup f$ and $m \coloneqq \inf f$. We want to show that there is at least one $d \in [a,b]$ with $f(d) = M$. Towards a contradiction, suppose that $f(x) \neq M$ for all $x \in [a,b]$. Then, the function $g = \frac{1}{M-f}$ is defined and continuous on all of $[a,b]$ (see Theorem 5.1.5). Then g is also bounded by Theorem 5.2.4. However, by a familiar property of the supremum (Theorem 1.9.1), for every $n \in \mathbb{N}$, there exists $x \in [a,b]$ such that $M - \frac{1}{n} < f(x) \leq M$. That is, $g(x) = \frac{1}{M-f(x)} > n$. Since n is arbitrary, this shows that g is unbounded and we have reached a contradiction. We conclude that $f(d) = M$ for some $d \in [a,b]$. The proof that $f(c) = m$ for some $c \in [a,b]$ proceeds similarly by considering the function $h = \frac{1}{f-m}$. $\square$

In situations where a function $f : X \to \mathbb{R}$ achieves its supremum (or infimum) we say that the function has a (global) maximum or (global) minimum, as appropriate. Notice this is just the same definition that was used for the maximum/minimum of a set (Definition 1.3.1), but now applied to the set of values of a function.

$$\max f \coloneqq \max\{f(x) : x \in X\} \qquad\qquad \min f \coloneqq \min\{f(x) : x \in X\}$$

Exercise 5.2.8. We saw in Example 5.2.6 above that Theorem 5.2.7 can fail if the closed interval $[a, b]$ is replaced by an open interval (a, b). Show that Theorem 5.2.7 can also fail if the function $f : [a, b] \to \mathbb{R}$ is not continuous by providing an example of a bounded discontinuous function on $[0, 1]$ which does not achieve its supremum or infimum.
Answer: Among various possible solutions, one would be

$$f : [0, 1] \to \mathbb{R} \qquad\qquad f(x) = \begin{cases} x & \text{if } 0 < x < 1 \\ 1/2 & \text{if } x = 0 \text{ or } x = 1. \end{cases}$$

5.3 The intermediate value theorem

Given a continuous function on a closed interval $f : [a, b] \to \mathbb{R}$, the intermediate value property guarantees that f takes on every value lying in between its values at the endpoints. This is intuitively quite clear if one is imagining the graph of f as an unbroken curve that can be drawn without lifting one's pencil. Suppose γ is such that $f(a) < \gamma < f(b)$. Since the graph of f starts out below γ, ends up above γ, and isn't permitted to jump anywhere, it ought to cross through the value γ somewhere in between. However, as we will eventually see, continuous objects can still be pretty pathological (try googling "space-filling curve" or "nowhere differentiable function") so we should be cautious about relying too much on intuition when it comes to continuous functions.

Theorem 5.3.1. *Suppose f is continuous on $[a, b]$ and γ is between $f(a)$ and $f(b)$ (that is $f(a) \leq \gamma \leq f(b)$ or $f(b) \leq \gamma \leq f(a)$). Then there exists $t \in [a, b]$ such that $f(t) = \gamma$.*

Proof. We only consider the case where $f(a) \leq \gamma \leq f(b)$ as the other case is easy to recover from this one. In fact, it does no harm to assume that $f(a) < \gamma < f(b)$ since if $\gamma = f(a)$ or $\gamma = f(b)$ we may accordingly take $t = a$ or $t = b$. Note it is possible that there are many points $t \in [a, b]$ with $f(t) = \gamma$. In this proof we will produce the largest possible t by defining

$$t \coloneqq \sup\{x \in [a, b] : f(x) \leq \gamma\}$$

(note $f(a) < \gamma$ so we are not taking the supremum of the empty set). It remains to prove that $f(t) = \gamma$. To prove this claim we use the strategy of giving ourselves an "ϵ of room", proving instead the equivalent statement that $\gamma - \epsilon < f(t) < \gamma + \epsilon$ every $\epsilon > 0$. Indeed, since f is continuous at t, there exists $\delta > 0$ such that $|x - t| < \delta$ implies $|f(x) - f(t)| < \epsilon$. From the definition of t, there exists $x_1 \in [a, b]$ satisfying $t - \delta < x_1 \leq t$ such that $f(x_1) \leq \gamma$. Referring again to the definition of t, we have that $f(x) > \gamma$ for $x > t$. Thus we can find $x_2 \in [a, b]$ satisfying $t \leq x_2 < t + \delta$ such that $f(x_2) > \gamma$ (one can show that $t = b$ cannot occur but, even if it did, we could still take $x_2 = b = t$ here). Now observe:

- since $f(t) - \epsilon < f(x_1) < f(t) + \epsilon$ and $f(x_1) \leq \gamma$, we obtain $f(t) < \gamma + \epsilon$, and
- since $f(t) - \epsilon < f(x_2) < f(t) + \epsilon$ and $f(x_2) > \gamma$, we obtain $\gamma - \epsilon < f(t)$.

We have shown $\gamma - \epsilon < f(t) < \gamma + \epsilon$ for all $\epsilon > 0$, and so $f(t) = \gamma$. □

Exercise 5.3.2. Show the assumption of continuity is important above by giving an example of a discontinuous function $f : [0, 1] \to \mathbb{R}$ with $f(0) = 0$, $f(1) = 1$ which never takes the value $1/2$.
Answer: Among many possible solutions, one would be $f(x) = 0$ for $0 \leq x \leq 1/2$ and $f(x) = 1$ for $1/2 < x \leq 1$.

5.4 Examples of discontinuous functions

We will now look at some examples of functions $f : \mathbb{R} \to \mathbb{R}$ which are discontinuous, meaning "not continuous". Remember that a function is continuous if and only if it is continuous at every point in its domain. Negating this statement, a function is discontinuous if and only if it is discontinuous at least one point in its domain (but possibly many points). In the exercises below, we will start to develop a sense for what the set of points where a function is continuous/discontinuous can look like.

Exercise 5.4.1. Give an example of a function $f : \mathbb{R} \to \mathbb{R}$ which is discontinuous at 0 but continuous everywhere else.
Answer: One possible example is the step function:

$$f(x) = \begin{cases} -1 & \text{if } x < 0 \\ 0 & \text{if } x = 0 \\ 1 & \text{if } x > 0. \end{cases}$$

Exercise 5.4.2. Give an example of a function $f : \mathbb{R} \to \mathbb{R}$ which is discontinuous at every integer but continuous everywhere else.
Answer: One possible example is the so-called "floor function", $f(x) = \lfloor x \rfloor$. Here, $\lfloor x \rfloor$ denotes the unique integer n such that $x \in [n, n + 1)$. For example, we have: $\lfloor 3.7 \rfloor = 3$, $\lfloor 5 \rfloor = 5$, $\lfloor -0.1 \rfloor = -1$.

Exercise 5.4.3. Give an example of a function $f : \mathbb{R} \to \mathbb{R}$ which is discontinuous at every real number.
Answer: One possible example is $f = \chi_{\mathbb{Q}}$, the indicator function of the rationals. In general, given a set $A \subseteq \mathbb{R}$, the indicator function $\chi_A : \mathbb{R} \to \mathbb{R}$ is defined by

$$\chi_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A. \end{cases}$$

Since both the rationals and the irrationals are dense subsets of $\mathbb{R}$, the function $\chi_{\mathbb{Q}}$ oscillates wildly between 1 and 0 on every open interval, causing $\chi_{\mathbb{Q}}$ to be discontinuous at every point. Let us prove that $\chi_{\mathbb{Q}}$ is discontinuous at every rational input. The proof that $\chi_{\mathbb{Q}}$ is discontinuous at irrational inputs is similar. Fix $c \in \mathbb{Q}$ and consider $\epsilon = 1/2$. We must show that, no matter how $\delta > 0$ is chosen, it is not true that $|x - c| < \delta$ implies $|\chi_{\mathbb{Q}}(x) - \chi_{\mathbb{Q}}(c)| < 1/2$. Indeed, for any $\delta > 0$, since the irrationals are dense in $\mathbb{R}$, there exists an irrational number $x \in (c - \delta, c + \delta)$ and we have $|\chi_{\mathbb{Q}}(x) - \chi_{\mathbb{Q}}(c)| = |0 - 1| = 1$, which is not smaller than $1/2$.

Exercise 5.4.4. Give an example of a function $f : \mathbb{R} \to \mathbb{R}$ which is continuous at 0 but discontinuous on $\mathbb{R} \setminus \{0\}$.
Answer: Building on the idea from Exercise 5.4.3 of using the indicator function of the rationals, we consider $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} x & \text{if } x \in \mathbb{Q} \\ -x & \text{if } x \notin \mathbb{Q}. \end{cases}$$

First let us prove that f is discontinuous at every point $c \in \mathbb{R} \setminus \{0\}$. Put $\epsilon = |c|$. We claim there is no $\delta > 0$ with the property that $|x - c| < \delta$ implies $|f(x) - f(c)| < |c|$. Indeed, for any $\delta > 0$, we can pick $x \in (c - \delta, c + \delta)$ to satisfy the following constraints:

- x is rational if c is irrational,
- x is irrational if c is rational,
- x has the same sign (positive or negative) as c. This condition is actually automatic if δ is sufficiently small, which we could assume without loss of generality if we want.

These conditions imply that $f(x)$ and $f(c)$ have opposite signs which leads to

$$|f(x) - f(c)| = |x| + |c| > |c|.$$

We leave to the reader the exercise of proving that f is continuous at 0.

Exercise 5.4.5. Give an example of a function $f : \mathbb{R} \to \mathbb{R}$ which is continuous at every irrational number and discontinuous at every rational number.
Answer: A well-known example of a function with this property is Thomae's function, also known as the "popcorn function". Recall that every rational number $x \in \mathbb{Q}$ has a unique lowest-terms representation $x = p/q$ where $p \in \mathbb{Z}$, $q \in \mathbb{N}$ and $\gcd(p, q) = 1$. The popcorn function is defined as follows:

$$f(x) = \begin{cases} \frac{1}{q} & \text{if } x \in \mathbb{Q} \text{ and } \frac{p}{q} \text{ is its lowest-terms representation} \\ 0 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

So, for instance $f(\pi) = f(\frac{\sqrt{2}}{10}) = 0$, $f(\frac{5}{3}) = \frac{1}{3}$, and $f(\frac{44}{-77}) = f(\frac{-4}{7}) = \frac{1}{7}$. An approximation to the graph of f on the interval $(0, 1)$ is shown in Figure 5.1.

It is fairly easy to show f is discontinuous at every rational (near every rational, there are arbitrarily close-by irrationals at which f vanishes). Proving that f is continuous at every irrational requires more effort. Instead of making a detailed study of the popcorn function, we shift our attention to a different function $g : \mathbb{R} \to \mathbb{R}$ with the same continuity properties that is less famous, but will be a bit less work to deal with rigorously. To define g, first fix any sequence $\{q_n\}_{n=1}^{\infty}$ listing all the rational numbers once each. Define $g : \mathbb{R} \to \mathbb{R}$ as follows:

$$g(x) = \begin{cases} \frac{1}{n} & \text{if } x \in \mathbb{Q} \text{ and } x = q_n \\ 0 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

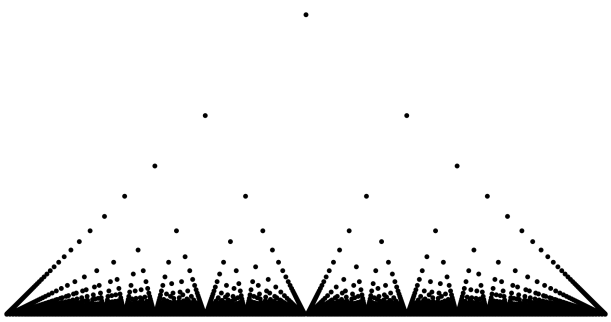


Figure 5.1: The popcorn function (Smithers888, Public domain, via Wikimedia Commons).

It is straightforward to show that g is discontinuous at every rational. Now let us show that, for any irrational number $c \in \mathbb{R} \setminus \mathbb{Q}$, the function g is continuous at c. Since $g(c) = 0$, we must show that for every $\epsilon > 0$ there exists $\delta > 0$ such that $|x - c| < \delta$ implies $|g(x)| < \epsilon$. Indeed, let $N \in \mathbb{N}$ be chosen such that $\frac{1}{N} < \epsilon$ and let $\delta \coloneqq \min\{|c - q_n| : n = 1, \ldots, N\}$. Then, if $|x - c| < \delta$ then either x is irrational in which case $|g(x)| = 0 < \epsilon$ or else $x = q_n$ for some $n > N$ in which case $|g(x)| = \frac{1}{n} < \frac{1}{N} < \epsilon$.

Exercise 5.4.6. Give an example of a function $f : \mathbb{R} \to \mathbb{R}$ which is continuous at every rational number and discontinuous at every irrational number.
Answer: This is a trick question. It is impossible to give an example of a function with these properties, though it does take some work to prove this. Briefly, one first shows that the set of points of continuity of any function $f : \mathbb{R} \to \mathbb{R}$ must be always be a countable intersection of open sets. Next one shows it is impossible to express $\mathbb{Q}$ as the intersection of countably many open sets using something called the Baire category theorem. We may return to this later on if time permits.

5.5 Uniform continuity

Next we will study a strengthening of the definition of continuity called uniform continuity. A function $f : X \to \mathbb{R}$ where $X \subseteq \mathbb{R}$ is continuous if it is continuous at every $x \in X$. This means that, given $\epsilon > 0$, each $x \in X$ can be assigned a $\delta_x > 0$ such that $|y - x| < \delta_x$ implies $|f(y) - f(x)| < \epsilon$. The definition of uniform continuity is exactly the same, except that the quantity δ_x is no longer dependent on x. The same δ needs to work for all basepoints.

Definition 5.5.1. Let $f : X \to \mathbb{R}$ where $X \subseteq \mathbb{R}$. We say that f is **uniformly continuous** if, for every $\epsilon > 0$, there exists $\delta > 0$ such that $|x - y| < \delta$ implies $|f(x) - f(y)| < \epsilon$.

It is clear that every uniformly continuous function is also continuous.

Example 5.5.2. The function $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 5x$ is uniformly continuous. Indeed, let $\epsilon > 0$ be given. Put $\delta = \epsilon/5$. Then, if $|x - y| < \delta$, we have $|f(x) - f(y)| = |5x - 5y| = 5|x - y| < 5\delta = \epsilon$. Since f is uniformly continuous, it is also continuous.

Example 5.5.3. The function $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$ is continuous. However, f is not uniformly continuous. Indeed, suppose for contradiction it was. Let $\epsilon = 1$. If f were uniformly continuous, there would exist $\delta > 0$ such that $|y-x| < \delta$ implied $|f(y)-f(x)| < 1$. However, given any $\delta > 0$, we can consider $x = \frac{2}{\delta}$ and $y = x + \frac{\delta}{2}$. Then $|x - y| = \frac{\delta}{2} < \delta$ but

$$|f(y) - f(x)| = y^2 - x^2 = (y + x)(y - x) > 2x(y - x) = 2 \cdot \frac{2}{\delta} \cdot \frac{\delta}{2} = 2.$$

Even though uniform continuity is a stronger condition than continuity in general, when the domain is a closed interval, continuity automatically implies uniform continuity:

Theorem 5.5.4. *If $f : [a, b] \to \mathbb{R}$ is a continuous function on a closed interval, then f is uniformly continuous.*

Remark 5.5.5. This same theorem holds more generally if $[a, b]$ is replaced by any compact set. Exactly the same proof works.

Proof. Fix $\epsilon > 0$. We must find $\delta > 0$ such that $|x - y| < \delta$ implies $|f(x) - f(y)| < \epsilon$. Since f is continuous, for each $x \in [a, b]$ there exists $\delta_x > 0$ such that $|x - y| < \delta_x$ implies $|f(x) - f(y)| < \epsilon/2$. Define an open cover of $[a, b]$ as follows:

$$\mathscr{U} := \{(x - \delta_x/2, x + \delta_x/2) : x \in [a, b]\}.$$

Since $[a, b]$ has the Heine-Borel property (Theorem 4.4.1), there exists finitely-many points $x_1, \ldots, x_n \in [a, b]$ such that

$$[a, b] \subseteq (x_1 - \delta_{x_1}/2, x_1 + \delta_{x_1}/2) \cup \ldots \cup (x_n - \delta_{x_n}/2, x_n + \delta_{x_n}/2).$$

We claim that taking $\delta := \min\{\delta_1/2, \ldots, \delta_n/2\}$ does the job. Indeed, suppose that $x, y \in [a, b]$ and $|x - y| < \delta$. There must exist $i \in \{1, \ldots, n\}$ such that $|x - x_i| < \delta_{x_i}/2$. Then, we also have

$$|y - x_i| \leq |y - x| + |x - x_i| < \delta + \delta_{x_i}/2 \leq \delta_{x_i}/2 + \delta_{x_i}/2 = \delta_{x_i}.$$

Using that both x, y are within δ_{x_i} of x_i, we obtain the required estimate:

$$|f(x) - f(y)| \leq |f(x) - f(x_i)| + |f(y) - f(x_i)| < \epsilon/2 + \epsilon/2 = \epsilon.$$

□

It is important that the domain interval is taken to be closed in the above theorem. For open intervals, the theorem fails.

Exercise 5.5.6. Give examples of continuous functions $f : (0, 1) \to \mathbb{R}$ which are not uniformly continuous. Give one example where f is unbounded and another where f is bounded.
Answer: For an unbounded example, one may define $f(x) = \frac{1}{x}$. For a bounded example, one may define $f(x) = \sin(\frac{1}{x})$. Let us verify that this second function f is not uniformly continuous. Indeed, let $\epsilon = 1$. We claim that there is no $\delta > 0$ such that $|x - y| < \delta$ implies $|f(x) - f(y)| < 1$. Indeed, for $n \in \mathbb{N}$, let

$$x = \frac{1}{2\pi n + \frac{\pi}{2}} \qquad\qquad y = \frac{1}{2\pi n + \frac{3\pi}{2}}.$$

For sufficiently large n, we have $|x|, |y| < \delta/2$ so that $|x - y| < |x| + |y| < \delta$. However, $f(x) = 1$ and $f(y) = -1$ so that $|f(x) - f(y)| = 2$ which is larger than $\epsilon = 1$.

5.6 Limits of functions

Previously (Definition 2.3.1) we studied limits of sequences: $\lim_{n\to\infty} a_n$. We now consider limits of real functions: $\lim_{x\to c} f(x)$. In the sequence case, the index n never actually takes on the value $n = \infty$. Rather the limit attempts to capture the trend of a_n as n gets arbitrarily "close to infinity", i.e. arbitrarily large. Similarly, in the function case, the variable x is not permitted to take on the value $x = c$. Rather, the limit attempts to capture the trend of $f(x)$ as x gets arbitrarily close to c without ever actually achieving the value $x = c$. For this reason, we need to require c to be what we call an "accumulation point" of the domain of f to ensure that arbitrarily close inputs x with $x \neq c$ can be found.

Definition 5.6.1. Suppose that $X \subseteq \mathbb{R}$. We say that a point $c \in \mathbb{R}$ is an **accumulation point** of X if, for every $\epsilon > 0$, the interval $(c - \epsilon, c + \epsilon)$ contains a point $x \in X$ with $x \neq c$.

Remark 5.6.2. Unfortunately, there is not a global consensus among mathematicians concerning the above terminology. There are several closely-related concepts and terminologies (accumulation point, limit point, cluster point, adherent point, condensation point...) at play and no universal agreement about which words refer to which concepts. One point of consistency in the literature is that, given a set X, the set X' of accumulation points (as defined above) of X is called the **derived set** of X.

Exercise 5.6.3. For each given set $X \subseteq \mathbb{R}$, determine its set X' of accumulation points.

(i) An open interval $X = (a, b)$. *Answer:* $X' = [a, b]$.

(ii) A single point $X = \{x\}$. *Answer:* $X' = \varnothing$.

(iii) $X = \{0\} \cup \{\frac{1}{n} : n \in \mathbb{N}\}$. *Answer:* $X' = \{0\}$.

Definition 5.6.4. Consider a function $f : X \to \mathbb{R}$ with domain $X \subseteq \mathbb{R}$. Let $c \in \mathbb{R}$ be an accumulation point of X. We say that the **limit** as x approaches c of f exists and equals the number $L \in \mathbb{R}$ if for all $\epsilon > 0$, there exists $\delta > 0$ such that $0 < |x - c| < \delta$ implies $|f(x) - L| < \epsilon$. In this case, we write $\lim_{x\to c} f(x) = L$.

Remark 5.6.5. As in the case of sequences, one can show that the limit of a real function is unique. One never has $\lim_{x\to c} f(x) = L_1$ and $\lim_{x\to c} f(x) = L_2$ for two different real numbers L_1, L_2.

Remark 5.6.6. It may or may not be the case in the above definition that c belongs to the domain of f. All we require is that c is an accumulation point of the domain. The familiar limit $\lim_{x\to 0} \frac{\sin x}{x} = 1$ from calculus classes is a natural example of a function having a limit at a point outside of its domain.

Both the limit concept for functions just introduced and the limit concept for sequences studied earlier are closely related to continuity. The theorem below shows that we may think of continuous functions as functions which "preserve limits" in either sense of the word.

Theorem 5.6.7. *Consider a function $f : X \to \mathbb{R}$ with domain $X \subseteq \mathbb{R}$ and fix $c \in X$. Then the following are equivalent:*

(i) f is continuous at c.

(ii) For every sequence $\{x_n\}$ in X such that $\lim_{n\to\infty} x_n = c$ one has $\lim_{n\to\infty} f(x_n) = f(c)$.

Furthermore, if c is an accumulation point of X, then (i) and (ii) are also equivalent to:

(iii) $\lim_{x\to c} f(x) = f(c)$.

The proof of this theorem is left as an exercise for the reader.

Chapter 6

Differential calculus

6.1 The derivative

We have finally arrived in the world of calculus! Before giving the definition of the derivative, we recall a topological concept that appeared on the homework.

Definition 6.1.1. Suppose $X \subseteq \mathbb{R}$ and $x \in X$. We say that x is an **interior point** of X if there exists $\epsilon > 0$ such that $(x - \epsilon, x + \epsilon) \subseteq X$.

Informally, the interior points of a set have a bit of "wiggle room" inside the set. Notice that every interior point is automatically an accumulation point (Definition 5.6.1). For this reason, the limit in the definition below makes sense:

Definition 6.1.2. A real function f is **differentiable at a point** $c \in \operatorname{dom} f$ if:

(i) c is an interior point of $\operatorname{dom} f$, and

(ii) $f'(c) \coloneqq \lim_{x \to c} \dfrac{f(x) - f(c)}{x - c}$ exists.

We say that f is **differentiable** if it is differentiable at every point of its domain (which is necessarily open). The function f' whose domain consists of all points at which f is differerentiable and which is defined by the above limit is called the **derivative** of f.

Remark 6.1.3. In principle, the definition of differentiability of f at c given above still makes sense if c is just an accumulation point of $\operatorname{dom} f$ instead of an interior point. However, for a variety of reasons, one would prefer f to be defined on an interval containing c. For example, we would like to say that $f + g$ is differentiable at c if f and g both are. However, if c is only an accumulation point of $\operatorname{dom} f$ and $\operatorname{dom} g$, it could occur that c is not an accumulation point of $\operatorname{dom}(f + g) = \operatorname{dom} f \cap \operatorname{dom} g$ (exercise: give an example!).

Remark 6.1.4. For functions defined on closed intervals, it is sometimes useful to also define left/right derivatives at the endpoints of the interval, but let us not worry about such technicalities for the moment.

To warm ourselves up to the definition, we prove that differentiability of f at a point c is a stronger condition than continuity at c.

Theorem 6.1.5. *If $f : X \to \mathbb{R}$ is differentiable at $c \in X$, then f is also continuous at c.*

Proof. Note differentiability at c (Definition 6.1.2) implies in particular that c is an accumulation point of the domain of f. Thus, by Theorem 5.6.7, f is continuous at c if and only if $\lim_{x\to c} f(x) = f(c)$. In other words, given $\epsilon > 0$, we must show that there exists $\delta > 0$ such that $0 < |x - c| < \delta$ implies $|f(x) - f(c)| < \epsilon$. Note we can write

$$|f(x) - f(c)| = \left|\frac{f(x) - f(c)}{x - c}\right| |x - c|$$

for $x \neq c$. Since $\lim_{x\to c} \frac{f(x)-f(c)}{x-c} = f'(c)$, there exists $\delta_1 > 0$ such that $0 < |x - c| < \delta_1$ implies $\left|\frac{f(x)-f(c)}{x-c} - f'(c)\right| < 1$. In particular, $0 < |x - c| < \delta_1$ implies $\left|\frac{f(x)-f(c)}{x-c}\right| < M$ where $M \coloneqq |f'(c)| + 1$. Next put $\delta_2 \coloneqq \epsilon/M$ and put $\delta \coloneqq \min\{\delta_1, \delta_2\}$. If $0 < |x - c| < \delta$, we find that

$$|f(x) - f(c)| = \left|\frac{f(x) - f(c)}{x - c}\right| |x - c| \leq M|x - c| < M \cdot \epsilon/M = \epsilon.$$

□

Remark 6.1.6. Similarly to how one establishes limit laws for sequences (Theorem 2.6.2), one may establish analogous limit laws for functions. We did not do this (yet), but such limit laws would allow for the above theorem to be proved a bit more cleanly. Using standard limit laws for functions, we have:

$$\begin{aligned}
\lim_{x\to c}(f(x) - f(c)) &= \lim_{x\to c}\left(\frac{f(x) - f(c)}{x - c} \cdot (x - c)\right) \\
&= \lim_{x\to c} \frac{f(x) - f(c)}{x - c} \cdot \lim_{x\to c}(x - c) \\
&= f'(c) \cdot 0 \\
&= 0.
\end{aligned}$$

and so $\lim_{x\to c} f(x) = f(c)$. However, getting some practice using the ϵ-δ definition of differentiability is not a bad thing.

6.2 Differentiation of rapidly oscillating functions

There is a surprising amount one can learn about differentiation by studying the behaviour of functions of the form $x^m \sin(\frac{1}{x^n})$ near $x = 0$. We have not rigorously developed the theory of trigonometric functions in this course, but the student should feel free to use their knowledge about the behaviour and derivatives of these functions from calculus courses.

Exercise 6.2.1. Show that the function $h : \mathbb{R} \to \mathbb{R}$ defined by

$$h(x) = \begin{cases} x \sin(\frac{1}{x}) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$$

is continuous at 0 but not differentiable 0.

Answer: To see h is continuous at 0, we should argue that $\lim_{x\to 0} h(x) = h(0) = 0$. Indeed, given $\epsilon > 0$, let $\delta = \epsilon$. If $0 < |x| < \delta$, then $|h(x)| = |x||\sin(\frac{1}{x})| \leq |x| < \epsilon$. One may also think of the continuity of h at $x = 0$ as a consequence of the squeeze law for limits of real functions, relying on the fact that $-|x| \leq h(x) \leq |x|$. Note, however, that we have not yet formally discussed the limit laws for real functions.

To see h is not differentiable at 0, we should argue that $\lim_{x\to 0} \frac{h(x)-h(0)}{x-0} = \lim_{x\to 0} \frac{h(x)}{x}$ does not exist. Indeed, suppose towards a contradiction that this limit existed. Let $x_n = \frac{1}{2\pi n + \frac{\pi}{2}}$ for $n \in \mathbb{N}$ so that $x_n \to 0$. Then we have:

$$h'(0) = \lim_{x\to 0} \frac{f(x)}{x} = \lim_{n\to\infty} \frac{h(x_n)}{x_n} = \lim_{n\to\infty} \frac{x_n \sin(2\pi n + \frac{\pi}{2})}{x_n} = \lim_{n\to\infty} 1 = 1.$$

On the other hand, putting $y_n = \frac{1}{2\pi n + \frac{3\pi}{2}}$, we have $y_n \to 0$ and

$$h'(0) = \lim_{x\to 0} \frac{h(x)}{x} = \lim_{n\to\infty} \frac{h(y_n)}{y_n} = \lim_{n\to\infty} \frac{y_n \sin(2\pi n + \frac{3\pi}{2})}{y_n} = \lim_{n\to\infty} -1 = -1.$$

Exercise 6.2.2. Show that the function $F : \mathbb{R} \to \mathbb{R}$ defined by

$$F(x) = \begin{cases} x^2 \sin(\frac{1}{x}) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$$

is differentiable at 0. Furthermore (assuming standard differentiation rules from introductory calculus) show that the deriviative F' is not continuous at 0.

Answer: We claim that F is differentiable at 0 with derivative $F'(0) = 0$. Indeed,

$$\lim_{x\to 0} \frac{F(x) - F(0)}{x - 0} = \lim_{x\to 0} \frac{F(x)}{x} = \lim_{x\to 0} \frac{x^2 \sin(\frac{1}{x})}{x} = \lim_{x\to 0} x \sin(\tfrac{1}{x}) = 0,$$

as was shown in the previous example.

Next, using usual differentiation rules, the derivative function F' is given by

$$F'(x) = \begin{cases} 2x \sin(\frac{1}{x}) - \cos(\frac{1}{x}) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}.$$

Similarly to what was done in Exercise 6.2.1 above, one may argue that $\lim_{x\to 0} F'(x)$ does not exist by examining the behaviour of F' along appropriately chosen sequences.

6.3 Differentiation laws

The student will already be familiar with the standard differentiation laws (product rule, chain rule, etc) used for symbolic differentiation in calculus courses. In some cases, the student will also have seen proofs of these laws. In other cases, proofs may have been skipped to save time. In this section, we will review the basic differentiation laws and go over some of the proofs. The remaining proofs should be taken as exercises or looked up in other references.

It would be possible, but somewhat labour intensive, to prove all the differentiation laws directly from the definition of the derivative (Definition 6.1.2) and the definition of the limit of a function (Definition 5.6.4). However, as already pointed in Remark 6.1.6, we can streamline our arguments by first establishing limit laws for functions. The limit laws allow us to hide the ϵ-δ details of proofs "under the hood" and argue in a more formal way.

Theorem 6.3.1 (Limit laws - function versions)**.** *Consider two functions $f, g : X \to \mathbb{R}$ where $X \subseteq \mathbb{R}$. Let $c \in \mathbb{R}$ be an accumulation point of X (it may or may not be true that $c \in X$). If $\lim_{x\to c} f(x)$ and $\lim_{x\to c} g(x)$ both exist, then:*

(i) $\lim_{x\to c} [f(x) + g(x)] = \lim_{x\to c} f(x) + \lim_{x\to c} g(x)$

(ii) $\lim_{x\to c} [f(x)g(x)] = \lim_{x\to c} f(x) \cdot \lim_{x\to c} g(x)$

(iii) $\lim_{x\to c} \dfrac{f(x)}{g(x)} = \dfrac{\lim_{x\to c} f(x)}{\lim_{x\to c} g(x)}$*, provided the bottom limit is nonzero.*

The proofs of these statements are very similar to the proofs of the analogous limit laws for sequences (Theorem 2.6.2). For purpose of illustration, we include only the proof of *(iii)* below, since it is the most involved.

Proof. For brevity, put $A \coloneqq \lim_{x\to c} f(x)$ and $B \coloneqq \lim_{x\to c} g(x)$. By hypothesis, $B \neq 0$. First we argue that there exists $\delta_0 > 0$ such that

$$0 < |x - c| < \delta_0 \Longrightarrow \left| \frac{f(x)}{g(x)} - \frac{A}{B} \right| \leq \lambda_1 |f(x) - A| + \lambda_2 |g(x) - B|$$

for some constants $\lambda_1, \lambda_2 > 0$ to be defined shortly. To see this, we use that $|B|/2 > 0$ and $\lim_{x\to c} g(x) = B$. Thus, there exists $\delta_0 > 0$ such that $0 < |x - c| < \delta_0$ implies $|g(x) - B| <$

$|B|/2$ which implies $|g(x)| \geq |B|/2$ which implies

$$\begin{aligned}
\left|\frac{f(x)}{g(x)} - \frac{A}{B}\right| &= \frac{|Bf(x) - Ag(x)|}{|B||g(x)|} \\
&\leq \frac{|Bf(x) - Ag(x)|}{|B| \cdot |B|/2} \\
&= \frac{|Bf(x) - BA + BA - Ag(x)|}{|B|^2/2} \\
&\leq \frac{|Bf(x) - BA| + |BA - Ag(x)|}{|B|^2/2} \\
&= \frac{2}{|B|}|f(x) - A| + \frac{2|A|}{|B|^2}|g(x) - B| \\
&\leq \lambda_1|f(x) - A| + \lambda_2|g(x) - B| \qquad \text{where } \lambda_1 \coloneqq \frac{2}{|B|}, \lambda_2 \coloneqq \frac{2|A|}{|B|^2} + 1.
\end{aligned}$$

Technical comment: it is convenient to add 1 *in the definition of* λ_2 *to make sure it is always positive, even in the edge case* $A = 0$.

Now we prove $\lim_{x\to c} \frac{f(x)}{g(x)} = \frac{A}{B}$. Let $\epsilon > 0$ be arbitrary. Since $\lim_{x\to c} f(x) = A$ and $\lim_{x\to c} g(x) = B$, there exists $\delta_1, \delta_2 > 0$ such that:

$$\begin{aligned}
0 < |x - c| < \delta_1 &\implies |f(x) - A| \leq \frac{\epsilon}{2\lambda_1} \\
0 < |x - c| < \delta_2 &\implies |g(x) - B| \leq \frac{\epsilon}{2\lambda_2}.
\end{aligned}$$

Put $\delta \coloneqq \min\{\delta_0, \delta_1, \delta_2\}$. Then, $0 < |x - c| < \delta$ implies

$$\begin{aligned}
\left|\frac{f(x)}{g(x)} - \frac{A}{B}\right| &\leq \lambda_1|f(x) - A| + \lambda_2|g(x) - B| \\
&< \lambda_1\frac{\epsilon}{2\lambda_1} + \lambda_2\frac{\epsilon}{2\lambda_2} \\
&= \epsilon.
\end{aligned}$$

as desired. □

With the limit laws for functions at hand, we move on to the basic differentiation laws.

Theorem 6.3.2 (Basic differentiation laws)**.** *Suppose* f *and* g *are real functions that are both differentiable at* $c \in \mathbb{R}$*. Then:*

(i) $(f + g)'(c) = f'(c) + g'(c)$,

(ii) $(fg)'(c) = f'(c)g(c) + f(c)g'(c)$,

(iii) $\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{g(c)^2}$, *assuming* $g(c) \neq 0$.

Proof. For (i), note for $x \neq c$, we have:

$$\frac{(f+g)(x)-(f+g)(c)}{x-c} = \frac{f(x)-f(c)}{x-c} + \frac{g(x)-g(c)}{x-c}.$$

Taking the limit as $x \to c$ and using the limit law for sums, we get that $f+g$ is differentiable at c and $(f+g)'(c) = f'(c) + g'(c)$.

Similarly, for (ii), note that for $x \neq c$, we have:

$$\begin{aligned}\frac{f(x)g(x)-f(c)g(c)}{x-c} &= \frac{f(x)g(x)-f(c)g(x)+f(c)g(x)-f(c)g(c)}{x-c} \\ &= g(x)\frac{f(x)-f(c)}{x-c} + f(c)\frac{g(x)-g(c)}{x-c}.\end{aligned}$$

Taking the limit as $x \to c$ and using the limit laws for sums/products, we get that fg is differentiable at c and $(fg)'(c) = f'(c)g(c) + f(c)g'(c)$. Note here we also used the fact that $\lim_{x\to c} g(x) = g(c)$. This holds because g is differentiable at c hence also continuous c (see Theorem 6.1.5 and Theorem 5.6.7 (iii)).

The proof of (iii) is similar and we leave it to the reader. □

Remark 6.3.3. In the proofs of the differentiation laws above, we glossed over some technical (and frankly slightly boring) details relating to domains. If f is differentiable at $c \in \mathbb{R}$ then, referring to Definition 6.1.2, it is required that the domain of f includes an open interval I_1 with $c \in I_1$. Likewise, if g is differentiable at c, then the domain of g must include an open interval I_2 with $c \in I_2$. By definition, the domain of $f+g$ is the intersection of the domains of f and g. Thus, the domain of $f+g$ includes the open interval $I_1 \cap I_2$ with $c \in I_1 \cap I_2$. The existence of such an interval is a prerequisite to talking about the differentiability of $f+g$ at c. The story for fg is the same. For f/g, the situation is slightly more complicated as its domain also excludes the points where g vanishes. However, since g is continuous at c by Theorem 6.1.5 and $g(c) \neq 0$ by hypothesis, it is simple to argue that there is a $\delta > 0$ such that $|x-c| < \delta$ implies $g(x) \neq 0$. Thus, the domain of f/g includes the open interval $I_1 \cap I_2 \cap (c-\delta, c+\delta)$ which contains the point c. These domain considerations are not strictly theoretical issues; they show up all the time in calculations, often without our notice. For example, consider the differentiability of the functions

$$\begin{aligned} f &: \mathbb{R} \setminus \{-2, 2\} \to \mathbb{R} & g &: [0, \infty) \to \mathbb{R} \\ f(x) &= \frac{1}{x^2-4} & g(x) &= \sqrt{x}. \end{aligned}$$

at the point $c = 1/2$. The domain of the sum $f+g$ is $[0,2) \cup (2,\infty)$, the intersection of the domains of f and g. The largest open interval containing $1/2$ that is included in the domain of f is $(-2,2)$. The largest open interval containing $1/2$ that is included in the domain of g is $(0,\infty)$. The largest open interval containing $1/2$ that is included in the domain of $f+g$ is $(0,2) = (-2,2) \cap (0,\infty)$.

Next we come to the chain rule: if f is differentiable at c and g is differentiable at $f(c)$, then $g \circ f$ is differentiable at c and

$$(g \circ f)'(c) = g'(f(c))f'(c).$$

An informal proof of this is as follows:

$$(g \circ f)'(c) = \lim_{x \to c} \frac{g(f(x)) - g(f(c))}{x - c} = \lim_{x \to c} \frac{g(f(x)) - g(f(c))}{f(x) - f(c)} \cdot \lim_{x \to c} \frac{f(x) - f(c)}{x - c}.$$

Since f is continuous at c, we have that $f(x) \to f(c)$ as $x \to c$. Thus (really??) the first limit on the right goes to $g'(f(c))$. The second limit on the right goes to $f'(c)$, so we get the result. Unfortunately, making this sketch into a precise proof requires some distracting detours. The issue is that $f(x) = f(c)$ can potentially occur for $x \neq c$ arbitrarily close to c leading to division by zero above. One common strategy for overcoming this issue is to switch tacks and use sequential limits instead of function limits. The reason for doing this is that the sequence formalism is a bit better suited to dividing the argument into cases. The connection between sequential limits and function limits is something that appeared on the homework previously. Let's revisit it here in order to have a precise statement handy:

Theorem 6.3.4. *Suppose $f : X \to \mathbb{R}$ where $X \subseteq \mathbb{R}$ and $c \in \mathbb{R}$ is an accumulation point of X (it may or may not be the case that $c \in X$). Let $L \in \mathbb{R}$. Then the following are equivalent:*

(i) $\lim_{x \to c} f(x) = L$,

(ii) $\lim_{n \to \infty} f(x_n) = L$ *for every sequence* $\{x_n\}_{n=1}^{\infty}$ *in* $X \setminus \{c\}$ *with* $x_n \to c$.

Proof. First we prove $(i \Rightarrow ii)$. Assume (i) holds and let $\{x_n\}_{n=1}^{\infty}$ be any sequence in $X \setminus \{c\}$ such that $x_n \to c$. We must show that $f(x_n) \to L$. To this end, let $\epsilon > 0$ be given. By (i), there exists $\delta > 0$ such that $0 < |x - c| < \delta$ implies $|f(x) - L| < \epsilon$. Since $x_n \to c$, there exists $N \in \mathbb{N}$ such that $n \geq N$ implies $|x_n - c| < \delta$. Automatically, $0 < |x_n - c|$ for all $n \in \mathbb{N}$ because $\{x_n\}$ is a sequence in $X \setminus \{c\}$, so it follows that $n \geq N$ implies $|f(x_n) - L| < \epsilon$, as required.

For the reverse implication $(ii \Rightarrow i)$, we instead prove the contrapositive. Suppose that (i) does not hold. That is, there exists $\epsilon > 0$ such that, for all $\delta > 0$, there exists $x \in X$ with $0 < |x - c| < \delta$ such that $|f(x) - L| \geq \epsilon$. In particular, this is the case when $\delta = \frac{1}{n}$ for $n \in \mathbb{N}$, and so there exists a sequence $\{x_n\}_{n=1}^{\infty}$ in $X \setminus \{c\}$ such that $|x_n - c| < \frac{1}{n}$ and $|f(x_n) - L| \geq \epsilon$ for all $n \in \mathbb{N}$. It follows (why?) that $x_n \to c$ but $f(x_n) \not\to L$, i.e. statement (ii) does not hold. □

We now use the above characterization of limits to prove the chain rule.

Theorem 6.3.5 (Chain rule)**.** *Let f and g be real functions. Suppose that f is differentiable at c and g is differentiable at $f(c)$. Then the composition $g \circ f$ is differentiable at c and $(g \circ f)'(c) = g'(f(c))f'(c)$.*

Proof. Let $X = \text{dom}(g \circ f) = \{x \in \mathbb{R} : x \in \text{dom}(f) \text{ and } f(x) \in \text{dom}(g)\}$. Applying Theorem 6.3.4 above, the goal is to prove that, for every sequence $\{x_n\}$ in $X \setminus \{c\}$ such that $x_n \to c$, we have

$$\lim_{n\to\infty} \frac{g(f(x_n)) - g(f(c))}{x_n - c} = g'(f(c))f'(c).$$

We consider two cases. First, suppose there exists $\delta > 0$ such that $0 < |x - c| < \delta$ implies $f(x) \neq f(c)$. Let $\{x_n\}$ be any sequence in $X \setminus \{c\}$ such that $x_n \to c$. The assumption of this case implies that for sufficiently large $n \in \mathbb{N}$ we have $f(x_n) \neq f(c)$ and so:

$$\frac{g(f(x_n)) - g(f(c))}{x_n - c} = \frac{g(f(x_n)) - g(f(c))}{f(x_n) - f(c)} \cdot \frac{f(x_n) - f(c)}{x_n - c}.$$

Since f is continuous at c, we have $f(x_n) \to f(c)$. Since g is differentiable at $f(c)$ and f is differentiable at c, the product rule for limits gives:

$$\lim_{n\to\infty} \frac{g(f(x_n)) - g(f(c))}{x_n - c} = \lim_{n\to\infty} \frac{g(f(x_n)) - g(f(c))}{f(x_n) - f(c)} \cdot \lim_{n\to\infty} \frac{f(x_n) - f(c)}{x_n - c} = (g \circ f)'(c)f'(c).$$

It remains to consider the case where, for all $\delta > 0$, there exists x with $0 < |x - c| < \delta$ such that $f(x) = f(c)$. In particular, this applies to $\delta = \frac{1}{n}$ for every $n \in \mathbb{N}$. Thus, there exists a sequence $\{y_n\}$ such that $0 < |y_n - c| < \frac{1}{n}$ and $f(y_n) = f(c)$ for all $n \in \mathbb{N}$. By design, $y_n \to c$ (why?) and so we have

$$f'(c) = \lim_{n\to\infty} \frac{f(y_n) - f(c)}{y_n - c} = \lim_{n\to\infty} \frac{0}{y_n - c} = 0.$$

Thus, since $g'(f(c))f'(c) = g'(f(c)) \cdot 0 = 0$, our goal becomes to show that for any sequence $\{x_n\}_{n=1}^{\infty}$ in $X \setminus \{c\}$ with $x_n \to c$ we have:

$$\lim_{n\to\infty} \frac{g(f(x_n)) - g(f(c))}{x_n - c} = 0. \tag{6.1}$$

Note that whenever $n \in \mathbb{N}$ is such that $f(x_n) = f(c)$, we obtain:

$$\frac{g(f(x_n)) - g(f(c))}{x_n - c} = \frac{0}{x_n - c} = 0.$$

In particular, if $f(x_n) = f(c)$ for all sufficiently large n, then (6.1) is satisfied. Otherwise, $f(x_n) \neq f(c)$ for infinitely many $n \in \mathbb{N}$ and we may consider the subsequence $\{x_{n_k}\}$ consisting exactly of the terms for which $f(x_{n_k}) \neq f(c)$. Since $x_{n_k} \to c$ and $f'(c) = 0$, we have (reading carefully the statement of Theorem 6.3.4) that

$$\lim_{k\to\infty} \frac{f(x_{n_k}) - f(c)}{x_{n_k} - c} = f'(c) = 0.$$

Thus, by the product rule for limits:

$$\begin{aligned}\lim_{k\to\infty}\frac{g(f(x_{n_k}))-g(f(c))}{x_{n_k}-c} &= \lim_{k\to\infty}\frac{g(f(x_{n_k}))-g(f(c))}{f(x_{n_k})-f(c)}\cdot\lim_{k\to\infty}\frac{f(x_{n_k})-f(c)}{x_{n_k}-c}\\ &= g'(f(c))\cdot 0\\ &= 0.\end{aligned}$$

We have shown the sequence $\{\frac{g(f(x_n))-g(f(c))}{x_n-c}\}_{n=1}^{\infty}$ converges to 0 along a certain subsequence and is equal to 0 away from that subsequence. Thus (why?) the whole sequence converges to 0 which is to say (6.1) holds as desired. □

Remark 6.3.6. As before, there are some technicalities having to do with domains that we glossed over in the proof of Theorem 6.3.5. If f is differentiable at c, there is required to exist $\epsilon_1 > 0$ such that $(c-\epsilon_1, c+\epsilon_1)$ is contained in the domain of f. If g is differentiable at $f(c)$, there is required to exist $\epsilon_2 > 0$ such that $(f(c)-\epsilon_2, f(c)-\epsilon_2)$ is contained in the domain of g. Then, since f is continuous at c (Theorem 6.1.5), there exists $\delta > 0$ such that $|x-c| < \delta$ implies $|f(x)-f(c)| < \epsilon_2$. Putting $\epsilon \coloneqq \min\{\epsilon_1, \delta\}$, one sees that $(c-\epsilon, c+\epsilon)$ is contained in the domain of $g \circ f$. The existence of such an interval is a prerequisite to talking about the differentiability of $g \circ f$ at c.

The proof of the chain rule given above is quite technical and sadly not terribly enlightening. Things become much more conceptually clear if one swaps out the limit-based definition of the derivative we have been using for an equivalent definition designed to capture the idea that the derivative is a "best linear approximation": $f(c+x) \approx f(c) + f'(c)x$ when $x \approx 0$. The purpose of the following exercise is to implement this idea precisely.

Exercise 6.3.7.

(a) Prove that f is differentiable at c if and only if there exists a function $\varepsilon : (-\delta, \delta) \to \mathbb{R}$ where $\delta > 0$ which vanishes continuously at 0 (meaning $\varepsilon(0) = 0$ and $\lim_{x\to 0}\varepsilon(x) = 0$) and a constant $m \in \mathbb{R}$ such that $f(c+x) = f(c) + mx + x\varepsilon(x)$ for all $x \in (-\delta, \delta)$. Moreover, when these equivalent conditions hold, prove that $f'(c) = m$.

(b) Use this alternative characterization of the derivative to give a more streamlined proof of the chain rule not requiring any casework or arguments involving sequences.

Remark 6.3.8. What is written above can be expressed more compactly using what is called "little o notation". In terms of that formalism, one would write:

$$f(c+x) = f(c) + f'(c)x + o(x).$$

You may find it enlightening to read up on "little o notation" and rewrite your proof of the chain rule in that language.

6.4 Maximums and minimums

The student will already be intimately familiar with the following principle: the local extrema of a real function can only occur at critical points: points where the derivative is zero or does not exist. When the graph of f is a nicely-behaved curve this captures the idea that the tangent line to the graph becomes horizontal whenever a local maximum or minimum occurs. Let's now go over the definitions and proofs.

Definition 6.4.1. Suppose that $f : X \to \mathbb{R}$ where $X \subseteq \mathbb{R}$. We say that f has a **local maximum** at $c \in X$ if there exists $\epsilon > 0$ such that $f(x) \leq f(c)$ for all $x \in X$ satisfying $|x - c| < \epsilon$. Similarly, we say that f has a **local minimum** at $c \in X$ if there exists $\epsilon > 0$ such that $f(x) \geq f(c)$ for all $x \in X$ satisfying $|x - c| < \epsilon$.

Theorem 6.4.2. *Suppose f is a real function that is differentiable at c. If f has either a local maximum or a local minimum at c then $f'(c) = 0$.*

Proof. We only consider the case where f has a local maximum at c, as the case of a local minimum is similar. We again find it convenient to leverage the sequential characterization of limits of functions (Theorem 6.3.4). Since f is differentiable at c, for any sequence $\{x_n\}$ with $x_n \neq c$ for all $n \in \mathbb{N}$ such that $x_n \to c$, we have:

$$\lim_{n\to\infty} \frac{f(x_n) - f(c)}{x_n - c} = f'(c).$$

Using the fact that f has a local maximum at c, one may find sequences $\{x_n\}$ and $\{y_n\}$ such that:

- $x_n, y_n \to c$,
- $x_n < c < y_n$ for all $n \in \mathbb{N}$,
- $f(x_n), f(y_n) \leq f(c)$ for all $n \in \mathbb{N}$.

Then it is easy to check:

$$\frac{f(x_n) - f(c)}{x_n - c} \geq 0 \qquad \frac{f(y_n) - f(c)}{y_n - c} \leq 0 \qquad \forall n \in \mathbb{N}.$$

Taking the limit as $n \to \infty$ and using standard properties of limits, we obtain $f'(c) \geq 0$ and $f'(c) \leq 0$. Thus $f'(c) = 0$ as claimed. □

Note: As stated in the introduction, the following sections on the mean value theorem and Taylor's theorem were partly written by Nicolae Strungaru who was kindly covering my class at the time. Any errors which survived or were introduced when incorporating his materials into the present document are the sole responsibility of the author.

6.5 The mean value theorem

The statement below is a special case of the mean value theorem, to be stated shortly. It is traditional (and efficient) to prove this statement first and then use it as a lemma to deduce the more general mean value theorem.

Theorem 6.5.1 (Rolle's theorem)**.** *Suppose f is a real function that is continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b). If $f(a) = f(b)$, then there exists t with $a < t < b$ such that $f'(t) = 0$.*

Intuitively, if $f(a) = f(b)$, then the graph of f must have a horizontal tangent line at some point strictly in between.

Proof. From Theorem 5.2.7, a continuous function on a closed interval must achieve a maximum and a minimum value. That is, there exist $c, d \in [a, b]$ such that $f(c) \leq f(x) \leq f(d)$ for all $x \in [a, b]$. If $f(c) = f(d)$, then the function f is constant on $[a, b]$ and it follows that $f'(x) = 0$ for all $x \in (a, b)$. Thus we may take t to be any point in (a, b). Otherwise, $f(c) < f(d)$. Then, since $f(a) = f(b)$, we must have either $c \in (a, b)$ or $d \in (a, b)$ (oftentimes both belong to the open interval). Let us assume $d \in (a, b)$ as the other case is similar. Since f has a local maximum at d and f is differentiable at d, Theorem 6.4.2 gives $f'(d) = 0$ and we may take $t = d \in (a, b)$. □

Using Rolle's theorem as a stepping stone, we now state and prove the (more general) mean value theorem.

Theorem 6.5.2 (Mean value theorem)**.** *Let f be a real function which is continuous on a closed interval $[a, b]$ and differentiable on the open interval (a, b). Then there exists t with $a < t < b$ such that $f'(t) = \frac{f(b)-f(a)}{b-a}$.*

Geometrically (see Figure 6.1) the mean value theorem expresses the fact that there exists a point t between a and b such that the tangent line at t is parallel to the secant line connecting the points $(a, f(a))$ and $(b, f(b))$. Note that if we rotate the graph to make the x-axis parallel to this secant line, we are reduced to the case of Rolle's theorem. We did not study how rotating the axes affects derivatives, but we can instead subtract from f any straight line function parallel to the secant and achieve the same reduction. This is what we do below.

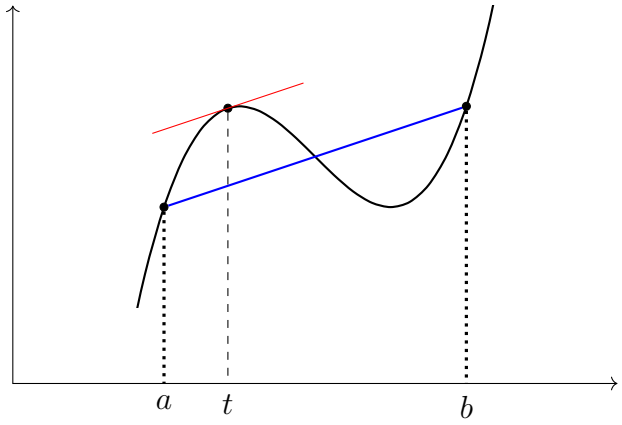


Figure 6.1: Mean value theorem (TikZ code adapted from tex.stackexchange.com).

Proof. Put $m := \frac{f(b)-f(a)}{b-a}$ for brevity (this is the slope of the secant line in Figure 6.1). Let $\ell : \mathbb{R} \to \mathbb{R}$ be the unique straight line function which takes the same values as f at a and b. Explicitly, $\ell(x) = m(x-a) + f(a)$ for all $x \in \mathbb{R}$. Put $g = f - \ell$ so that $g(a) = g(b) = 0$. By Rolle's theorem, there exists $t \in (a,b)$ such that $g'(t) = 0$. Since $g'(t) = f'(t) - \ell'(t) = f'(t) - m$, we get $f'(t) = m$ as desired. □

Remark 6.5.3. Although it does occasionally occur that f is differentiable on the open interval (a,b) while only being continuous at the endpoints a and b, in practice it is more likely that f is differentiable on the whole closed interval $[a,b]$. For a simple example of a function to which only the more general statement applies, consider $f : [-1,1] \to \mathbb{R}$ defined by $f(x) = \sqrt{1-x^2}$. Then, f is differentiable in the interior of its interval of definition but only continuous at its endpoints where one has $f(-1) = f(1) = 0$. Indeed, as assured by the mean value theorem (or indeed Rolle's theorem) one has $f'(0) = 0$ where $-1 < 0 < 1$.

Exercise 6.5.4. Suppose the police have two radar controls at a highway, 20 km apart. The speed limit is 110 km/h. They measure a truck going through the first control at 11:11am at 105 km/h, and going through the second control at 11:20am at 107 km/h. They issue a speeding ticket. Why?

Proof. Note that it took at most 10 minutes (could be more than 9 minutes!) for the truck to travel between the two controls. Therefore, the average speed between the two controls was at least

$$s \geq \frac{20 \text{ km}}{10 \text{ min}} = \frac{20 \text{ km}}{1/6 \text{ h}} = 120 \text{ km/h}\,.$$

By the mean value theorem, there exists a point between the controls where the truck was going at least 120 km/h.

□

Exercise 6.5.5. Let f be a real function such that f is differentiable on $[a,b]$ and f' is continuous on $[a,b]$. Prove that for each $\epsilon > 0$ there exists some $\delta > 0$ such that for all

$x, y \in [a, b]$ with $0 < |x - y| < \delta$ we have

$$\left| \frac{f(x) - f(y)}{x - y} - f'(y) \right| < \epsilon \, .$$

Remark 6.5.6. Note that the definition of the derivative implies that for each $y \in [a, b]$ and each $\epsilon > 0$ there exists some $\delta > 0$, **which depends on both y and ϵ**, such that for all $x \in [a, b]$ with $0 < |x - y| < \delta$ we have $\left| \frac{f(x)-f(y)}{x-y} - f'(y) \right| < \epsilon$. The above problem shows that, if f' is continuous on $[a, b]$, this δ can be chosen to depend only on ϵ. This is a form of "uniform differentiability". Our solution will be a simple application of the mean value theorem. It is difficult to solve the problem without this tool.

Proof. Let $\epsilon > 0$. Since f' is continuous on $[a, b]$, it is uniformly continuous on $[a, b]$ by Theorem 5.5.4. Therefore, there exists some $\delta > 0$ so that for all $x, y \in [a, b]$ with $|x - y| < \delta$ we have

$$|f'(x) - f'(y)| < \epsilon \, .$$

Let $x, y \in [a, b]$ with $0 < |x - y| < \delta$ be arbitrary. By the mean value theorem there exists some t between x and y so that

$$\frac{f(x) - f(y)}{x - y} = f'(t) \, .$$

Note that since t is between x and y we have

$$|t - y| \leq |x - y| < \delta \Longrightarrow |f'(t) - f'(y)| < \epsilon \, .$$

Therefore

$$\left| \frac{f(x) - f(y)}{x - y} - f'(y) \right| = |f'(t) - f'(y)| < \epsilon \, .$$

□

We now briefly discuss the following generalization of the mean value theorem:

Theorem 6.5.7 (Cauchy's mean value theorem)**.** *Let f, g be real functions satisfying the following:*

- *f, g are continuous on $[a, b]$.*
- *f, g are differentiable on (a, b).*
- *$g(a) \neq g(b)$.*

Then, there exists some $t \in (a, b)$ such that

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(t)}{g'(t)} \, .$$

Remark 6.5.8. This theorem has the following famous wrong proof: By the usual mean value theorem (Theorem 6.5.2) there exists $t \in (a, b)$ such that

$$\frac{f(b) - f(a)}{b - a} = f'(t)\,.$$

Again by the mean value theorem, there exists $t \in (a, b)$ so that

$$\frac{g(b) - g(a)}{b - a} = g'(t)\,.$$

Since $0 \neq \frac{g(b)-g(a)}{b-a} = g'(t)$ we can divide the two equations and get the desired result.

Of course, the issue with the above "proof" is that we abuse notation and denote by the same letter t two potentially different quantities. Let us now give a correct proof of Theorem 6.5.7.

Proof. Define $h(x) = [f(b) - f(a)]g(x) - [g(b) - g(a)]f(x)$. The given conditions imply that h is continuous on $[a, b]$ and differentiable on (a, b). Moreover,

$$h(b) - h(a) = [f(b) - f(a)][g(b) - g(a)] - [g(b) - g(a)][f(b) - f(a)] = 0\,.$$

By Rolle's theorem (Theorem 6.5.1), there exists some $t \in (a, b)$ such that

$$0 = h'(t) = [f(b) - f(a)]g'(t) - [g(b) - g(a)]f'(t)\,.$$

The desired claim follows. □

Remark 6.5.9. One nice application of Cauchy's generalized version of the mean value theorem is l'Hôpital's rule. Try using Theorem 6.5.7 to prove at least one of the versions of l'Hôpital's rule.

6.6 Applications of the mean value theorem

We will now use the mean value theorem to rigorously establish various familiar facts concerning the connection between the sign of the derivative and the increasing/decreasing behaviour of a real function. As a warm-up, we show that a function whose derivative vanishes on an interval is constant on that interval. Of course, the converse statement holds as well: constant functions have zero derivative.

Theorem 6.6.1. *Let f be a real function that is differentiable on an interval $I \subseteq \mathbb{R}$. If $f'(x) = 0$ for all $x \in I$ then f is constant on I.*

Proof. If f is not constant, then there exist $x, y \in I$ with $x < y$ such that $f(x) \neq f(y)$. The mean value theorem then implies there exists $t \in (x, y)$ such that $f'(t) = \frac{f(y)-f(x)}{y-x} \neq 0$. Since I is an interval, $t \in I$ and we have reached a contradiction. □

Next, let us define precisely what we mean when we say that a real function is (strictly) increasing/decreasing. Note the close parallels with the corresponding terminology used for sequences (Definition 2.6.7).

Definition 6.6.2. Let f be a real function.

- If $x < y$ implies $f(x) < f(y)$, we say f is **strictly increasing**.
- If $x < y$ implies $f(x) \leq f(y)$, we say f is **increasing**.
- If $x < y$ implies $f(x) > f(y)$, we say f is **strictly decreasing**.
- If $x < y$ implies $f(x) \geq f(y)$, we say f is **decreasing**.

Remark 6.6.3.

- f is (strictly) decreasing if and only its negative $-f$ is (strictly) increasing.
- If f is strictly increasing or strictly decreasing, then f is one-to-one. This is one strategy for proving that a given function is one-to-one.

Having warmed up with Theorem 6.6.1, we now prove in the same spirit a theorem connecting the increasing/decreasing behaviour of a function to the sign of its derivative.

Theorem 6.6.4. *Suppose f is a real function that is defined and differentiable on an interval $I \subseteq \mathbb{R}$. Then, the following statements hold:*

(a) If f is increasing on I then $f' \geq 0$ on I.

(b) If f is decreasing on I then $f' \leq 0$ on I.

(c) If $f' > 0$ on I then f is strictly increasing on I.

(d) If $f' < 0$ on I then f is strictly decreasing on I.

(e) If $f' \geq 0$ on I then f is increasing on I.

(f) If $f' \leq 0$ on I then f is decreasing on I.

Proof. Note that, by the first part of Remark 6.6.3, we have:

$$(a) \iff (b) \qquad (c) \iff (d) \qquad (e) \iff (f).$$

Accordingly, we need only prove one of each pair of implications. For (a), fix $c \in I$. Then,

$$f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c} = \lim_{x \to c+} \frac{\overbrace{f(x) - f(c)}^{\geq 0}}{\underbrace{x - c}_{>0}} \geq 0\,.$$

For (c), suppose $x < y$ belong to I. By the mean value theorem, there exists t satisfying $x < t < y$ such that

$$\frac{f(y) - f(x)}{y - x} = f'(t) > 0\,.$$

Since $y - x > 0$, we get $f(y) - f(x) > 0$.

Similarly, for (e), suppose $x < y$ belong to I. By the mean value theorem, there exists t satisfying $x < t < y$ such that

$$\frac{f(y) - f(x)}{y - x} = f'(t) \geq 0\,.$$

Since $y - x > 0$, we get $f(y) - f(x) \geq 0$. □

Remark 6.6.5. Notice that a differentiable function $f : \mathbb{R} \to \mathbb{R}$ can be strictly increasing without having a strictly positive derivative at all points, as the example $f(x) = x^3$ shows.

Example 6.6.6. Suppose $f : \mathbb{R} \to \mathbb{R}$ is a differentiable function such that $f'(x) \geq 0$ for all x and $f'(x) = 0$ only when $x = 0$ (for instance, $f(x) = x^5 + x^3$ is such a function). Carefully use Theorem 6.6.4 to show f is strictly increasing.

Proof. We must exploit the fact that f' doesn't have too many zeroes. Let $x_1 < x_2$. We split the problem into cases.

Case 1: $0 \notin [x_1, x_2]$. Then, $f' > 0$ on $[x_1, x_2]$ and hence f is strictly increasing on $[x_1, x_2]$. Thus $f(x_1) < f(x_2)$.

Case 2: $0 \in [x_1, x_2]$. We could have $x_1 = 0$, or $x_2 = 0$, or $x_1 < 0 < x_2$. Rather than split this case into three subcases, we cover all three at once. Note that in all three situations there exists some a, b so that $x_1 < a < b < x_2$ and $0 \notin [a, b]$. Then $f' > 0$ on $[a, b]$ implies that f is strictly increasing on $[a, b]$. Thus

$$f(a) < f(b)\,.$$

Also, $f' \geq 0$ on $[x_1, a]$ and $[b, x_2]$ implies that f is increasing on those intervals. Therefore

$$f(x_1) \leq f(a) \qquad\qquad f(b) \leq f(x_2)\,.$$

Putting everything together:

$$f(x_1) \leq f(a) < f(b) \leq f(x_2)\,.$$

□

We close out this section by studying the following question:

Question 6.6.7. Let f be a differentiable function. Suppose $f'(c) > 0$. Must there be a small open interval J containing c such that f is increasing on J?

While intuitively the answer may appear to be "yes", it is actually "no", as Example 6.6.8 below will show. We will also show that the answer becomes "yes" under the extra assumption that f' is continuous at c.

Example 6.6.8. Let

$$f(x) = \begin{cases} \frac{x}{4} + x^2 \sin(\frac{1}{x}) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}.$$

Then, f is differentiable on $\mathbb{R}$ and $f'(0) > 0$, but for any open interval (a, b) containing 0 the function f is not increasing on (a, b).

Proof. Using familiar facts about trig functions and their derivatives, f is differentiable on $(-\infty, 0)$ and $(0, \infty)$. Moreover, its derivative is given by $f'(x) = \frac{1}{4} + 2x \sin(x) - \cos(\frac{1}{x})$ on those intervals. Also, by a standard application of the squeeze theorem,

$$\lim_{x \to 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \to 0} \tfrac{1}{4} + x \sin(\tfrac{1}{x}) = \tfrac{1}{4},$$

so f is differentiable at $x = 0$ too. In summary, f is differentiable on all of $\mathbb{R}$ and

$$f'(x) = \begin{cases} \frac{1}{4} + 2x \sin(\frac{1}{x}) - \cos(\frac{1}{x}) & \text{if } x \neq 0 \\ \frac{1}{4} & \text{if } x = 0 \end{cases}.$$

In particular, $f'(0) > 0$.

Suppose now that $a < 0 < b$. Pick some positive integer n such that

$$c \coloneqq \frac{1}{2\pi n} \in (0, b)$$

and observe that

$$f'(c) = \tfrac{1}{4} + \tfrac{1}{\pi n} \sin(2\pi n) - \cos(2\pi n) = -\tfrac{3}{4}.$$

Since $f'(c) < 0$ and f' is continuous at $c \in (0, b)$ we can find (why?) some interval $J \subseteq (0, b)$ containing c such that $f'(x) < 0$ for all $x \in J$. But then f is decreasing on $J \subseteq (a, b)$ by Theorem 6.6.4 and hence f is not increasing on (a, b). □

Remark 6.6.9. Example 6.6.8 also provides an example of a function f which is differentiable but whose derivative is **not** continuous.

Example 6.6.10. Let f be a differentiable function. Suppose c is a point in the domain of f such that $f'(c) > 0$ and that f' is continuous at c. Then there exists an open interval J containing c such that f is increasing on J.

Proof. Since $f'(c) > 0$ and f' is continuous at c there exists (why?) some open interval J containing c such that $f'(x) > 0$ for all $x \in J$. Then, f is increasing on J by Theorem 6.6.4. □

6.7 The intermediate value property of the derivative

By now we have learned by studying examples that the derivative f' of a differentiable function f may fail to be a continuous. Even though derivatives are not guaranteed to be continuous, they do always enjoy a weaker property known as the "intermediate value property" or "Darboux continuity". The intermediate value theorem tells us that all continuous functions have this property.

Theorem 6.7.1 (Darboux's theorem)**.** *If f is a differentiable function on an open interval I, its derivative $f' : I \to \mathbb{R}$ has the* ***intermediate value property****: given $a, b \in I$ with $a < b$ and γ any number between $f'(a)$ and $f'(b)$, there exists some $t \in [a, b]$ such that $f'(t) = \gamma$.*

Proof. If $\gamma = f'(a)$ or $\gamma = f'(b)$ the claim is obvious. Otherwise, up to replacing f by $-f$, we can assume without loss of generality that $f'(a) < \gamma < f'(b)$. Put $g(x) = f(x) - \gamma x$ so that $g : I \to \mathbb{R}$ is differentiable and, by design, $g'(a) < 0$ and $g'(b) > 0$. By Theorem 5.2.7, g has an absolute minimum in $[a, b]$. We claim that neither a nor b can be an absolute minimum. Indeed, if we assume to contrary that a is an absolute minimum, then

$$g'(a) = \lim_{x \to a^+} \frac{f(x) - f(a)}{x - a} \geq 0\,.$$

Similarly, if we assume to contrary that b is an absolute minimum, then

$$g'(b) = \lim_{x \to b^-} \frac{f(x) - f(b)}{x - b} \leq 0\,.$$

Therefore, g has an absolute minimum at some $c \in (a, b)$ and, by Theorem 6.4.2,

$$f'(c) - \gamma = g'(c) = 0\,.$$

□

Darboux's theorem says that every function admitting an antiderivative has the intermediate value property. Let us look at an interesting example of such a function. To make the discussion simpler, we "cheat" and use a result that will be proved later.

Example 6.7.2. Let

$$g(x) = \begin{cases} \cos(\frac{1}{x}) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}.$$

Then there exists some function G which is differentiable on $\mathbb{R}$ such that $G' = g$. Notice as well that g has the intermediate value property (indeed this is forced by Theorem 6.7.1).

Proof. In Exercises 6.2.1 and 6.2.2, we looked at the functions $h, F : \mathbb{R} \to \mathbb{R}$ defined by

$$h(x) = \begin{cases} x \sin(\frac{1}{x}) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases} \qquad F(x) = \begin{cases} x^2 \sin(\frac{1}{x}) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}.$$

We determined h is continuous and is F is differentiable with derivative:

$$f(x) = \begin{cases} 2x\sin(\frac{1}{x}) - \cos(\frac{1}{x}) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}.$$

Note the function g we are presently analyzing decomposes as $g = 2h - f$. Later, we will prove a result (Theorem 7.8.6) to the effect that all continuous functions have antiderivatives. Thus, there exists some differentiable function H on $\mathbb{R}$ such that $H' = h$. Then $G \coloneqq 2H - F$ satisfies $G' = (2H - F)' = 2h - f = g$ and we have found an antiderivative for g. □

By slightly modifying the above example, we can also obtain functions which do not admit antiderivatives, but which still have the intermediate value property.

Example 6.7.3. Fix $\lambda \in [-1, 1]$ and define $g_\lambda : \mathbb{R} \to \mathbb{R}$ by

$$g_\lambda(x) = \begin{cases} \cos(\frac{1}{x}) & \text{of } x \neq 0 \\ \lambda & \text{if } x = 0 \end{cases}.$$

Then, g_λ satisfies the intermediate value property for all $\lambda \in [-1, 1]$. However, g_λ has an antiderivative if and only if $\lambda = 0$.

Proof. First we prove that g_λ has the intermediate value property for every $\lambda \in [-1, 1]$. Fix $a, b \in \mathbb{R}$ with $a < b$. Let γ be a number between $g_\lambda(a)$ and $g_\lambda(b)$.

- *Case 1:* $0 \notin [a, b]$. Then g_λ is continuous on $[a, b]$ and the intermediate value theorem (Theorem 5.3.1) implies the existence of $t \in [a, b]$ such that $g_\lambda(t) = \gamma$.

If $0 \in [a, b]$, we are in one of two overlapping cases treated below. First, note that $\gamma \in [-1, 1]$ must occur simply because $[-1, 1]$ is the range of g_λ. Now, for every $n \in \mathbb{Z}$, note the cosine function maps $[n\pi, (n+1)\pi]$ bijectively onto $[-1, 1]$. Accordingly, for any positive integer n, g_λ maps $\left[\frac{1}{(n+1)\pi}, \frac{1}{n\pi}\right]$ bijectively onto $[-1, 1]$. Since the endpoints of $\left[\frac{1}{(n+1)\pi}, \frac{1}{n\pi}\right]$ converge to 0 as $n \to \infty$, it follows that the image of $(0, \epsilon)$ under g_λ is $[-1, 1]$ for any $\epsilon > 0$. Similarly, for any $\epsilon > 0$, the image of $(-\epsilon, 0)$ under g_λ is $[-1, 1]$ as well.

- *Case 2:* $a \leq 0 < b$. Then, since the image of $(0, b)$ under g_λ is $[-1, 1]$, we can choose t with $a \leq 0 < t < b$ such that $g(t) = \gamma$.

- *Case 3:* $a < 0 \leq b$. Then, since the image of $(a, 0)$ under g_λ is $[-1, 1]$, we can choose t with $a < t < 0 \leq b$ such that $g(t) = \gamma$.

Now we prove that g_λ has an antiderivative if and only if $\lambda = 0$. From Example 6.7.2, we know that g_0 has an antiderivative G_0. Conversely, suppose that g_λ has an antiderivative G_λ. Then $G_\lambda - G_0$ is an antiderivative for $g_\lambda - g_0$ so, by Darboux's theorem, $g_\lambda - g_0$ has the intermediate value property. However, it is obvious that

$$(g_\lambda - g_0)(x) = \begin{cases} 0 & \text{if } x \neq 0 \\ \lambda & \text{if } x = 0 \end{cases}$$

only has the intermediate value property if $\lambda = 0$. □

6.8 Taylor Polynomials

Given a real function f and a basepoint $c \in \operatorname{dom} f$, the following question is important:

Question 6.8.1. Does there exist a sequence of real numbers $a_0, a_1, a_2, \ldots$ and an open interval I containing c such that for all $x \in I$, one can express f by the convergent series:

$$f(x) = \sum_{n=0}^{\infty} a_n (x-c)^n \, ?$$

Informally, can f be "expanded" into an infinitely long polynomial near c?

The precise name for such a series representation is a **power series** and, given more time, we would study them in detail. Instead, let us simply assert that most familiar functions admit power series expansions and these representations are extremely useful for both theoretical and practical reasons. Several examples are given below, taking $c = 0$ as the basepoint for simplicity:

$$\begin{aligned}
e^x &= \sum_{n=0}^{\infty} \frac{1}{n!} x^n && x \in \mathbb{R} \\
\cos(x) &= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} && x \in \mathbb{R} \\
\sin(x) &= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} && x \in \mathbb{R} \\
\arctan(x) &= \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1} && x \in (-1,1) \\
\frac{1}{1-x} &= \sum_{n=0}^{\infty} x^n && x \in (-1,1) \\
\sqrt{1+x} &= \sum_{n=0}^{\infty} \binom{1/2}{n} x^n && x \in (-1,1).
\end{aligned}$$

In general, if the answer to Question 6.8.1 is "yes", the function f is said to be **real analytic** on the interval I. In this case, it can be shown that f is differentiable infinitely-many times on I and that

$$a_n = \frac{f^{(n)}(c)}{n!} \, .$$

Thus, the only power series that can approximate f in this way is its Taylor series:

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!} (x-c)^n = f(c) + \frac{f'(c)}{1!}(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \ldots + \frac{f^{(n)}(c)}{n!}(x-c)^n + \ldots$$

In other words, Question 6.8.1 above reduces to the following:

Question 6.8.2. Suppose that a real function f is differentiable infinitely-many times on an open interval I containing c. Is the following expansion valid for all $x \in I$?

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!}(x-c)^n$$

The meaning of this is that the partial sum

$$P_n(x) \coloneqq f(c) + \frac{f'(c)}{1!}(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \ldots + \frac{f^{(n)}(c)}{n!}(x-c)^n$$

converges to $f(x)$ as $n \to \infty$ for all $x \in I$ or, equivalently, that

$$R_n(x) \coloneqq f(x) - P_n(x)$$

converges to 0 as $n \to \infty$ for all $x \in I$.

Definition 6.8.3. Fix a real function f and a basepoint $c \in \operatorname{dom} f$. Assume f is n times differentiable at c. The functions

$$P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(c)}{k!}(x-c)^k = f(c) + \frac{f'(c)}{1!}(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \ldots + \frac{f^{(n)}(c)}{n!}(x-c)^n$$
$$R_n(x) = f(x) - P_n(x)$$

are called the **nth Taylor polynomial** and the **nth remainder** of f at c.

The main virtue of the definition of the Taylor polynomial given above is that it is an explicit formula. A formula alone can be a rather uninformative thing, however. The exercises below explore some important equivalent definitions of $P_n(x)$ in terms of its more essential properties. Induction is recommended for both exercises.

Exercise 6.8.4. Suppose that f is n times differentiable at c. Prove that the Taylor polynomial $P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(c)}{k!}$ is the unique polynomial of degree $\leq n$ whose value and first n derivatives at c match those of f, i.e. $P^{(k)}(c) = f^{(k)}(c)$ for $k = 0, 1, \ldots, n$.

Exercise 6.8.5. Suppose f is a real function and $c \in \operatorname{dom} f$ is an interior point. Prove that there is at most one polynomial $P(x)$ of degree $\leq n$ such that $P(c) = f(c)$ and

$$\lim_{x \to c} \frac{f(x) - P(x)}{(x-c)^n} = 0.$$

Prove that, if f is n times differentiable at c, then the Taylor polynomial $P_n(x)$ of f at c fulfills these conditions.

Remark 6.8.6. Using "little o notation" (cf. Remark 6.3.8) one may write

$$f(x) = P(x) + o((x-c)^n)$$

to express more compactly that $P(c) = f(c)$ and $\lim_{x \to c} \frac{f(x)-P(x)}{(x-c)^n} = 0$.

Example 6.8.7. Here is a somewhat technical point: having an nth order polynomial approximation for f at c of the type in Exercise 6.8.5 does not force nth order differentiability of f at c. Indeed, consider

$$f : \mathbb{R} \to \mathbb{R} \qquad\qquad f(x) = \begin{cases} 0 & x \in \mathbb{Q} \\ e^{-\frac{1}{x^2}} & x \notin \mathbb{Q}\,. \end{cases}$$

While f is in fact differentiable at 0, its second and higher derivatives do not exist at 0 because f is not even continuous away from 0. Nonetheless, one does have $f(x) = o(x^n)$ for every positive integer n so that the zero polynomial may be considered an nth order polynomial approximation to f at 0 for every n. However, one would typically not use the word "Taylor polynomial" here due to the lack of higher differentiability.

Because Taylor expansions are so useful, it would be great if they always converged to f. Unfortunately, this is not always true, as the following important exercise illustrates.

Exercise 6.8.8. Define $f : \mathbb{R} \to \mathbb{R}$ by:

$$f(x) = \begin{cases} e^{-\frac{1}{x^2}} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0\,. \end{cases}$$

(a) Show that for each positive integer n there exists some polynomial S_n such that

$$f^{(n)}(x) = \frac{S_n(x)}{x^{3n}} e^{-\frac{1}{x^2}} \qquad\qquad x \neq 0.$$

(b) Show $f(x)$ is infinitely-many times differentiable on $\mathbb{R}$ and $f^{(n)}(0) = 0$ for all $n \in \mathbb{N}$.

(c) Does the Taylor expansion of f at 0 converge to f near 0?

In spite of Exercise 6.8.8, one can show that in the "right circumstances" the Taylor series of f does converge to f near the base point. We prove one possible theorem of this type below (Theorem 6.9.1). Typically, to show convergence of $P_n(x)$ to $f(x)$, one needs an explicit expression or estimate for the remainder $R_n(x)$. There are a wide variety of "remainder theorems" to choose from and all are more-or-less equally suited to this purpose. We will cover Lagrange's expression for the remainder because it is a natural extension of the mean value theorem covered earlier.

Theorem 6.8.9 (Lagrange)**.** *Let f be a real function that is $(n+1)$-times differentiable at and between given points $c, x \in \mathbb{R}$. Consider the nth Taylor polynomial and remainder:*

$$P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(c)}{k!}(x-c)^k \qquad\qquad R_n(x) = f(x) - P_n(x)\,.$$

Then, there exists t between c and x such that

$$R_n(x) = \frac{f^{n+1}(t)}{(n+1)!}(x-c)^{n+1}\,.$$

Remark 6.8.10. When $n = 0$, Lagrange's theorem reduces, essentially, to the mean value theorem. Technically, the statement we gave for the mean value theorem is mildly stronger since it only demanded continuity (as opposed to differentiability) at the endpoints.

Proof. We replace the symbol x in the statement by y in the proof so that x is free to act as a variable. If $y = c$ the result is obvious, so assume $y \neq c$. Assume in fact that $y > c$ as the case $y < c$ can be handled similarly. We will show there exists t such that

$$c < t < y \qquad \text{and} \qquad R_n(y) = \frac{f^{(n+1)}(t)}{(n+1)!}(y-c)^{n+1}.$$

Towards this end, we first prove a lemma analogous to Rolle's theorem:

Lemma. *Let g be $n+1$ times differentiable on $[c, y]$ and vanish at the endpoints:*

$$g(c) = g(y) = 0.$$

Suppose as well that the first n derivatives of g vanish at c:

$$g'(c) = g''(c) = \ldots = g^{(n)}(c) = 0.$$

Then, there exists t with $c < t < y$ such that $g^{(n+1)}(t) = 0$.

Proof. Applying Rolle's theorem to g on $[c, y]$ yields t_1 with $c < t_1 < y$ such that $g'(t_1) = 0$. Applying Rolle's theorem to g' on $[c, t_1]$ yields t_2 with $c < t_2 < t_1$ such that $g''(t_2) = 0$. Repeating this process eventually yields $t \coloneqq t_{n+1}$ with $c < t_{n+1} < t_n < \ldots < t_1 < y$ such that $g^{(n+1)}(t) = 0$. □

It follows from this lemma (why?) that, if an $n+1$ times differentiable function Q on $[c, y]$ has the same values as f at c and y and the same first n derivatives as f at c, i.e.

$$f^{(k)}(c) = Q^{(k)}(c), k = 0, 1 \ldots, n \qquad \text{and} \qquad f(y) = Q(y)\,,$$

then there exist t with $c < t < y$ such that

$$f^{(n+1)}(c) = Q^{(n+1)}(c).$$

The Taylor polynomial $P_n(x)$ has the same value and first n derivatives as f at c (indeed, as explored in Exercise 6.8.4, this is one way to define P_n). Moreover, for any real constant M, the same is true of the polynomial:

$$Q(x) = P_n(x) + M(x-c)^{n+1}.$$

By making the right choice for M, namely $M \coloneqq \frac{f(y)-P_n(y)}{(y-c)^{n+1}}$, we may also arrange to have $Q(y) = f(y)$. Then, as noted above, there exists t with $c < t < y$ such that

$$f^{(n+1)}(t) = Q^{(n+1)}(t) = (n+1)!M$$

and this last equation proves the theorem. □

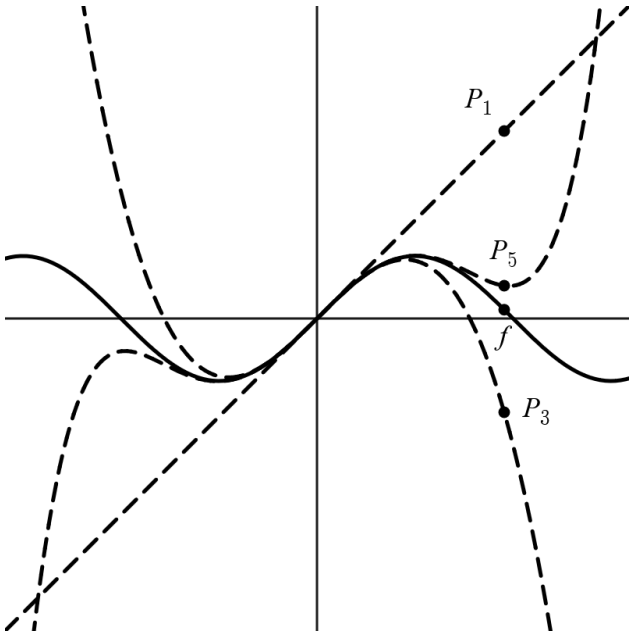


Figure 6.2: $f(x) = \sin(x)$ and its Taylor polynomials $P_n(x)$ at 0 for $n = 1, 3, 5$.

Remark 6.8.11. Suppose $x \neq c$ in the above statement of Lagrange's theorem. Then, the attentive reader will have noticed from the proof that the point t can always be chosen to lie *strictly* between c and x. The *very* attentive reader may also have noticed that the same proof goes through under slightly weaker differentiablility hypotheses on f. It is enough to assume that:

- f is n times differentiable at c and $f^{(n)}$ is continuous at c.
- f is continuous at x.
- f is $n+1$ times differentiable at every point strictly between x and c.

However, it is difficult to imagine any situation where slightly generalizing Theorem 6.8.9 while complicating its statement in this way would be a worthwhile trade-off. This remark is related to Remarks 6.5.3 and 6.8.10 appearing earlier.

Generally speaking (see Figure 6.2) one expects $P_n(x)$ to be a good approximation of $f(x)$ when n is large and $x \approx c$. One can apply the Lagrange remainder theorem to obtain a concrete estimate of how good this approximation is. Let us look at one example of such an estimate.

Example 6.8.12. Consider the infinitely-differentiable function $f : (0, \infty) \to \mathbb{R}$ defined by $f(x) = \frac{1}{x}$. It is easy to show (induction) that the kth derivative of f is given by:

$$f^{(k)}(x) = \frac{(-1)^k \cdot k!}{x^{k+1}}.$$

Thus, the nth Taylor polynomial of f at a given basepoint $c \in (0, \infty)$ is:

$$P_n(x) \coloneqq \sum_{k=0}^{n} \frac{f^{(k)}(c)}{k!}(x-c)^k = \sum_{k=0}^{n} \frac{(-1)^k}{c^{k+1}}(x-c)^k.$$

According to Lagrange's theorem, for every $x \in (0, \infty)$ there exists a point t between c and x such that the remainder $R_n(x) \coloneqq f(x) - P_n(x)$ is given by:

$$R_n(x) = \frac{f^{(n+1)}(t)}{(n+1)!}(x-c)^{n+1} = \frac{(-1)^{n+1}}{t^{n+2}}(x-c)^{n+1}.$$

We can use Lagrange's expression for the remainder to get a concrete estimate of how well $P_n(x)$ approximates $f(x)$ on a given closed interval $[c-r, c+r] \subseteq (0, \infty)$ where $0 < r < c$. Indeed:

- Since t is between x and c, we have $x \in [c-r, c+r] \Rightarrow t \in [c-r, c+r] \Rightarrow \frac{1}{t^{n+2}} \leq \frac{1}{(c-r)^{n+2}}$.
- Since $x \in [c-r, c+r]$ we have $|x-c|^{n+1} \leq r^{n+1}$.
- Combining the above two observations, we have:

$$|R_n(x)| = \frac{1}{t^{n+2}}|x-c|^{n+1} \leq \frac{r^{n+1}}{(c-r)^{n+2}} \text{ for all } x \in [c-r, c+r].$$

Consider in particular the case $n = 1$, $c = 100$, $r = 1$. The first order Taylor polynomial (also known as the best linear approximation) of $f(x) = \frac{1}{x}$ at $c = 100$ is:

$$P_1(x) = f(c) + f'(c)(x-c) = \frac{1}{100} - \frac{1}{10000}(x-100).$$

From the discussion above, we get that the remainder $R_1(x) = f(x) - P_1(x)$ satisfies the bound:

$$|R_1(x)| = \frac{1}{t^3}|x-c|^2 \leq \frac{r^2}{(c-r)^3} = \frac{1}{99^3} \approx 0.00000103 \text{ for all } x \in [99, 101].$$

Indeed, one can calculate that the true maximum of $|R_1(x)|$ on $[99, 101]$ is $\frac{1}{990000} \approx 0.00000101$ which is only slightly smaller than our estimate.

The following exercise is a good tool to assess one's understanding of the proof of Lagrange's theorem:

Exercise 6.8.13. Suppose f is 5 times differentiable on $[a, b]$ and satisfies:

$$f(a) = f'(a) = f^{(3)}(a) = 0 \qquad\qquad f(b) = f''(b) = f^{(4)}(b) = 0.$$

Prove that there exists $t \in (a, b)$ such that $f^{(5)}(t) = 0$.

6.9 Conditions for analyticity

Since we do not have time to make an organized study of power series in this course, the following section should be considered as supplemental material. By leveraging Lagrange's remainder theorem (or indeed any of a number of closely-related remainder theorems), one can prove that, as long as the derivatives of f do not grow "too fast" near a chosen basepoint, the Taylor series of f will converge to f near the basepoint. Here is one possible theorem of this type:

Theorem 6.9.1. *Let f be infinitely differentiable on an interval I and define:*

$$A_n \coloneqq \sup_{t \in I} \left| \frac{f^{(n)}(t)}{n!} \right| \qquad\qquad \rho \coloneqq \frac{1}{\limsup_{n\to\infty} A_n^{1/n}} \in [0, \infty].$$

Then, for all $c, x \in I$ satisfying $|x - c| < \rho$, the Taylor series

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!}(x-c)^n$$

converges to $f(x)$.

Proof. With no harm done, assume $\rho > 0$ or else there is nothing to prove. Suppose $c, x \in I$ satisfy $|x - c| < \rho$. Choose r, s satisfying $|x - c| < r < s < \rho$. Since $\frac{1}{s} > \limsup_{n\to\infty} A_n^{1/n}$, there exists N such that $n \geq N$ implies $A_n^{1/n} < \frac{1}{s}$. Thus, $\left|\frac{f^{(n)}(t)}{n!}\right| < \frac{1}{s^n}$ for all $t \in I$, $n \geq N$. By Lagrange's theorem, the nth remainder $R_n(x) = f(x) - \sum_{k=0}^{n} \frac{f^{(k)}(c)}{k!}(x-c)^k$ satisfies $R_n(x) = \frac{f^{(n+1)}(t)}{(n+1)!}(x-c)^{n+1}$ for some t between c and x. Since $t \in I$, we have that, for $n + 1 \geq N$,

$$|R_n(x)| = \left| \frac{f^{(n+1)}(t)}{(n+1)!} \right| |x-c|^{n+1} < \frac{1}{s^{n+1}} r^{n+1} = (r/s)^{n+1}.$$

Since $r/s < 1$, the above tends to 0 as $n \to \infty$. □

Example 6.9.2. Let $f : \mathbb{R} \to \mathbb{R}$ be the exponential function $f(x) = e^x$. Crucially, f has the property that $f = f'$. Thus, all higher derivatives of f are equal to f as well. We shall see that:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!}(x-c)^n = \sum_{n=0}^{\infty} \frac{e^c}{n!}(x-c)^n \qquad\qquad \text{for all } c, x \in \mathbb{R}.$$

That is, every Taylor series of f converges everywhere. Restricting attention to an interval $I = [-r, r]$ where $r > 0$ is arbitrary, we have:

$$A_n \coloneqq \sup_{|t| \leq r} \frac{f^{(n)}(t)}{n!} = \frac{e^r}{n!}.$$

One may check (exercise) that $(n!)^2 \geq n^n$ from which it follows that $(n!)^{1/n} \geq \sqrt{n}$. Thus,

$$\limsup_{n\to\infty} A_n^{1/n} \leq \limsup_{n\to\infty} \frac{e^{r/n}}{\sqrt{n}} = 0$$

and we obtain

$$\rho \coloneqq \frac{1}{\limsup_{n\to\infty} A_n^{1/n}} = \infty.$$

Thus, Theorem 6.9.1 says that, for all $c, x \in [-r, r]$, the Taylor series of f at c converges to $f(x)$. Since $r > 0$ was arbitrary, we indeed have that $P_n(x) \to f(x)$ for all $c, x \in \mathbb{R}$.

Remark 6.9.3. The above example is a bit misleading. Earlier, in Example 2.9.3, we discussed the limit $e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n$. This classical limit affords us one possible definition for the constant e and so, eventually, the exponential function $x \mapsto e^x$. In principle, one could alternatively define the exponential function as the unique solution to the ordinary differential equation $f'(x) = f(x)$, $f(0) = 1$, but this still leaves the problem of existence. Ultimately, it is much more common and much more useful to construct the exponential function (and in particular the constant $e = e^1$) by way of its power series $e^x = \sum_{n=0}^{\infty} \frac{1}{n!} x^n$. If this series is being used to define the exponential function, one may quite legitimately complain that Example 6.9.2 is circular.

We conclude this supplementary section by proving that Theorem 6.9.1 above is, in a sense, the best sort of result one could hope for. If f is analytic at a point c then, near c, the derivatives of f necessarily satisfy a growth condition of the type in Theorem 6.9.1. Some familiarity with the theory of power series is assumed of the reader. In particular:

- We utilize the radius of convergence R of a power series $\sum_{n=0}^{\infty} a_n x^n$, which is given explicitly by the Cauchy-Hadamard formula:

$$\frac{1}{R} = \limsup_{n\to\infty} |a_n|^{1/n}.$$

- We assume it makes sense to differentiate a power series "term-by-term".

- We use the expansion

$$\left(\frac{1}{1-x}\right)^{n+1} = \sum_{k=0}^{\infty} \binom{n+k}{k} x^k \qquad\qquad |x| < 1.$$

  Formally, the above can be obtained by starting with the geometric series

$$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k \qquad\qquad |x| < 1,$$

  raising both sides to the power $n+1$, expanding the right hand side, and collecting like powers of x. These formal manipulations are rigorously justified by the Cauchy product formula.

Theorem 6.9.4. *Let $\sum_{n=0}^{\infty} a_n x^n$ be a power series whose radius of convergence R is positive. Define a (real analytic) function $f : (-R, R) \to \mathbb{R}$ by $f(x) = \sum_{n=0}^{\infty} a_n x^n$. Fix r with $0 < r < R$ and define*

$$A_n := \sup_{|t| \leq r} \frac{f^{(n)}(t)}{n!}.$$

Then,

$$\limsup_{n\to\infty} A_n^{1/n} \leq \frac{1}{R-r}.$$

Moreover, if $c, x \in [-r, r]$ and $|x - c| < R - r$, then

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!}(x-c)^n.$$

Proof. Choose s such that $r < s < R$. Since $1/s > 1/R = \limsup |a_n|^{1/n}$, there exists N such that $|a_k|^{1/k} \leq 1/s$ for all $k \geq N$. That is, $|a_k| \leq 1/s^k$ for all $k \geq N$. Now, for any $t \in (-R, R)$, we have

$$\frac{f^{(n)}(t)}{n!} = \frac{1}{n!} \sum_{k=n}^{\infty} k(k-1)\cdots(k-n+1) a_k t^{k-n} = \sum_{k=0}^{\infty} \binom{n+k}{n} a_{n+k} t^k.$$

Thus, if $|t| \leq r$ and $n \geq N$ so that $|a_{n+k}| \leq 1/s^{n+k}$ for all $k \geq 0$, we have

$$\left|\frac{f^{(n)}(t)}{n!}\right| \leq \sum_{k=0}^{\infty} \binom{n+k}{n} \frac{1}{s^{n+k}} r^k = \frac{1}{s^n}\left(\frac{1}{1-r/s}\right)^{n+1} = \frac{s}{s-r}\left(\frac{1}{s-r}\right)^n.$$

Thus,

$$A_n := \max_{|t|\leq r} \left|\frac{f^{(n)}(t)}{n!}\right| \leq \frac{s}{s-r}\left(\frac{1}{s-r}\right)^n$$

for all $n \geq N$. From this bound, we obtain

$$\limsup_{n\to\infty} A_n^{1/n} \leq \limsup_{n\to\infty} \left(\frac{s}{s-r}\right)^{1/n} \frac{1}{s-r} = \frac{1}{s-r}.$$

Since the above holds whenever $r < s < R$, we may now take the limit as s approaches R from below to obtain the stronger bound:

$$\limsup_{n\to\infty} A_n^{1/n} \leq \frac{1}{R-r}$$

which we set out to prove.

The "moreover" statement follows directly from Theorem 6.9.1. □

Chapter 7

Integral calculus

7.1 Overview

In introductory calculus, one learns that the Riemann integral $\int_a^b f(x)\ dx$ of a function $f : [a, b] \to \mathbb{R}$ is defined as the limit of Riemann sums. Subdivide $[a, b]$ into small intervals $[x_{i-1}, x_i]$ where $a = x_0 < x_1 < \ldots < x_n = b$ and choose a sample input $t_i \in [x_{i-1}, x_i]$ for $i = 1, \ldots, n$. The associated Riemann sum is:

$$\sum_{i=1}^{n} f(t_i)(x_i - x_{i-1}).$$

Roughly speaking, the Riemann integral is defined as:

$$\int_a^b f(x)\ dx \coloneqq \lim \sum_{i=1}^{n} f(t_i)(x_i - x_{i-1})$$

where the limit refers to the limiting value (if it exists) of the Riemann sums as the subdivision intervals $[x_{i-1}, x_i]$ approach 0 in length.

We associate the name of Bernhard Riemann (1826-1866) with this integral because of its appearance in his 1853 Habilitationsschrift thesis (a sort of second PhD thesis required to assume a professorship in Germany). However, in spite of the terminologies "Riemann integral" and "Riemann sum", this definition for the definite integral $\int_a^b f(x)\ dx$ was in use long before the work of Riemann. Mathematicians had been integrating functions essentially since the invention of calculus. Cauchy in particular used the same approach as Riemann to define definite integrals of continuous functions. What Riemann contributed were results and examples that clarified which kinds of functions can be integrated using this definition. The class of Riemann integrable functions is much larger than the class of continuous functions, but how much larger was less well understood before Riemann's work.

There is an equivalent approach to the Riemann integral introduced by the mathematican Gaston Darboux (1842–1917) which is based on the concepts of upper and lower sums instead of Riemann sums. Because the Darboux and Riemann approaches to the definite integral are

equivalent, it is very common to refer to both of them as the "Riemann integral". Indeed, that is what I did in class. In these notes, I will try to use separate terminologies "Darboux integral" and "Riemann integral" to keep the concepts straight, at least until we have proven the two definitions to be equivalent. However, to reiterate, it is common to use these terms interchangeably and the student is welcome to do so.

As a final note, the trend established by Riemann of trying to broaden the class of functions which can be integrated as much as possible turned out to be a very important one. As the subject of analysis developed further, it became necessary to have definitions that worked for more and more general classes of functions. The most significant development along these lines was the invention of the Lebesgue integral and the associated subject of measure theory. Every Riemann integrable function is also Lebesgue integrable and the Lebesgue integral has much better properties with respect to taking limits. For these reasons, the Riemann integral has essentially been replaced by the Lebesgue integral in more advanced analysis.

7.2 Partitions of intervals

In the context of Riemann integration, the word "partition" is used in a way that's a bit different from the way it is used when we talk about partitioning a set.

Definition 7.2.1. A **partition** P of a closed interval $[a, b]$ is a finite set $P \subseteq [a, b]$ with $a, b \in P$. When we list the points of $P = \{x_0, x_1, x_2, \ldots, x_n\}$, we will always assume the indexing respects the order of the points, i.e. $a = x_0 < x_1 < x_2 < \ldots < x_n = b$.

Remark 7.2.2. We think of a partition $P = \{x_0, x_1, \ldots, x_n\}$ of $[a, b]$ as determining a subdivision of $[a, b]$ into n subintervals $[x_{i-1}, x_i]$ for $i = 1, \ldots, n$. Note that the decomposition

$$[a, b] = [x_0, x_1] \cup [x_1, x_2] \cup \ldots \cup [x_{n-1}, x_n]$$

is not a partition in the usual sense of set theory because the intervals are not quite disjoint (they overlap at endpoints). Note as well that due to the usual "fencepost problem", the number of points in the partition is one greater than the number of intervals.

Example 7.2.3. The simplest strategy for partitioning an interval $[a, b]$ is to divide it into n subintervals of equal lengths. In more detail, given $n \in \mathbb{N}$, we put $\Delta x := \frac{b-a}{n}$ and $x_i := a + i \cdot \Delta x$ for $i = 0, \ldots, n$, Then $P = \{x_0, x_1, \ldots, x_n\}$ is a partition of $[a, b]$ and each interval $[x_{i-1}, x_i]$, $i = 1, \ldots, n$ has length equal to Δx. In general, there is no requirement for the spacing between the consecutive points of partition to be constant. So, for example $P = \{1, 2, 3, 8, 9.9, 9.999, 10\}$ is a legitimate partition of $[1, 10]$.

Definition 7.2.4. If P and Q are partitions of $[a, b]$, we say that Q is a **refinement** of P if $P \subseteq Q$.

Remark 7.2.5. In general, given two partitions P, Q, it is possible that neither one of them is a refinement of the other. However, there is always a simple way to produce a third partition which is a refinement of both of them, namely by considering their union $P \cup Q$.

The relationship of refinement gives a way of quantifying how fine a partition P is in a relative sense, by comparing it to other partitions. There is also a way of quantifying how fine P is in an absolute sense, by considering the largest gap between consecutive points. This is called the norm of the partition.

Definition 7.2.6. The **norm** of a partition $P = \{x_0, x_1, \ldots, x_n\}$ of $[a, b]$ is:

$$\|P\| \coloneqq \max\{x_i - x_{i-1} : i = 1, \ldots n\}.$$

That is, the largest length of the intervals $[x_{i-1}, x_i]$, $i = 1, \ldots, n$.

For example the partition $P = \{1, 2, 6, 7, 9.9, 10\}$ of $[1, 10]$ has norm $\|P\| = 4$, this being the length of the interval $[2, 6]$. Note that if Q is a refinement of P then $\|Q\| \leq \|P\|$.

7.3 The Darboux integral

Recall that Riemann sums involve the choice of a sample input from each interval. In Darboux's approach to integration, one instead considers the infimum or supremum of the function on each interval so that no sample points are involved.

Definition 7.3.1. Given a bounded function $f : [a, b] \to \mathbb{R}$ and partition $P = \{x_0, x_1, \ldots, x_n\}$ of $[a, b]$ the **upper sum** $U(f, P)$ and **lower sum** $L(f, P)$ are defined by:

$$U(f, P) \coloneqq \sum_{i=1}^{n} M_i(x_i - x_{i-1}) \qquad \text{where } M_i \coloneqq \sup\{f(x) : x \in [x_{i-1}, x_i]\} \text{ for } i = 1, \ldots, n.$$

$$L(f, P) \coloneqq \sum_{i=1}^{n} m_i(x_i - x_{i-1}) \qquad \text{where } m_i \coloneqq \inf\{f(x) : x \in [x_{i-1}, x_i]\} \text{ for } i = 1, \ldots, n.$$

Remark 7.3.2. Note that the assumption that f be bounded is essential in the above definition in order to ensure that $U(f, P)$ and $L(f, P)$ are real numbers. If f is not bounded, at least one of M_i, m_i, $i = 1, \ldots, n$ will be equal to $\pm\infty$.

The following proposition quite straightforward and a good exercise in understanding the definitions.

Proposition 7.3.3. *Suppose that $f : [a, b] \to \mathbb{R}$ is bounded.*

(i) If P and is a partition of $[a, b]$ and P' is a refinement of P, then

$$U(f, P') \leq U(f, P) \qquad\qquad L(f, P') \geq U(f, P).$$

(ii) For any partition P of $[a, b]$, one has

$$m(b - a) \leq L(f, P) \leq U(f, P) \leq M(b - a)$$

where $m \coloneqq \inf\{f(x) : x \in [a, b]\}$, $M \coloneqq \sup\{f(x) : x \in [a, b]\}$. □

Proof. To prove (i), it suffices (why?) to consider the case where the refinement contains exactly one more point than the original partition. Say $P = \{x_0, \ldots, x_n\}$ and $P' = P \cup \{y\}$ where $y \notin P$. We only prove the desired comparison of upper sums, since the proof for upper sums is analogous. Put $M_i \coloneqq \sup_{x \in [x_{i-1}, x_i]} f(x)$, as usual. Let $j \in \{1, \ldots, n\}$ be the unique index such that $x_{j-1} < y < x_j$. Put $M' \coloneqq \sup_{x \in [x_{j-1}, y]} f(x)$, $M'' \coloneqq \sup_{x \in [y, x_j]} f(x)$. Since $[x_{j-1}, y], [y, x_j] \subseteq [x_{j-1}, x_j]$, we have $M', M'' \leq M_j$ so that

$$\begin{aligned} U(f, P') &= M'(y - x_{j-1}) + M''(x_j - y) + \sum_{\substack{i=1 \\ i \neq j}}^{n} M_i(x_i - x_{i-1}) \\ &\leq M_j(y - x_{j-1}) + M_j(x_j - y) + \sum_{\substack{i=1 \\ i \neq j}}^{n} M_i(x_i - x_{i-1}) \\ &= M_j(x_j - x_{j-1}) + \sum_{\substack{i=1 \\ i \neq j}}^{n} M_i(x_i - x_{i-1}) \\ &= U(f, P), \end{aligned}$$

as claimed. We leave the proof of (ii) as an exercise to the reader. □

Intuitively, statement (i) above captures the idea that as a partition of $[a, b]$ is refined, the corresponding upper sums (resp. lower sums) hug the graph of f more closely from above (resp. below). As the partition gets finer the upper/lower sums are able to get further into the nooks and crannies of the graph of f.

Statement (ii) above says that the lower sum for a partition is always less than or equal to the upper sum for the same partition. In fact more is true: the lower sum for a partition is always less than or equal to the upper sum for some other possibly different partition.

Corollary 7.3.4. *Suppose $f : [a, b] \to \mathbb{R}$ is bounded and P, Q are both partitions of $[a, b]$ (we do not assume that P refines Q or vice versa). Then $L(f, P) \leq U(f, Q)$.*

Proof. We have

$$\begin{aligned} L(f, P) &\leq L(f, P \cup Q) && \text{Proposition 7.3.3 (1), noting } P \cup Q \text{ refines } P \\ &\leq U(f, P \cup Q) && \text{Proposition 7.3.3 (2)} \\ &\leq U(f, Q) && \text{Proposition 7.3.3 (1), noting } P \cup Q \text{ refines } Q. \end{aligned}$$

□

Darboux's upper and lower sums naturally lead to an upper and lower integral for every bounded function.

Definition 7.3.5. Given a bounded function $f : [a, b] \to \mathbb{R}$, the **upper integral** $U(f)$ and **lower integral** $L(f)$ of f on $[a, b]$ are defined by:

$$\begin{aligned} U(f) &\coloneqq \inf\{U(f, P) : P \text{ is a partition of } [a, b]\} \\ L(f) &\coloneqq \sup\{L(f, P) : P \text{ is a partition of } [a, b]\}. \end{aligned}$$

Remark 7.3.6. Technically the notations $U(f)$ and $L(f)$ are a bit imprecise as they do not indicate to the reader which interval is being integrated over. Nonetheless, to avoid cluttering our notation with additional symbols, we trust that the relevant interval will be apparent from the context. Note that some people use the alternative notations $\overline{\int}_a^b f(x)\,dx$ and $\underline{\int}_a^b f(x)\,dx$ for the upper and lower Darboux integrals. These alternative notations do not suffer from the same ambiguity.

Proposition 7.3.7. *For any bounded function $f : [a, b] \to \mathbb{R}$, the upper integral $U(f)$ and lower integral $L(f)$ exist and satisfy:*

$$m(b-a) \leq L(f) \leq U(f) \leq M(b-a)$$

where $m \coloneqq \inf\{f(x) : x \in [a, b]\}$, $M \coloneqq \sup\{f(x) : x \in [a, b]\}$.

Proof. For any partitions P and Q of $[a, b]$ we have combining Proposition 7.3.3 (2) and Corollary 7.3.3 that

$$m(b-a) \leq L(f, P) \leq U(f, Q) \leq M(b-a).$$

Taking the supremum over all partitions P of $[a, b]$, it follows that:

$$m(b-a) \leq L(f) \leq U(f, Q) \leq M(b-a).$$

Then, taking the infimum over all partitions Q of $[a, b]$, we obtain the desired bound:

$$m(b-a) \leq L(f) \leq U(f) \leq M(b-a).$$

□

In general, it is possible that the lower integral of a bounded function is strictly smaller than the upper integral. When these quantities do agree, we will declare the function to be Darboux integrable. The following proposition clarifies when that happens.

Proposition 7.3.8. *Let $f : [a, b] \to \mathbb{R}$ be bounded. Then, the following are equivalent:*

(i) For every $\epsilon > 0$, there exists a partition P of $[a, b]$ such that $U(f, P) - L(f, P) < \epsilon$.

(ii) $L(f) = U(f)$.

Proof. Suppose that (i) holds. Given $\epsilon > 0$, let P be a corresponding partition of $[a, b]$ as in (i). Noting that $L(f, P) \leq L(f) \leq U(f) \leq U(f, P)$, we have that

$$0 \leq U(f) - L(f) \leq U(f, P) - L(f, P) < \epsilon.$$

We have shown $0 \leq U(f) - L(f) < \epsilon$ for every $\epsilon > 0$ and it follows that $L(f) = U(f)$.

Conversely, suppose that (ii) holds. Let $\epsilon > 0$ be given. By a standard property of the supremum (Proposition 1.9.1), there exists a partition P_1 of $[a, b]$ such that

$$L(f) - \epsilon/2 < L(f, P_1) \leq L(f).$$

Similarly, by the analogous property of the infimum, there exists a partition P_2 of $[a,b]$ such that

$$U(f) \leq U(f,P_2) < U(f) + \epsilon/2.$$

Now, put $P \coloneqq P_1 \cup P_2$ so that

$$L(f) - \epsilon/2 < L(f,P_1) \leq L(f,P) \leq L(f) \leq U(f) \leq U(f,P) \leq U(f,P_2) < U(f) + \epsilon/2.$$

Since $L(f) = U(f)$ by assumption, we have shown that $L(f,P)$ and $U(f,P)$ both belong to an open interval of radius $\epsilon/2$ centred on $L(f) = U(f)$. Thus, $U(f,P) - L(f,P) < \epsilon$ as desired. □

The above proposition is the basis for the definition of the Darboux integral.

Definition 7.3.9. Let $f : [a,b] \to \mathbb{R}$ be bounded. We say that f is **Darboux integrable** on the interval $[a,b]$ if the equivalent conditions of Proposition 7.3.8 hold. This being the case, we define the **Darboux integral** of f over $[a,b]$ to be the real number $L(f) = U(f)$.

Remark 7.3.10. As mentioned above, the Darboux integral and Riemann integral turn out to be equivalent. For that reason, it is more common to refer to them both as the Riemann integral. Indeed, once we have proven their equivalence we will drop the extra jargon.

7.4 Examples of integrable/nonintegrable functions

Next we study examples and nonexamples of Darboux integrable (or, equivalently, Riemann integrable) functions.

Theorem 7.4.1. *If f is continuous on $[a,b]$, then it is Darboux integrable on $[a,b]$.*

Proof. Fix $\epsilon > 0$. We must prove that there exists a partition $P = \{x_0, x_1, \ldots, x_n\}$ of $[a,b]$ with the property that $U(f,P) - L(f,P) < \epsilon$. Since the interval $[a,b]$ is closed, we have for free that f is uniformly continuous on $[a,b]$ (see Theorem 5.5.4). Thus, given $\epsilon_1 > 0$ (to be selected shortly), we can find $\delta > 0$ such that $x, y \in [a,b]$, $|x-y| < \delta$ implies $|f(x)-f(y)| < \epsilon_1$. Choose $n \in \mathbb{N}$ such that $(b-a)/n < \delta$ (possible by the Archimedean property) and take the partition P to consist of equally spaced points so that $\|P\| = (b-a)/n < \delta$. For this choice of P, we have for any $x, y \in [x_{i-1}, x_i]$ where $i = 1, \ldots, n$ that $|x-y| < \delta$ so that $|f(x) - f(y)| < \epsilon_1$. It follows (why?) that $M_i - m_i \leq \epsilon_1$ for $i = 1, \ldots, n$ where $M_i \coloneqq \sup\{f(x) : x \in [x_{i-1}, x_i]\}$ and $m_i \coloneqq \inf\{f(x) : x \in [x_{i-1}, x_i]\}$. Thus,

$$U(f,P) - L(f,P) = \sum_{i=1}^{n} (M_i - m_i)(x_i - x_{i-1}) = \frac{b-a}{n} \sum_{i=1}^{n} (M_i - m_i) \leq \frac{b-a}{n} \cdot n\epsilon_1 = \frac{\epsilon_1}{b-a}.$$

Choosing $\epsilon_1 = \frac{\epsilon}{2(b-a)}$, we get $U(f,P) - L(f,P) \leq \frac{\epsilon}{2} < \epsilon$ as required. □

Example 7.4.2. As previously discussed, the function $f : [0,1] \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} 0 & \text{if } x = 0 \\ x \sin(\frac{1}{x}) & \text{if } x \neq 0 \end{cases}$$

is continuous on the interval $[0,1]$, and so it is Darboux integrable on $[0,1]$. Note: there is not an elementary expression for the value of the integral $\int_0^1 f(x)\ dx$.

Example 7.4.3. Let $f : [a,b] \to \mathbb{R}$ be the characteristic function of the rational numbers:

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

For any partition $P = \{x_0, x_1, \ldots, x_n\}$ of $[a,b]$ one has that

$$\begin{aligned} M_i &:= \sup\{f(x) : x \in [x_{i-1}, x_i]\} = 1 \\ m_i &:= \inf\{f(x) : x \in [x_{i-1}, x_i]\} = 0 \qquad\qquad \text{for } i = 1, \ldots, n \end{aligned}$$

the reason being that the sets of rational and irrational numbers are both dense, so that $[x_{i-1}, x_i]$ contains both types of numbers. It follows that

$$\begin{aligned} U(f,P) &= \sum_{i=1}^{n} M_i(x_i - x_{i-1}) = \sum_{i=1}^{n} (1)(x_i - x_{i-1}) = b - a \\ L(f,P) &= \sum_{i=1}^{n} m_i(x_i - x_{i-1}) = \sum_{i=1}^{n} (0)(x_i - x_{i-1}) = 0. \end{aligned}$$

Taking the infimum/supremum over all partitions P, we obtain $L(f) = 0$ and $U(f) = b - a$. Since $L(f) \neq U(f)$, this function is not Darboux integrable.

Example 7.4.4. Recall the so-called "popcorn function" $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} \frac{1}{q} & \text{if } x \in \mathbb{Q} \text{ and } x = \frac{p}{q} \text{ is its lowest-terms representation} \\ 0 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

As discussed earlier (see Exercise 5.4.5) this function is discontinuous at all rational inputs and continuous at all irrational inputs. In spite of having a dense set of discontinuities, the popcorn function is Darboux integrable on every closed interval and has integral 0. Rather than work out this example in detail, we consider a simpler example in a similar spirit below.

Example 7.4.5. Fix a sequence $\{q_k\}_{k=1}^{\infty}$ which lists all the rational numbers in the interval $[0,1]$ once each. Define a function $f : [0,1] \to \mathbb{R}$ as follows:

$$f(x) = \begin{cases} \frac{1}{k} & \text{if } x \in \mathbb{Q} \text{ and } x = q_k \\ 0 & \text{if } x \notin \mathbb{Q}. \end{cases}.$$

As discussed earlier (see Exercise 5.4.5) this function is discontinuous at all rational inputs and continuous at all irrational inputs. We will show that, in spite of having a dense set of discontinuities, this function is Darboux integrable with integral 0.

Let $P = \{x_0, x_1, \ldots, x_n\}$ be any partition of $[0,1]$. Since the irrational numbers are dense in $\mathbb{R}$, each interval $[x_{i-1}, x_i]$ contains irrational numbers. Consequently,

$$m_i \coloneqq \inf\{f(x) : x \in [x_{i-1}, x_i]\} = 0 \text{ for } i = 1, \ldots, n.$$

It follows that $L(f, P) = 0$, regardless of how the partition P was chosen. Therefore, the lower Darboux integral of f vanishes, $L(f) = 0$.

It remains to show that the upper Darboux integral also vanishes, $U(f) = 0$. To prove this, it suffices to show that $U(f) < \epsilon$ for fixed but arbitrary $\epsilon > 0$. First we note that $f(x) < \epsilon/2$ for most inputs x, the only exceptions being a finite set of rational inputs. More precisely, by taking $N \in \mathbb{N}$ such that $1/N \leq \epsilon/2$ (possible by the Archimedean property) and putting $E = \{q_1, q_1, \ldots, q_N\}$, we have $f(x) < \epsilon/2$ for all $x \in [0,1] \setminus E$.

The next step is to choose a partition $P = \{x_0, x_1, \ldots, x_n\}$ of $[0,1]$ in such a way that the the total length of the subintervals which contain one of the exceptional inputs $q_1, \ldots, q_N$ is less than $\epsilon/2$. There are several ways to arrange this, but a simple one is to take P to be an equally-spaced partition with n chosen sufficiently large (possible by the Archimedean property) that $2N \cdot \frac{b-a}{n} < \frac{\epsilon}{2}$. This works because every interval $[x_{i-1}, x_i]$ has length $\frac{b-a}{n}$ and at most $2N$ of them contain a point of $E = \{q_1, q_2, \ldots, q_N\}$ (the factor of 2 is necessary because it is possible for a point to lie on the boundary of two adjacent intervals). With P chosen in this way, we let:

$$I = \{i \in \{1, \ldots, n\} : E \cap [x_{i-1}, x_i] \neq \varnothing\} \qquad J \coloneqq \{1, \ldots, n\} \setminus I.$$

If $i \in I$, then $M_i \leq 1$ since $f \leq 1$ everywhere. If $j \in J$, then $M_j < \epsilon/2$ since $f(x) < \epsilon/2$ for all $x \in [0,1] \setminus E$. Thus, we find that

$$\begin{aligned} U(f, P) &= \sum_{i \in I} M_i(x_i - x_{i-1}) + \sum_{j \in J} M_j(x_j - x_{j-1}) \\ &\leq \sum_{i \in I}(x_i - x_{i-1}) + \frac{\epsilon}{2}\sum_{j \in J}(x_j - x_{j-1}) \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} \\ &= \epsilon. \end{aligned}$$

as we claimed.

We saw above that all continuous functions are Darboux integrable. Now we look at another nice class of Darboux integrable functions: monotone functions. Note that continuous functions may fail to be monotone (e.g. $x \mapsto \sin(x)$) and monotone functions may fail to be continuous (e.g. the floor function $x \mapsto \lfloor x \rfloor$), so these are quite different classes of functions.

Theorem 7.4.6. *Let $f : [a,b] \to \mathbb{R}$ be any monotone function. Then, f is Darboux integrable on $[a,b]$.*

Proof. Recall that to say f is monotone is to say that f is either increasing or decreasing. We only consider the case where f is increasing, as the case where f is decreasing is similar. Firstly, we should verify that f is bounded as this is a requirement for f to be Darboux integrable. Indeed, $f(a) \leq f(x) \leq f(b)$ for all $x \in [a, b]$, i.e. all values of f lie in the interval $[f(a), f(b)]$.

Now let $\epsilon > 0$ be given. We should show there exists a partition $P = \{x_0, x_1, \ldots, x_n\}$ of $[a, b]$ such that $U(f, P) - L(f, P) < \epsilon$. We claim that taking P to be an equally-spaced partition with n sufficiently large does the job. Thus, we take $x_i = a + i \cdot \Delta_x$ for $i = 1, \ldots, n$ where $\Delta x \coloneqq \frac{b-a}{n}$. Observe that, because f is increasing, we have

$$\begin{aligned} M_i &= \sup\{f(x) : x \in [x_{i-1}, x_i]\} = f(x_i) \\ m_i &= \inf\{f(x) : x \in [x_{i-1}, x_i]\} = f(x_{i-1}) \qquad\qquad i = 1, \ldots, n. \end{aligned}$$

Thus,

$$\begin{aligned} U(f, P) - L(f, P) &= \sum_{i=1}^{n} (M_i - m_i)(x_i - x_{i-1}) \\ &= \sum_{i=1}^{n} (f(x_i) - f(x_{i-1}))\Delta x \\ &= \Delta x \cdot [(f(x_1) - f(x_0) + (f(x_2) - f(x_1)) + \ldots + (f(x_n) - f(x_{n-1}))] \\ &= \Delta x (f(b) - f(a)) \\ &= \frac{(b-a)(f(b) - f(a))}{n}. \end{aligned}$$

By the Archimedean property, taking $n \in \mathbb{N}$ sufficiently large gives $U(f, P) - L(f, P) < \epsilon$ as required. □

7.5 Pathological monotone functions

At this point we have seen two "nice" classes of integrable functions: continuous functions and monotone functions (Theorems 7.4.1 and 7.4.6). We have also looked at some strange functions (Examples 7.4.3, 7.4.4, 7.4.5) that are neither continuous nor monotone: the characteristic function of the rationals is not continuous at any point, and it is not integrable; the popcorn function is continuous only at the irrational numbers, and it is integrable.

Although the classes of continuous functions and monotone functions are nice in some ways, there is still lots of room for pathological behaviour even inside these classes. When we think of a continuous function, we usually picture a function with a nice smooth graph, mostly differentiable with perhaps the occasional vertical tangent line, sharp angle or cusp (as in the examples $f(x) = x^{1/3}$, $f(x) = |x|$, $f(x) = \sqrt{|x|}$). However, as we may discuss later, there exist continuous functions which are not differentiable at *any* point; functions whose graphs appear "rough" at arbitrarily small scales.

Monotone functions can also be somewhat more ill-behaved than one might initially suspect. When one thinks of a monotone function, one usually pictures a function which is mostly continuous, perhaps with some well-controlled set of jump discontinuities. For example, the floor function $f(x) = \lfloor x \rfloor$ is increasing with a jump discontinuity at every integer. The function:

$$f(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ \frac{1}{n} & \text{if } \frac{1}{n+1} \leq x < \frac{1}{n} \text{ for } n \in \mathbb{N} \\ 1 & \text{if } x \geq 1 \end{cases}$$

has a jump discontinuity at every element of the set $\{\frac{1}{n} : n \in \mathbb{N}\}$. It may come as something of a surprise that a monotone function can actually have discontinuties at every rational input.

Example 7.5.1. Let $\{q_n\}_{n=1}^{\infty}$ be a sequence which lists each rational number in the interval $[0, 1]$ exactly once. Define a function $f : [0, 1] \to \mathbb{R}$ in the following way:

$$f(x) = \sum_{\substack{n \in \mathbb{N} \\ q_n \leq x}} \frac{1}{2^n}.$$

On may think of this function as follows: each rational number q_n has been assigned a weight $\frac{1}{2^n}$ in such a way that the sum of all these weights is equal to 1, i.e. $\sum_{n=1}^{\infty} \frac{1}{2^n} = 1$. We define $f(x)$ to be the sum of the weights of all rational numbers lying to the left of x. The function f is monotone because, as x gets larger, more rational numbers are included in the set of rationals to the left of x, and so the value of the sum can only increase. Using the fact that the rationals are dense, one may prove that f is in fact strictly increasing (as opposed to merely increasing). One may furthermore prove that f is continuous at every irrational input and discontinuous at every rational input.

7.6 The Riemann integral

We now return to Riemann's definition of the integral and establish its equivalence to Darboux's definition. We began with Darboux's definition because it is a bit more convenient to prove that a function is Darboux integrable than it is to prove it is Riemann integrable. Consequently, starting with Darboux's definition gave us more immediate access to examples.

Definition 7.6.1. Let $f : [a, b] \to \mathbb{R}$ be a bounded function. Let $P = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a, b]$. Choose a sequence $T = \{t_i\}_{i=1}^{n}$ of sample points (sometimes called "tags") $t_i \in [x_{i-1}, x_i]$ for $i = 1, \ldots, n$. The associated **Riemann sum** is:

$$R(f, P, T) \coloneqq \sum_{i=1}^{n} f(t_i)(x_i - x_{i-1}).$$

We say that f is **Riemann integrable** on $[a, b]$ if there is a number $S \in \mathbb{R}$ with the following property: for every $\epsilon > 0$ there is a $\delta > 0$ such that for every partition P of $[a, b]$ with norm $\|P\| < \delta$ and selection of sample points T, one has

$$|R(f, P, T) - S| < \epsilon.$$

In this case, the number S is unique (check this!) and we define the **Riemann integral** of f over $[a, b]$ to be:

$$\int_a^b f(x)\, dx \coloneqq S.$$

Remark 7.6.2. Above, we required that Riemann integrable functions be bounded as part of the definition. In fact, even if we left this requirement out, it would follow automatically that every Riemann integrable function is bounded. If $f : [a, b] \to \mathbb{R}$ was unbounded then, for any partition $P = \{x_0, x_1, \ldots, x_n\}$, the function f would be unbounded on at least one of the intervals $[x_{i-1}, x_i]$, $i = 1, \ldots, n$. As a result, it would be possible to choose the tags $T = \{t_i\}_{i=1}^n$ in such a way that $|R(f, P, T)|$ is arbitrarily large, precluding the possibility that the Riemann sums converge to any specific real value as $\|P\| \to 0$.

The following lemma will be useful in proving that Darboux integrability and Riemann integrability are equivalent. It also provides an answer to a natural question:

> We know (Proposition 7.3.3 (i)) that, if P' is a refinement of P, then $U(f, P')$ is smaller than or equal to $U(f, P)$. In the most extreme cases, how much smaller than $U(f, P)$ could $U(f, P')$ be?

To answer this question, the main thing is to think through how much $U(f, P)$ could possibly decrease in the case where P' is obtained from P by adding a single point, as illustrated in Figure 7.1.

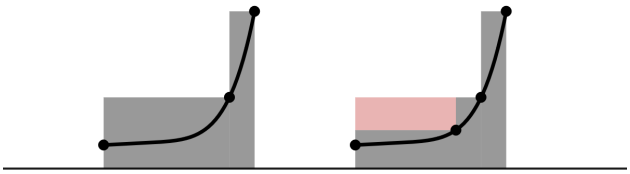


Figure 7.1: Adding a new point to the partition decreases the upper Darboux sum.

Lemma 7.6.3. *Let $f : [a, b] \to \mathbb{R}$ be a bounded function. Let P be a partition of $[a, b]$ and let P' be a refinement of P. Then, the following inequalities are satisfied:*

$$U(f, P') \geq U(f, P) - k(M - m)\|P\| \tag{7.1}$$
$$L(f, P') \leq L(f, P) + k(M - m)\|P\|. \tag{7.2}$$

Here, k is the cardinality of $P' \setminus P$, i.e. the number of added points in the refinement. As usual, $M \coloneqq \sup\{f(x) : x \in [a, b]\}$, $m \coloneqq \inf\{f(x) : x \in [a, b]\}$, and $\|P\|$ is the norm of P.

Proof. We only prove (7.1) as the proof of (7.2) is analogous. We also suppose at first that $k = 1$ so that $P' = P \cup \{y\}$ where $y \notin P$. Define $P = \{x_0, x_1, \ldots, x_n\}$ and let $j \in \{1, \ldots, n\}$ be the unique index such that $x_{j-1} < y < x_j$. Let:

$$M_i \coloneqq \sup\{f(x) : x \in [x_{i-1}, x_i]\} \qquad i = 1, \ldots, n$$
$$M' \coloneqq \sup\{f(x) : x \in [x_{j-1}, y]\}$$
$$M'' \coloneqq \sup\{f(x) : x \in [y, x_j]\}.$$

Noting that $[x_{j-1}, x_j] = [x_{j-1}, y] \cup [y, x_j]$, at least one of the following must be true:

$$M_j = M' \qquad\qquad M_j = M''.$$

Let us assume the first of these alternatives holds as the proof in the other case proceeds similarly. Then, we have

$$\begin{aligned} U(f, P) - U(f, P') &= M_j(x_j - x_{j-1}) - M'(y - x_{j-1}) - M''(x_j - y) \\ &= M'(x_j - x_{j-1}) - M'(y - x_{j-1}) - M''(x_j - y) \\ &= (M' - M'')(x_j - y) \\ &\leq (M - m)(x_j - y) \\ &\leq (M - m)(x_j - x_{j-1}) \\ &\leq (M - m)\|P\|. \end{aligned}$$

The above calculation proves (7.1) in the case $k = 1$. More generally, if P' is a refinement of P containing $k \geq 2$ additional points, we may add them one at a time to arrive at the desired bound, noting that the norm of any refinement of P is less than or equal to $\|P\|$. □

Now we prove the equivalence of the Darboux and Riemann integrals.

Theorem 7.6.4. *Let $f : [a, b] \to \mathbb{R}$ be a bounded function. Then f is Riemann integrable on $[a, b]$ if and only if f is Darboux integrable on $[a, b]$. Moreover, in this case, the Riemann integral and Darboux integral are equal.*

Proof. Suppose that f is Riemann integrable with Riemann integral $S \coloneqq \int_a^b f(x)\, dx$. To show f is Darboux integrable, we must show that, given $\epsilon > 0$, there is a partition P of $[a, b]$ such that $U(f, P) - L(f, P) < \epsilon$. Indeed, using the Riemann integrability, there is a number

$\delta > 0$ such that if $\|P\| < \delta$ then $|R(f, P, T) - S| < \epsilon/4$ for any choice of tags T. It follows (why?) that $|L(f, P) - S| \leq \epsilon/4$ and $|U(f, P) - S| \leq \epsilon/4$ and so:

$$U(f, P) - L(f, P) = |U(f, P) - S| + |S - L(f, P)| \leq \epsilon/2 < \epsilon,$$

as was to be shown.

The converse requires more effort. Suppose that f is Darboux integrable with Darboux integral $S \coloneqq L(f) = U(f)$. We will show that f is Riemann integrable and that the Riemann integral is also S. To this end, let $\epsilon > 0$ be arbitrary. We must find $\delta > 0$ such that for any partition $P = \{x_0, x_1, \ldots, x_n\}$ of $[a, b]$ with norm $\|P\| < \delta$ and any sequence of tags $T = \{t_i\}_{i=1}^n$ for P, the associated Riemann sum $R(f, P, T) = \sum_{i=1}^n f(t_i)(x_i - x_{i-1})$ satisfies $|R(f, P, T) - S| < \epsilon$.

First, since $S = U(f) = L(f)$ is the infimum of all upper Darboux sums as well as the supremum of all lower Darboux sums, we can find a partitions Q_1 and Q_2 of $[a, b]$ such that

$$S - \epsilon/2 < L(f, Q_2) \leq S \leq U(f, Q_1) < S + \epsilon/2.$$

Then, putting $Q \coloneqq Q_1 \cup Q_2$, Proposition 7.3.3 (i) gives:

$$S - \epsilon/2 < L(f, Q) \leq S \leq U(f, Q) < S + \epsilon/2.$$

Let $M \coloneqq \sup\{f(x) : x \in [a, b]\}$, $m \coloneqq \inf\{f(x) : x \in [a, b]\}$, and let N be the number of elements of Q. Note that $P \cup Q$ is a refinement of P with $k \leq N$ additional elements. Indeed, we even have $k \leq N - 2$ since, at minimum, the elements $a, b \in Q$ already belonged to P. Thus, we have

$$\begin{aligned} U(f, P) &\leq U(f, P \cup Q) + k(M - m)\|P\| && \text{by Lemma 7.6.3} \\ &\leq U(f, Q) + N(M - m)\|P\| && P \cup Q \text{ refines } Q \text{ and } k \leq N. \end{aligned}$$

In the same way,

$$\begin{aligned} L(f, P) &\geq L(f, P \cup Q) - k(M - m)\|P\| && \text{by Lemma 7.6.3} \\ &\geq L(f, Q) - N(M - m)\|P\| && P \cup Q \text{ refines } Q \text{ and } k \leq N. \end{aligned}$$

Putting $\delta \coloneqq \frac{\epsilon}{2N(M-m)}$, we obtain that $\|P\| < \delta$ implies:

$$\begin{aligned} U(f, P) &< U(f, Q) + \epsilon/2 < S + \epsilon/2 + \epsilon/2 = S + \epsilon \\ L(f, P) &> L(f, Q) - \epsilon/2 > S - \epsilon/2 - \epsilon/2 = S - \epsilon. \end{aligned}$$

We have produced a $\delta > 0$ such that $\|P\| < \delta$ implies

$$S - \epsilon < L(f, P) \leq R(f, P, T) \leq U(f, P) < S - \epsilon,$$

so f is Riemann integrable on $[a, b]$ with Riemann integral $\int_a^b f(x)\, dx = S$. $\square$

7.7 Properties of the Riemann integral

Next we take a look at various standard properties of the Riemann integral.

Theorem 7.7.1. *Suppose $f, g : [a, b] \to \mathbb{R}$ are Riemann integrable. Then the following hold:*

(i) $f + g$ is Riemann integrable and $\int_a^b [f(x) + g(x)]\ dx = \int_a^b f(x)\ dx + \int_a^b g(x)\ dx$.

(ii) cf is Riemann integrable for any $c \in \mathbb{R}$ and $\int_a^b cf(x)\ dx = c \cdot \int_a^b f(x)\ dx$.

(iii) fg is Riemann integrable.

(iv) If $f \leq g$ then $\int_a^b f(x)\ dx \leq \int_a^b g(x)\ dx$.

(v) $|f|$ is Riemann integrable and $|\int_a^b f(x)\ dx| \leq \int_a^b |f(x)|\ dx$.

Since we have already proven the Darboux integral and Riemann integral are equivalent (Theorem 7.6.4) we are free to use whichever definition is more convenient to establish the above properties. Properties (i), (ii) and (iv) are quite straightforward; of these we prove only (i) and leave the other two as exercises.

Proof of Theorem 7.7.1 (i). Define for convenience $S_1 \coloneqq \int_a^b f(x)\ dx$ and $S_2 \coloneqq \int_a^b g(x)\ dx$. We must show $f + g$ has Riemann integral $S_1 + S_2$. Indeed, let $\epsilon > 0$ be given. We know there exists $\delta_1 > 0$ such that, for every partition P of $[a, b]$ with $\|P\| < \delta_1$ and every sequence of sample points T, the inequality $|R(f, P, T) - S_1| < \epsilon/2$ is satisfied. Likewise, there exists $\delta_2 > 0$ such that, for every partition P of $[a, b]$ with $\|P\| < \delta_2$ and every sequence of sample points T, $|R(g, P, T) - S_2| < \epsilon/2$. Put $\delta \coloneqq \min(\delta_1, \delta_2)$. Suppose that $\|P\| < \delta$ and T is a sequence of sample points for P. Observe that $R(f + g, P, T) = R(f, P, T) + R(g, P, T)$ by linearity of ordinary finite sums. Then, using the triangle inequality, we obtain:

$$\begin{aligned} |R(f + g, P, T) - (S_1 + S_2)| &\leq |R(f, P, T) - S_1 + R(g, P, T) - S_2| \\ &\leq |R(f, P, T) - S_1| + |R(g, P, T) - S_2| \\ &< \epsilon/2 + \epsilon/2 \\ &= \epsilon \end{aligned}$$

as was to be shown. □

Exercise 7.7.2. Prove (ii) and (iv) of Theorem 7.7.1 along the same lines.

For Theorem 7.7.1 (iii), a nice strategy is to first prove the following more general result about composition with a continuous function. This same result also helps with Theorem 7.7.1 (v), though admittedly this approach is probably overkill for dealing with taking absolute values.

Theorem 7.7.3. *Suppose that $f : [a, b] \to \mathbb{R}$ is Riemann integrable and $g : [m, M] \to \mathbb{R}$ is continuous, where $m \coloneqq \inf f$, $M \coloneqq \sup f$. Then, $g \circ f$ is Riemann integrable.*

Proof. It will be helpful to set up some additional notations:

$$\begin{aligned} m_i &\coloneqq \inf\{f(x) : x \in [x_{i-1}, x_i]\} & M_i &\coloneqq \sup\{f(x) : x \in [x_{i-1}, x_i]\} & i &= 1, \ldots, n \\ k_i &\coloneqq \inf\{g(f(x)) : x \in [x_{i-1}, x_i]\} & K_i &\coloneqq \sup\{g(f(x)) : x \in [x_{i-1}, x_i]\} & i &= 1, \ldots, n \\ k &\coloneqq \min g & K &\coloneqq \max g. \end{aligned}$$

Let $\epsilon > 0$ be given. Since g is uniformly continuous on $[m, M]$, we may choose $\delta > 0$ to be such that:

$$y_1, y_2 \in [m, M] \text{ and } |y_1 - y_2| < \delta \qquad \Longrightarrow \qquad |g(y_1) - g(y_2)| < \frac{\epsilon}{2(b-a)}.$$

Next, because f is Darboux integrable, there is a partition $P = \{x_0, x_1, \ldots, x_n\}$ of $[a, b]$ such that:

$$U(f, P) - L(f, P) < \frac{\epsilon \cdot \delta}{2(K-k)}.$$

Let I denote the set of $i \in \{1, \ldots, n\}$ with $M_i - m_i < \delta$. Let J be the complementary set of $j \in \{1, \ldots, n\}$ with $M_j - m_j \geq \delta$. Observe that:

$$\delta \cdot \sum_{j \in J} (x_j - x_{j-1}) \leq \sum_{j \in J} (M_j - m_j)(x_j - x_{j-1}) \leq U(f, P) - L(f, P) < \frac{\epsilon \cdot \delta}{2(K-k)}.$$

Cancelling δ on both sides, we get that the sum of the lengths of the intervals $[x_{j-1}, x_j]$ for which $M_j - m_j \geq \delta$ satisfies

$$\sum_{j \in J} (x_j - x_{j-1}) < \frac{\epsilon}{2(K-k)}.$$

Putting everything together:

$$\begin{aligned} U(g \circ f, P) - L(g \circ f, P) &= \sum_{i \in I} (K_i - k_i)(x_i - x_{i-1}) + \sum_{j \in J} (K_j - k_j)(x_j - x_{j-1}) \\ &< \frac{\epsilon}{2(b-a)} \sum_{i \in I} (x_i - x_{i-1}) + (K-k) \sum_{j \in J} (x_j - x_{j-1}) \\ &< \frac{\epsilon}{2(b-a)} (b-a) + (K-k) \frac{\epsilon}{2(K-k)} \\ &= \epsilon. \end{aligned}$$

□

Taking the absolute value is a continuous operation, so Theorem 7.7.3 immediately gives:

Corollary 7.7.4. *If $f : [a, b] \to \mathbb{R}$ is Riemann integrable, then $|f|$ is also Riemann integrable.*

Exercise 7.7.5. Complete the proof of (v) from Theorem 7.7.1 by establishing the inequality $|\int_a^b f(x)\, dx| \leq \int_a^b |f(x)|\, dx$.

Though it may not appear to be the case at first glance, the fact that products of Riemann integrable functions are Riemann integrable is also a corollary of Theorem 7.7.3. The trick is to use a technique known as "polarization" which, in certain situations, allows one to reduce claims about the operation of multiplication to the simpler operation of squaring.

Corollary 7.7.6. *Statement (iii) from Theorem 7.7.1 holds: if $f, g : [a, b] \to \mathbb{R}$ are Riemann integrable, then so is their product fg.*

Proof. Since squaring is a continuous operation, Theorem 7.7.3 shows in particular that the square of a Riemann integrable function is Riemann integrable. Since sums and scalar multiples of Riemann integrable functions are Riemann integrable (Theorem 7.7.1 (i), (ii)), the identity:

$$fg = \tfrac{1}{4}\left[(f+g)^2 - (f-g)^2\right]$$

implies that fg is Riemann integrable. □

7.8 The fundamental theorem of calculus

Next we come to the fundamental theorem of calculus, a coarse paraphrase of which is: "the operations of integration and differentiation are inverses of one another". In fact there are two parts to the fundamental theorem of calculus to be addressed. These may be roughly described as follows:

FTC 1. Every continuous function has an antiderivative which can be constructed using definite integration.

FTC 2. Antiderivatives can be used to calculate values of definite integrals.

The following lemma contains the basic construction relevant for FTC 1, namely taking the "integral so far" of an integrable function.

Lemma 7.8.1. *Suppose that f is Riemann integrable on $[a, b]$. Then, $F : [a, b] \to \mathbb{R}$ defined by $F(x) = \int_a^x f(y)\, dy$ is continuous.*

This lemma already exposes a basic point about differentiation and integration, namely that integrating creates regularity whereas differentiation destroys it. Integrating a function makes it smoother/more regular. The function f above only needs to be integrable whereas the integrated function F is continuous (a stronger property).

Before proceeding to the proof of this lemma, we record one more property of the Riemann integral which was previously omitted. The proof of this property is left as an exercise.

Proposition 7.8.2. *Suppose that $a < c < b$ and $f : [a, b] \to \mathbb{R}$ is bounded. Then, f is Riemann integrable on $[a, b]$ if and only if it is Riemann integrable on both $[a, c]$ and $[c, b]$. Moreover, when this is the case,*

$$\int_a^b f(x)\, dx = \int_a^c f(x)\, dx + \int_c^b f(x)\, dx$$

is satisfied. □

We now return to the proof of the above lemma.

Proof of Lemma 7.8.1. Since f is bounded, there exists $M > 0$ such that $|f(x)| \leq M$ for all $x \in M$. Suppose $x, x_0 \in [a, b]$ and (without loss of generality) $x_0 \leq x$. Then, using Proposition 7.8.2 above, we have:

$$F(x) - F(x_0) = \int_a^x f(y)\ dy - \int_a^{x_0} f(y)\ dy = \int_{x_0}^x f(y)\ dy$$

so that, using Theorem 7.7.1 (i), (iv), (v), we have

$$|F(x) - F(x_0)| = \left| \int_{x_0}^x f(y)\ dy \right| \leq \int_{x_0}^x |f(y)|\ dy \leq \int_{x_0}^x M\ dy = M|x - x_0|.$$

We leave to the reader the task of deducing continuity of F from the bound:

$$|F(x) - F(x_0)| \leq M|x - x_0| \text{ for all } x, x_0 \in [a, b].$$

In fact, this bound exactly shows that F has a stronger property than continuity known as "Lipschitz continuity". □

Theorem 7.8.3 (FTC 1)**.** *Suppose that f is Riemann integrable on $[a, b]$. Define a continuous function $F : [a, b] \to \mathbb{R}$ by $F(x) = \int_a^x f(y)\ dy$ (see Lemma 7.8.1). Then F is differentiable at every point $x \in [a, b]$ where f is continuous and $F'(x) = f(x)$.*

Again, notice this demonstrates the general principle that integration improves the regularity of a function. If f is continuous on $[a, b]$, then F has the stronger property of being differentiable on $[a, b]$.

Proof. Let $x \in [a, b]$ be a point where f is continuous. Let $\epsilon > 0$. We must show there is a $\delta > 0$ such that $0 < |h| < \delta$ implies

$$\left| \frac{F(x + h) - F(x)}{h} - f(x) \right| < \epsilon. \tag{7.3}$$

Since f is continuous at x, we may choose $\delta > 0$ to be such that $|x - y| < \delta$ implies $|f(x) - f(y)| < \epsilon$. For simplicity, we only prove (7.3) is satisfied when $0 < h < \delta$, leaving the reader to consider what happens when $-\delta < h < 0$. Using Proposition 7.8.2 (and our knowledge of how to integrate a constant function), we may write

$$\frac{F(x + h) - F(x)}{h} = \frac{1}{h} \left(\int_a^{x+h} f(y)\ dy - \int_a^x f(y)\ dy \right) = \frac{1}{h} \int_x^{x+h} f(y)\ dy$$

$$f(x) = \frac{1}{h} \int_x^{x+h} f(x)\ dy.$$

Thus, we have:

$$\begin{aligned}
\left|\frac{F(x+h)-F(x)}{h} - f(x)\right| &\leq \frac{1}{h}\left|\int_x^{x+h} [f(y)-f(x)]\ dx\right| && \text{Proposition 7.7.1 (i), (ii)} \\
&\leq \frac{1}{h}\int_x^{x+h} |f(y)-f(x)|\ dx && \text{Proposition 7.7.1 (v)} \\
&< \frac{1}{h}\int_x^{x+h} \epsilon\ dx && \text{Proposition 7.7.1 (iv)} \\
&= \epsilon,
\end{aligned}$$

as was to be shown. □

Remark 7.8.4. In Example 6.7.2], we "cheated" by assuming the continuous function

$$h(x) = \begin{cases} x\sin(\frac{1}{x}) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}.$$

had an antiderivative H. Now we can justify this assumption by using Theorem 7.8.3 to obtain an antiderivative by setting

$$H(x) = \int_0^x h(y)\ dy.$$

Take note that, although the theory tells us H is a differentiable function with $H' = h$, there is not a nice closed form for the function H.

Another nice application of Theorem 7.8.3 is the construction of a function f which is continuous everywhere but fails to be differentiable on a dense set.

Example 7.8.5. Recall the increasing function $f : [0,1] \to \mathbb{R}$ from Example 7.5.1 which was discontinuous at every rational input and continuous at every irrational input. Since f is monotone, it is Riemann integrable by Theorem 7.4.6. Define $F : [0,1] \to \mathbb{R}$ by $F(x) = \int_0^x f(y)\ dy$. Then, by Theorem 7.8.3, the function F is continuous everywhere and differentiable at every irrational input $x \in [0,1]$. With some additional effort (exercise), one may prove that F is *not* differentiable at every rational input. The function F is continuous and increasing on $[0,1]$, but its graph has a "sharp corner" at every rational input.

Now we come to the second part of the fundamental theorem of calculus which justifies the routine usage of antiderivatives in symbolic calculus to calculate definite integrals.

Theorem 7.8.6 (FTC 2)**.** *Suppose $f : [a,b] \to \mathbb{R}$ is Riemann integrable. Furthermore suppose that there exists an antiderivative for f, i.e. a function F which is differentiable on $[a,b]$ satisfying $f = F'$. Then,*

$$\int_a^b f(x)\ dx = F(b) - F(a).$$

Applications of this theorem are, of course, ubiquitous. Every time we use an antiderivative to calculate a definite integral, we are using this theorem. For example, the calculations $\int_0^1 x^2\ dx = \frac{1}{3}x^3|_0^1 = \frac{1}{3}$ and $\int_0^{\frac{\pi}{2}} \cos(x)\ dx = \sin(x)|_0^{\frac{\pi}{2}} = 1$ both rely on Theorem 7.8.6.

Proof of Theorem 7.8.6. Let $P = \{x_0, x_1, \ldots, x_n\}$ be any partition of $[a, b]$. Note the telescoping sum:

$$F(b) - F(a) = \sum_{i=1}^{n} F(x_i) - F(x_{i-1}).$$

Using the mean value theorem, there exist $t_i \in (x_{i-1}, x_i)$ such that

$$F(x_i) - F(x_{i-1}) = F'(t_i)(x_i - x_{i-1}) = f(t_i)(x_i - x_{i-1})$$

for $i = 1, \ldots, n$. Thus,

$$F(b) - F(a) = \sum_{i=1}^{n} f(t_i)(x_i - x_{i-1}).$$

It follows that

$$L(f, P) \leq F(b) - F(a) \leq U(f, P).$$

Taking the supremum/infimum over all partitions P, it follows that

$$L(f) \leq F(b) - F(a) \leq U(f).$$

However, since f is assumed Riemann integrable (or equivalently Darboux integrable) we have $\int_a^b f(x)\ dx = L(f) = U(f)$ and we get the desired result. □

7.9 Measure zero

Recall that the characteristic function of the rationals is discontinuous at every input and fails to be Riemann integrable (Example 7.4.3). Recall as well that the popcorn function is discontinuous at all rational inputs, continuous at all irrational inputs, and is Riemann integrable (Example 7.4.4). These examples suggest that Riemann integrability may have something to do with the "size" of the set of discontinuities, but what is the right concept of size?

Definition 7.9.1. A set $X \subseteq \mathbb{R}$ is said to have **measure zero** if, for every $\epsilon > 0$, there exists a sequence of open intervals $I_n = (a_n, b_n)$, $n \in \mathbb{N}$ such that:

- $X \subseteq \bigcup_{n=1}^{\infty} I_n$, i.e. the intervals cover X, and
- $\sum_{n=1}^{\infty}(b_n - a_n) < \epsilon$, i.e. the lengths of the intervals sum to less than ϵ.

Example 7.9.2. Every finite subset of $\mathbb{R}$ has measure zero. We leave the details to the reader as the next example will demonstrate something stronger.

Example 7.9.3. The set of rational numbers has measure zero. Indeed, every countably infinite set $X \subseteq \mathbb{R}$ has measure zero. To see this, let $\{x_n\}_{n=1}^{\infty}$ be a sequence listing each element of X exactly once. Let $\epsilon > 0$ be arbitrary. Define

$$I_n \coloneqq \left(x_n - \frac{\epsilon}{4} \cdot \frac{1}{2^n}, x_n + \frac{\epsilon}{4} \cdot \frac{1}{2^n} \right)$$

for $n \in \mathbb{N}$. Clearly this sequence of intervals covers X. Moreover, the sum of the lengths of these intervals is

$$\frac{\epsilon}{2} \sum_{n=1}^{\infty} \frac{1}{2^n} = \frac{\epsilon}{2} < \epsilon.$$

Conceptually, the idea is that, because a set such as $\mathbb{Q} \subseteq \mathbb{R}$ can be fit inside a sequence of intervals whose total length is arbitrarily small, it ought to be the case that the "length" of $\mathbb{Q}$ is zero. The set $\mathbb{Q}$, although big in the sense of being dense in $\mathbb{R}$, is actually negligible in the sense of length. The set of rationals is just "dust" with no substance to it.

Example 7.9.4. A closed interval $[a, b] \subseteq \mathbb{R}$ where $a < b$ is *not* a set of measure zero. Indeed, suppose that $[a, b]$ had measure zero. Then, taking $\epsilon = \frac{b-a}{2}$ (any positive number smaller than the length of $[a, b]$ will do), it must be the case that there exists a sequence of open intervals I_n, $n \in \mathbb{N}$ whose lengths sum to less than ϵ and which cover $[a, b]$. Since $[a, b]$ has the Heine-Borel property, it follows that only finitely-many of these intervals are necessary to cover $[a, b]$. In other words, for some $N \in \mathbb{N}$, there exist open intervals $I_1, \ldots, I_N$ such that

- $[a, b] \subseteq I_1 \cup \ldots \cup I_N$.
- $\text{length}(I_1) + \ldots + \text{length}(I_N) < \frac{b-a}{2}$.

This is a contradiction. We leave to the reader to contemplate how to prove the intuitively reasonable claim that an interval cannot be covered with finitely many intervals whose lengths total to less than the length of the interval being covered.

Exercise 7.9.5. Prove that it does not matter if the open intervals in Definition 7.9.1 are replaced by closed intervals. That is, prove that a set $X \subseteq \mathbb{R}$ has measure zero if and only if, for every $\epsilon > 0$, there exists a sequence of closed intervals $I_n = [a_n, b_n]$, $n \in \mathbb{N}$ such that:

- $X \subseteq \bigcup_{n=1}^{\infty} I_n$, i.e. the intervals cover X, and
- $\sum_{n=1}^{\infty} (b_n - a_n) < \epsilon$, i.e. the lengths of the intervals sum to less than ϵ.

The following characterization of which functions are Riemann integrable is due to Henri Lebesgue (1875-1941). We do not have time to go through the proof (though we do have all the tools to do so).

Theorem 7.9.6 (Lebegue's criterion)**.** *Let $f : [a, b] \to \mathbb{R}$ be bounded. Let $X \subseteq [a, b]$ be the set of points where f is discontinuous. Then, f is Riemann integrable if and only if X has measure zero.* □

Remark 7.9.7. It is traditional to say that a property holds "almost everywhere" if the set of points where the property fails has measure zero. In this language, one may rephrase Lebesgue's criterion as saying that a bounded function is Riemann integrable if and only if it is continuous almost everywhere.

Remark 7.9.8. It is worth revisiting some our examples from the perspective of Theorem 7.9.6:

- The characteristic function of the rationals is not Riemann integrable on any closed interval because its set of discontinuities does not have measure zero; it is discontinuous at every input!
- The popcorn function is Riemann integrable on every closed interval because its set of discontinuities is the rational numbers which has measure zero.
- Continuous functions are Riemann integrable on every closed interval because their set of discontinuities is empty and, in particular, has measure zero.
- One can show (exercise) that, if f is a monotone function, then the set of discontinuities of f is at worst countably infinite. In particular, the set of discontinuities of f has measure zero. This gives another explanation for why monotone functions are Riemann integrable on every closed interval.

We saw in Example 7.9.3 that every countable subset of $\mathbb{R}$ has measure zero. At first blush, one might wonder whether, conversely, every set of measure zero is countable. As our last order of business, we introduce the Cantor set $C \subseteq [0,1]$ and investigate its properties. The Cantor set is an extremely important example in analysis and topology. Among other things, it is an example of an uncountable set of measure zero.

In order to construct C, we use a sequence of progressively smaller "approximation sets" $C_0 \supseteq C_1 \supseteq C_2 \supseteq \dots$ whose intersection is C. Specifically, we consider the following sets:

$$\begin{aligned} C_0 &= [0,1] \\ C_1 &= [0, \tfrac{1}{3}] \cup [\tfrac{2}{3}, 1] \\ C_2 &= [0, \tfrac{1}{9}] \cup [\tfrac{2}{9}, \tfrac{1}{3}] \cup [\tfrac{2}{3}, \tfrac{7}{9}] \cup [\tfrac{8}{9}, 1] \\ &\vdots \end{aligned}$$

In general, C_n consists of 2^n pairwise disjoint closed intervals of length $(1/3)^n$. To continue the sequence, we apply the following recursive procedure: to obtain C_{n+1} from C_n, remove the open middle third of each of the intervals making up C_n, as illustrated in Figure 7.2.

Definition 7.9.9. The **Cantor set** C is defined as the intersection $C = \bigcap_{n=0}^{\infty} C_n$.

Since C_n is closed for every n and arbitrary intersections of closed sets are closed (Theorem 4.2.5), we have immediately that C is a closed set. Indeed, C is even compact (closed and bounded).

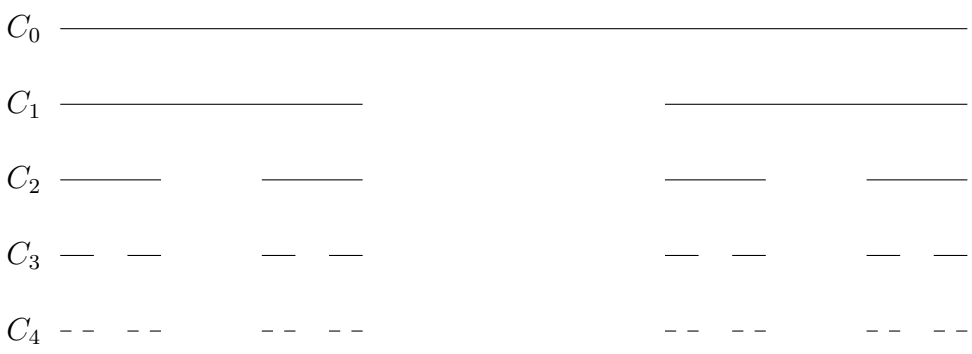


Figure 7.2: The construction of the Cantor set.

At first one might wonder whether there are any points in the Cantor set at all. Is it possible that, when forming this infinite intersection, everything is eventually removed? Note however that $0 \in C_n$ for every n, so $0 \in C$. In fact, with more thought, one sees that any $x \in [0,1]$ which is an endpoint of one the closed intervals involved in the construction of C satisfies $x \in C$. These endpoints constitute an infinite, but ultimately countable, collection of elements of C. To show that C is uncountable, we must show there are many more elements in C besides these endpoints.

Theorem 7.9.10. *The Cantor set is uncountable.*

Proof. We will show that there is a natural 1-1 correspondence between elements of the Cantor set C and infinite binary sequences. Since we proved earlier that the set of infinite binary sequences is uncountable (Theorem 2.2.6) it follows that C is uncountable as well.

In order to describe this correspondence, it will be helpful to devise a binary labelling scheme for the closed intervals involved in the construction of C. To this end, we introduce some notation. Given a closed interval $J = [a, b]$, let us denote the closed left third and right third of J by $J_0, J_1 \subseteq J$. More precisely, J_0 and J_1 are the closed intervals:

$$J_0 \coloneqq [a, \tfrac{2a+b}{3}] \qquad\qquad J_1 \coloneqq [\tfrac{a+2b}{3}, b].$$

We can iterate this process of taking the left/right third. So, for example, J_{001} denotes the right third of the left third of the left third of J. Armed with this notation, we can give a more granular description of the sets C_n, $n \geq 1$. Let $I = [0,1]$, the closed unit interval. We have:

$$\begin{aligned} C_1 &= I_0 \cup I_1 \\ C_2 &= I_{00} \cup I_{01} \cup I_{10} \cup I_{11} \\ C_3 &= I_{000} \cup I_{001} \cup I_{010} \cup I_{011} \cup I_{100} \cup I_{101} \cup I_{110} \cup I_{111} \\ &\vdots \end{aligned}$$

In general, C_n is the disjoint union of the closed intervals $I_{b_1 \cdots b_n}$ as $b_1 \cdots b_n$ runs over all 2^n binary strings of length n. The containments between these intervals can be illustrated by a binary tree. See Figure 7.3.

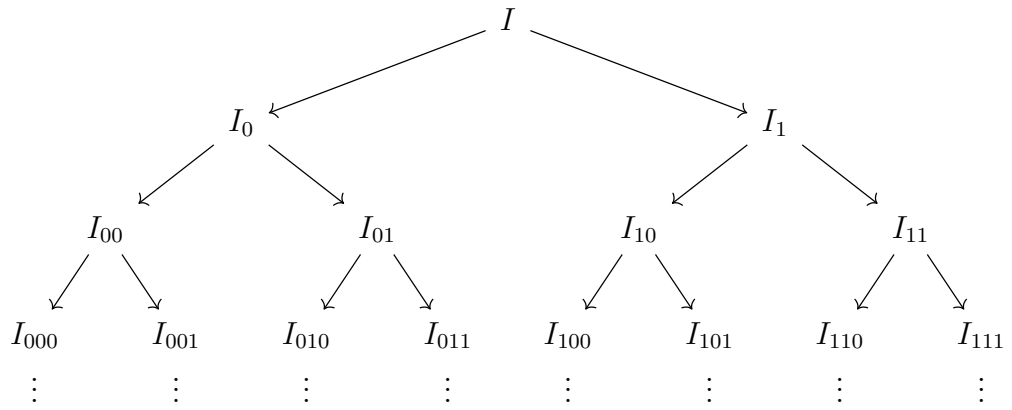


Figure 7.3: Containments between intervals involved in the construction of C.

Now, let us fix $x \in C$ and describe its corresponding infinite binary sequence $\{b_n\}_{n=1}^{\infty}$.

- Since $x \in C_1 = I_0 \cup I_1$ and I_0, I_1 are disjoint, there is a unique $b_1 \in \{0, 1\}$ such that $x \in I_{b_1}$.
- Since $x \in C_2 \cap I_{b_1} = I_{b_1 0} \cup I_{b_1 1}$ and $I_{b_1 0}, I_{b_1 1}$ are disjoint, there is a unique $b_2 \in \{0, 1\}$ such that $x \in I_{b_1 b_2}$.
- Since $x \in C_3 \cap I_{b_1 b_2} = I_{b_1 b_2 0} \cup I_{b_1 b_2 1}$ and $I_{b_1 b_2 0}, I_{b_1 b_2 1}$ are disjoint, there is a unique $b_3 \in \{0, 1\}$ such that $x \in I_{b_1 b_2 b_3}$.
- Continuing this process, we obtain a unique infinite binary sequence $\{b_n\}_{n=1}^{\infty}$ such that $x \in I_{b_1 \cdots b_n}$ for each positive integer n.

Conversely, given an infinite binary sequence $\{b_n\}_{n=1}^{\infty}$, we must describe its corresponding point $x \in C$. The sequence $\{b_n\}_{n=1}^{\infty}$ determines a nested sequence of closed intervals

$$I_{b_1} \supseteq I_{b_1 b_2} \supseteq I_{b_1 b_2 b_3} \supseteq \dots$$

whose lengths $1/3, (1/3)^2, (1/3)^3, \dots$ tend to zero. By the "nested interval theorem" (which you proved on a homework assignment) the intersection of this sequence of intervals consists of a single point x. Moreover, since $I_{b_1 \cdots b_n} \subseteq C_n$ for each n, we have

$$\{x\} = \bigcap_{n=1}^{\infty} I_{b_1 \cdots b_n} \subseteq \bigcap_{n=1}^{\infty} C_n = C$$

so that the point x also belongs to C.

We leave it to the reader to confirm that mappings between infinite binary sequences and points of C just described are inverses of one another. □

Lastly, we show that the Cantor set, despite being uncountable, has measure zero.

Theorem 7.9.11. *The Cantor set has measure zero.*

Proof. Referring to Exercise 7.9.5, we just need to show that, for every $\epsilon > 0$, it is possible to cover C by a sequence of *closed* intervals whose lengths total to less than ϵ. For every n, we have $C \subseteq C_n$ where C_n is the disjoint union of 2^n closed intervals of length $(1/3)^n$. Thus, for every n, we can cover C by a (finite) sequence of closed intervals whose lengths sum to $2^n(1/3)^n = (2/3)^n$. Since $\lim_{n\to\infty}(2/3)^n = 0$, it follows (why?) that C has measure zero. □

The following exercises investigate some further topological properties of the Cantor set.

Exercise 7.9.12. Prove that the Cantor set has no isolated points.

Exercise 7.9.13. Prove that the Cantor set has no interior points.

Exercise 7.9.14. Prove that the complement of the Cantor set is dense.

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